

Safety for intermediaries and supervisors



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1 Legislation

At the end of the 19th century, many children were working in dangerous and unhealthy factories. With the children's law of 'van Houten', this came to an end in 1874. In 1919, the Labour Act was introduced in the Netherlands. In 1934 the Security Act got introduced.

The Health and Safety Act replaced the Safety Act in 1981. The regulations of the Health and Safety Act relate to the rights and obligations of employers and employees. The Health and Safety Act is the Dutch interpretation of the European directives. It is a framework and only gives global rules.



The 'Tweede Kamer' in Den Haag

These rules are stated more detailed in the Health Decision, the Health and Safety Regulations and Policy Rules. The Policy Rules will be replaced the H&S catalog.

Inspection of work activities according to the Health and Safety Act is done by the Social Affairs Inspectorate (formerly the Labour Inspectorate). In the Netherlands, the government determines the rules and regulations. The 'Tweede Kamer' has an important role in this.

1.1 Basic principles of health and safety legislation

The Health and Safety Act provides the fundamental principles for the policy of health, safety and welfare of the employees at their work. The Health and Safety Act is a framework law.

The Health and Safety Act explains that working conditions is a joint responsibility of the employer and the employee. Both have to cooperate to achieve good working conditions.

The term "working conditions" is often abbreviated to "OSH" ("Arbo"). Working conditions are all conditions which are of influence on the occupational health and safety.

The objectives of the H&S legislation (de V&G wetgeving: the legislation of health and safety) are:

- * Protection when carrying out activities.
- * Improving the safety and health of employees at the workplace.
- * Working conditions that are as good as possible.

The Dutch Occupational Safety & Health Act describes the rights and obligations of both the employers and employees in the area of health and safety at the workplace and the conditions in which work is carried out.

The Dutch Occupational Safety & Health Act (H&S legislation) applies to all locations wherever work is carried out.

The Dutch Occupational Safety & Health Act applies to:

- * Employers.
- * Employees (including temporary staff and similar employees including trainees).

The Dutch Occupational Safety & Health Act is related to:

- * Safety.
- * Health.
- * Welfare.



The H&S legislation also applies to the harbour of Rotterdam.

The assumptions behind the Dutch Occupational Safety & Health Act are:

- * The employer protects the health and safety of the employees with respect to all aspects related to the work and adopts a policy to do so.
- * Work must not have a negative influence on the health and safety of employees.
- * Prevention of risks.
- * Support provided by experts or services.
- * Information for and instruction of employees.
- * Consulting and collaborating with the employees (representation).
- * Various employers at one working location collaborate and coordinate their actions.

1.2 The concept of welfare

An employer must guarantee the safety and health and the wellbeing of employees. The employer must pay attention to:

- * The layout of the workplace must be amended to suit the employee.
- * Account must be taken of the personal factors associated with the employee, such as age, gender, knowledge of the language used and skill.
- * Avoid monotonous work.



The workplace must be amended to suit the employee.

1.3 Expert support

Expert support to employer and employee is regulated by H&S experts. These experts can be employees of the H&S service, but also self-employed professionals like doctors or safety experts.

The expert support is also provided internally by the prevention manager or the HSE official. The emergency response team also provides expert support.

The employees who provide expert support internally, have the same legal protection as a member of the council.

The employer can choose multiple ways to ensure prevention and safety on the work floor. For instance:

- * Every employer establishes an internal prevention department.
- * If the company has insufficient in-house expertise, the employer calls upon external support.



The experts are there to help you carry out work activities in a safe way.

The most important task of the experts or services in the area of prevention is to support the employer and employees in the application of the H&S legislation.

The function of experts or services in the area of prevention and inspection is to support the employer and employees in the application of the HS legislation.

The most important tasks of the experts or services in the field of prevention are as follows:

- * To cooperate in the risk inventory and evaluation.
- * To cooperate in accident and incident investigation.
- * To advise and cooperate with the employees (representation).
- * To perform occupational health examinations
- * To supervise employees who because they are ill are incapable of carrying out their work.



The guidance of sick employees is an important task of the health and safety service.

The purpose of health monitoring is:

- * To propose measures to prevent damage to health during the execution of the occupation.
- * To prevent and track down occupational diseases in a timely way.

The employer is obliged to give employees the opportunity to participate in a medical examination to prevent and limit the risks to health. Such a check-up is called a PAGO (Periodiek ArbeidsGezondheidskundig Onderzoek) this is a periodic work-related medical check-up. It is also called PMO (Preventief Medisch Onderzoek; a preventive medical check-up).



The employer is obliged to offer the possibility to undergo a PAGO (Periodiek Arbeidsomstandigheden Onderzoek).

1.4 Care for people other than employees

The duties of the employer do not only apply to its own employees, but also to other people in the workplace.

People for whom the employer has the obligation to take effective measures in order to avoid risks as far as possible are:

- * Employees of other employers who are present at the workplace.
- * Suppliers, visitors, patients, etc.
- * Neighbours.

Multiple employers at a work place are obliged to collaborate in the execution of the measures related to the health and safety of the employees and to coordinate their actions in this respect.

1.5 Supervision in H&S legislation

The Labour Inspectorate supervises the implementation of the Health and Safety Act and the Working Hours Act in the Netherlands. The jurisdiction of the Labour Inspectorate is big. The Labour Inspectorate may, without a warrant, do research at companies.

The Labour Inspectorate is (or similar inspection services are) authorised:

- * To monitor compliance with the Health and Safety legislation.
- * To instigate an investigation into the cause of an accident.
- * To ask employees for valid proof of identity.
- * To set requirements with respect to the manner in which the Health and Safety legislation must be complied with.
- * To stop work activities if there is a serious danger to the people present.
- * To draw up an official report following an offence.
- * To give a warning if the regulations are not observed.



The logo of SZW (Sociale Zaken en Werkgelegenheid).

A general rule with respect to responsibility for working conditions: the employer who gives instructions, provides supervision and issues equipment is also responsible for compliance with the H&S legislation.

The government's Health and Safety Inspection Service is allowed to impose the following sanctions:

- * Following a sanctionable offence, to draw up a report concerning the fine for the employer/ employee.
- * Following a finable offence, to draw up a report concerning the fine for the employee.



Violations of the Health and Safety Act are covered by the Economic Offences Act (wet economische delicten).

1.6 Basic principles of environmental legislation

Controlling the climate is becoming increasingly important for us. Therefore one has to show some attention to the environment on his own workstation.

Ensure that due to poor order and tidiness the environment isn't polluted. A healthy environment starts with you!

The objectives of the environmental legislation are:

- * To protect man and the environment against the harmful effects of activities.
- * To limit emissions and waste flows.
- * To ensure the effective and correct removal of waste materials.

1.7 Basic principles of working hours legislation

The objectives of the Working Hours Legislation are:

- * To delineate maximum working hours and minimum rest hours to prevent safety and health being endangered.
- * To promote the possibility to combine work and care tasks/ family.



The assumptions behind the Working Hours Legislation are:

- * That the personal situation of the employee will be taken account of.
- * Coherence to the Health and Safety Policy.

The Working Hours Legislation applies to all employees below a certain salary level.

1.8 Basic principles of health and safety legislation related to agency work

The health and safety legislation ensures that temporary agency workers, with respect to health and safety at work, get the same level of protection as the other employees of the company that hires them.

A company that hires temporary agency workers is responsible for their safety and health during employment.

The supervisor of the hiring company is responsible to provide specific information and instructions to the temporary agency workers.

1.9 European Directives with respect to (Dutch) national legislation

The legislation for the Member States are determined from the European Community. Most legislation in our country is derived from European framework directives. Two major framework directives are:

- 100A for the safety of technical products.
- 118A for the safety of the users of the products.

The framework directives contain only the minimum requirements that the Member States must meet.

A European directive must be incorporated in national legislation within a certain period.

It is the mandatory responsibility of the manufacturer or the importer to attach CE labels. The CE label can be attached based on a technical file and a declaration of conformity, in which the manufacturer or importer confirms that the product conforms to the safety requirements specified in the Directive.

Only CE-marked equipment can be marketed.

The CE label on products or work equipment stands for:

- * The CE Label on a product indicates that one can assume that the product (for instance: work equipment, machines or PPE) meets the European directive with respect to safety.
- * Only CE labelled equipment may be sold and used in the EU (European Union).

The CE label indicates that work equipment meets the requirements of CE-labelling.



The flag of the European Union.



CE logo

Examination questions: Legislation

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 What measure can the Inspection Service take?

☐ C Hold a meeting to discuss working conditions.
☐ C Impose a fine.
☐ C Carry out a health check.
- ☐ 2 What is the objective of the Health and Safety legislation?

☐ A Providing the best possible working conditions.
☐ B Reducing the number of accidents at work.
☐ C Improving the atmosphere at work.
- ☐ 3 Which of the measures below can the Inspection Service take?

☐ A Have one or more employees fired.
☐ B Have the work stopped.
☐ C Carry out medical examinations.
- ☐ 4 What does the CE mark on an article indicate?

☐ A The article meets the minimum European requirements.
☐ B The article may only be traded within the EU.
☐ C The manufacturer of the article satisfies the European Health and Safety Act.
- ☐ 5 What is one of the objectives of the environmental management act?

☐ A Control waste flows during the entire production process.
☐ B Protect the environment against harmful business activities.
☐ C Protect human beings from hazardous substances.
- ☐ 6 What does 'ARBO' describe?

☐ A ARBO describes the working conditions (occupational health and safety).
☐ B ARBO describes the moments in which the Labour Inspectorate has to be informed.
☐ C ARBO describes the risks of working with hazardous substances.
- ☐ 7 What is the applicability of the Health and Safety legislation?

☐ A The legislation describes the rights and obligations of both employer and employee in the area of health and safety at home.
☐ B The legislation describes the rights and obligations of both employer and employee in the area of health and safety at the workplace.
☐ C The legislation describes the rights and obligations of the employee in the area of health and safety at.
- ☐ 8 At which locations does the H&S legislation apply?

☐ A All locations where work is carried out.
☐ B All locations where no work is carried out.
☐ C All locations with special working conditions.
- ☐ 9 The H&S legislation applies to:

☐ A The employer.
☐ B The employees.
☐ C Both answers are correct.

10 The H&S legislation does not relate to:

- A Health
- B Welfare
- C Security

11 The H&S legislation applies to:

- A The employer
- B The employee
- C Both answers are correct

12 What is not an assumption behind the H&S legislation?

- A Support provided by experts or services.
- B Various employees at one working location do not have to collaborate or coordinate their actions.
- C Consulting and collaborating with the employees (representation).

13 What is an important task of the experts or services in the area of prevention?

- A Cooperate in the risk inventory evaluation.
- B Supervise employers.
- C Perform medical examinations.

14 Important objectives of the working hours legislation are:

- A To delineate maximum working hours and minimum rest hours to prevent safety and health being endangered.
- B To promote the possibility to combine work and care tasks/family.
- C Both answers are right.

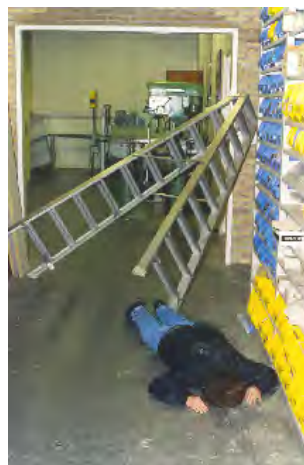
15 Principles of the CE identification symbols are:

- A A CE mark must be attached by the manufacturer and importer.
- B The CE mark is not required for every piece of equipment which is marketed and used.
- C A CE mark must be attached by the manufacturer or importer.

2 Dangers, risks and prevention

People say that "100% safety does not exist". Yet, no one wants to see accidents happen. Dangers are everywhere. We need to ensure that the risk of a potential hazard is as low as possible.

A door isn't dangerous, we think. But in a playgroup, a closing door can be a great risk! The risk of fingers of a toddler being jammed between the door! A danger can be a small risk in one situation, but a big risk in another.



There is always a risk of falling when working with ladders.

Assessing risks is also dependant on the individual: your perception determines the amount of your risk assessment. If you have experienced a fire, fire risks will always be estimated higher.

By taking preventive measures, the risks of an accident and the possible consequences become smaller.

2.1 Concepts: danger and risk

"A hazard" is a characteristic of a substance or product that can cause an incident. "A hazard" can also cause a situation that can result in an incident.

The risk is determined by the degree of probability of a hazard causing an accident. The extend of the risk is determined by the degree of damage (the consequence). To estimate the extend of the risk, a formula is made: $\text{Risk} = \text{probability} \times \text{effect}$

2.2 Recognising risks and dangers

A number of hazards at the workplace are:

- * Height (Falling).
- * Fire.
- * Explosions.
- * Poisonous substances.
- * Noise.
- * Radiation.
- * Moving parts (e.g. being trapped, hit).
- * Falling objects, loads.
- * Vehicles, traffic.

Possible sources of danger are:

- * The type of work.
- * The workplace or surroundings of the workplace.
- * Knowledge and expertise.
- * Products and materials.
- * Behaviour of employees.
- * Ability.



Working on ladders is dangerous.

Work or working conditions with possible dangers and risk are for example

- * Working with hazardous substances.
- * Working with dangerous machinery.
- * Working with high pressure.
- * Working with radiation sources.
- * Digging and working at or in excavations.
- * Working with electricity.
- * Working in cold.
- * Working in heat.
- * Working in noisy surroundings.
- * Working at height.

Factors that increase the risks at the workplace are:

- * Untidy workplace
- * Too much or too little light.
- * Vibrations.
- * Weather conditions.
- * Dust or contaminated air.
- * Working in the vicinity of radiation sources.
- * Traffic in the vicinity.
- * Working in a confined space.
- * Workplace that offers little room to manoeuvre.
- * A difficult-to-access workplace.
- * A workplace with few escape routes.
- * Hazardous activities in the vicinity.
- * Complex work activities.

2.3 Prevention hierarchy

Measures which can be taken to prevent or reduce risks are (preventive measures):

- * Preventing a risk from increasing.
- * Taking measures to reduce the risk.
- * Prevent an accident from occurring.
- * Prevent damage to health.
- * Prevent environmental damage.

The types of preventive measures and the order in which they must be taken are:

- * Firstly source-oriented measures: eliminating hazards, removing the cause, prevention, choosing another, safer working method.
- * Secondly limiting or reducing the risk: collective protection and guards, suitable working methods and equipment. limiting exposure in time, frequency, intensity, number of people.
- * Thirdly if risks are still present: personal protection equipment.

Prevention of unsafe actions and unsafe situation is the safest way of working.



Working with a high-pressure sprayer is risky.



Tripping hazard caused by a cluttered workspace.



Placing welding shields is also a preventive measure.

An unsafe action is an action that is not carried out in accordance with the safety instructions and that can lead to an accident.



Working with this hook is an unsafe action.

An unsafe situation is a situation in which work is carried out without complying with the conditions for safe working and that can lead to an accident.



If you find this hook, you have recorded an unsafe situation.

Examples of unsafe actions:

- * Working without work permits or permission in situations where this is imposed.
- * Not using the prescribed personal protection equipment.
- * Bypassing safeguards.
- * Using defect tools.
- * Using tools incorrectly.
- * Moving a load incorrectly.

Examples of unsafe situations:

- * No or insufficient light.
- * No or obstructed escape routes.
- * Untidy work floors.
- * Not safeguarded or insufficiently safeguarded equipment or machines.

The best way to solve and prioritise an unsafe situation is:

- * Remove the cause of the unsafe situation.
- * Screen off or safeguard the situation.
- * Warn others.
- * Calling in others.

When you see that someone is taking unsafe actions you should:

- * Report to a supervisor
- * Halt the action

Associated measures are:

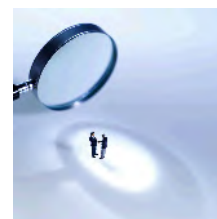
1. Training, information, instruction.
2. Identifying, warning, labelling.

Organisational measures are limiting exposure in time, frequency, intensity or number of people.

2.4 Risk management and implementing policy

Every employer is obliged to analyse the risks at the workplace. This is called a risk inventory (RI&E) or RA (Risk Analysis). The Risk Inventory and Evaluation (RI&E) consists therefore of three steps:

- * Identifying the hazards.
- * Making an inventory of the risks.
- * Evaluating the risks.



When an RI&E takes place, the organization will be underneath an H&S magnifying glass to detect risks.

A risk management policy is based on the RI&E and has the following parts:

- * Plan the policy (methods, distribution of tasks).
- * Planning of the policy (methods, task distribution) Details of the policy (objectives and resources) based on an RI&E: Risk Inventory and Evaluation.
- * Implement the policy.
- * Evaluate and adapt the policy.



One can find all RI&E's on the following website: www.rie.nl

2.5 (Task) Risk Analysis and LMRA

A task risk analysis is an analysis of the hazards that are associated with the execution of (high-risk) tasks with respect to the health and safety of the employees.

Carrying out a task risk analysis is very important because this is the time when all risks are investigated properly and in a structured manner.

LMRA = Last Minute Risk Analysis. The LMRA is a practical method to examine at the workplace itself whether the work can be safely executed.

The most important points of attention of the LMRA are:

- * Before starting work activities or executing a new task.
- * Execute with changes of working conditions.
- * Also execute LMRA with routine activities.
- * You will execute the LMRA on your working place.



When performing an LMRA, you record a bump hazard.

2.6 H&S action plan

The employer should draw up an H&S action plan following the RI&E. The H&S action plan is a written plan of execution to improve the working conditions. The employer should discuss the progress and execution of the action plan with the employees.

Components of the H&S annual action plan are:

- * The objectives to be achieved.
- * Concrete preventive measures.
- * Equipment/ resources to be used (organisational, financial, material).
- * Task distribution.
- * Responsible for the implementation.
- * Period.

2.7 Communication of risks and control measures for the work floor

Communication about risks and management should preferably take place at the workplace and prior to commencing the activities.

Examples of communication about risks and management are:

- * Meeting with client and contractors
- * Consultation between management and operational staff
- * Crew transfer, information transfer when changing teams
- * The VGM-meeting (SHE-meeting/toolbox meeting) with operational staff



Use different audio-visual means in communication.

The general guidelines with respect to communicating risks and control measures to temporary agency workers are:

- * General basic information related to the risks during the introduction given to the temporary agency worker by the intermediary: - regarding the activities and the circumstances - regarding the job and the workplace - the hiring company and the temporary employment agency have a shared duty with respect to providing information to the temporary agency worker, the temporary employment agency must provide the temporary agency worker with information about the workplace and the job requirements.
- * Specifically workplace-related: - during the initial introduction of the temporary agency worker by the hiring company - preferably at the workplace - preferably by the involved operational supervisor of the hiring company - prior to commencing the activities

2.8 Safety aspects in work planning

The aim of work planning is to establish a method that is safe and doesn't cause health damages.

During the work planning the following elements have to be considered to make sure work is safe:

- * Time planning for the entire work.
- * The order in which the activities can be safely carried out.
- * Which materials, tools and machines are required to perform the job safely.
- * The required time and measures that are needed to perform the work in a healthy and safe manner.

2.9 High-risk activities

Not all work activities are high-risk. High-risk activities are:

- * Work activities that take place in a high-risk environment.
- * Performing high-risk tasks.

Within a company high-risk activities are recorded based on the RI&E and (task-)risk analyses.

Before high-risk activities can be started, you must be sure that all necessary conditions have been fulfilled, these conditions are:

- * Availability and knowledge of the work permit.
- * Availability and knowledge of the (Job) Hazard Analysis.
- * References to precautionary measures in the company regulations of the customer.
- * Only starting work when all of the conditions are met.
- * Carrying out a Last Minute Risk Analysis (LMRA).



Degreasing of materials can be risky.

It can be that during the execution of the work situations occur which result in the conditions being different from those described in the work plan. The supervisor should:

- * Have the activities halted.
- * Modify the (job) hazard analysis and discuss it with all employees involved.
- * Discuss the modified (job) hazard analysis with the responsible supervisor and/or customer and if necessary the prevention officer/ safety expert.

During the execution of high-risk activities a supervisor should:

- * Regularly check whether the work is being performed in accordance with the agreements made.
- * Make use of the agreed control measures.
- * Be sufficiently present.
- * Intervene if the agreements that have been made are not adhered to.



The supervisor should set the example to his staff.

Examination questions: Dangers, risks and prevention

The correct answers of these questions can be found in annex 1 of this book.

1 How is the concept of risk described?

- A Risk = probability X effect.
- B Risk = hazard X probability.
- C Risk = effect X hazard.

2 Hazards at the workplace are:

- A Noise.
- B Radiation.
- C Both answers are correct.

3 What is not a source of dangers and risks at the workplace?

- A Equipment.
- B Behaviour of the employee.
- C Problems at home.

4 In which way can prevention be tackled practically?

- A Preventing unsafe situations
- B Preventing unsafe actions.
- C Both answers are correct.

5 What is an unsafe situation?

- A An unsafe situation is a situation in which work is carried out with complying to the conditions for safe working and that can lead to an accident
- B An unsafe situation is a situation in which people are not wearing personal protective equipment when there are mandatory signs.
- C An unsafe situation is a situation in which work is carried out without complying with the conditions for safe working and that can lead to an accident.

6 Which of the actions stated below is an unsafe action?

- A Working with permits or permission in situations where this is imposed.
- B Using tools incorrectly.
- C Using the prescribed personal protection equipment.

7 What is the best way you can deal with unsafe situations?

- A Removing the cause.
- B Calling others.
- C Warning others.

8 The various steps in an RI&E are:

- A
 1. Evaluating the risks
 2. Identifying the hazards
 3. Making an inventory of the risks
- B
 1. Identifying the hazards
 2. Making an inventory of the risks
 3. Evaluating the risks
- C
 1. Identifying the hazards
 2. Evaluating the risks
 3. Making an inventory of the risks

9 What does LMRA stand for?

- A Labour Market Regulatory Authority.
- B Last Minute Risk Analysis.
- C Last Minor Random Analysis.

10 What is the objective of the work planning?

- A To determine the time needed to finish the work activities.
- B To determine a working method that is safe and does not cause any health damage.
- C To determine a way to finish the work activities as quick as possible.

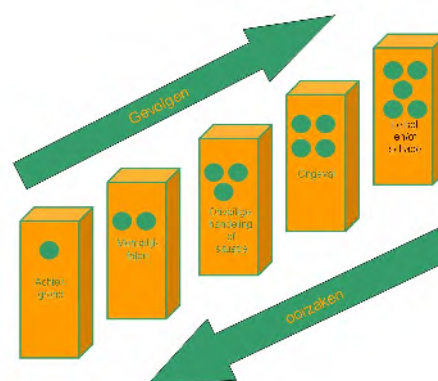
11 What is an approach of the intermediary in the event of posting employees for high-risk tasks?

- A Agreements with the hiring company about how the temporary agency worker will be informed (instructions, training, tests, etc).
- B Information to the temporary agency worker regarding the high-risk tasks.
- C Both answers are correct.

3 Accidents: causes and prevention

Most accidents are the result of one or several causes. American psychologists Heinrich and Lateiner have done research in the last century and developed the domino theory

If one domino falls, the next one may remain standing and therefore (thankfully) nothing happens. If the stone "unsafe action or situation" falls and the domino stone "accident" remains standing, then you have a situation without consequences. If the domino stone "accident" falls, it means you have an accident.



The domino theory for investigating accidents.

If the domino "injury or damage" remains standing, it means you have a near-accident. Will this stone drop you will have an injury or damage.

Accidents which need to be reported to the Labour inspection are:

- 1: Deadly accidents.
- 2: Accidents with permanent injury.
- 3: Accidents involving hospitalization.

3.1 Concepts

An accident is an undesired event that happens during the work resulting in damage (to material, the environment or loss of time) and/or injury.

A near-accident is an undesired event that does not result in damage or injury, which under somewhat different conditions could have led to a severe accident.

An incident is an undesired event without damage and/or injury.

3.2 Accident theory

The direct causes of accidents or incidents are:

- * Unsafe actions.
- * Unsafe situations.

The causes and consequences of an accident, using the cause-and-effect series are:

- * 1. Background (organisation).
- * 2. Human failure.
- * 3. Unsafe actions and/or situations.
- * 4. Accident.
- * 5. Injury and/or damage.

The possible points of attention of the cause-and-effect series for the policy to be adopted in order to reduce the number of accidents are:

- * An accident is the result of various causes at various levels within the company, the policy to be adopted must therefore focus on various levels (including management, supervisors, operational employees).
- * The causes can originate at various levels within the company (including management, supervisory and operational employees).

The accident pyramid is a diagram showing the statistic connection between the number of dangerous situations or actions and the number of accidents. This can also be shown in the shape of an iceberg. The accident pyramid is therefore sometimes called the iceberg theory.

Unsafe actions or situations are caused by:

- * Human error. Human error is promoted by: - Person-related factors - Task-related factors

Some examples of personal factors are:

- * Insufficient knowledge.
- * Insufficient experience.
- * Insufficient motivation.
- * Insufficient attention.



This unsafe act may cause you to fall.

In the case of task factors employees do not have the correct resources to properly carry out their task. Examples of task factors are:

- * Insufficient and/or incorrect management and supervision.
- * Incorrect working method.
- * Having insufficient time for the activities.
- * Required equipment and/or tools not being present.
- * Equipment and/or tools being not effectively maintained.



Working with good tools is a good start of the job.

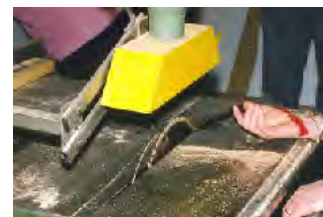
The accident pyramid for executing the policy to reduce the amount of accidents, has as a consequence that when preventing all unsafe actions and situations, this automatically leads to fewer accidents.

The policy in companies should therefore be aimed at tackling all unsafe actions and situations and at all levels within the company. Management is responsible for the policy and should therefore carry out the following action points:

- * Establishing a long-term plan to improve the working conditions.
- * Delegating tasks and powers of authority to departments and employees in the company.
- * Arranging the correct organisation of the working conditions policy.
- * Arranging an effective meeting structure with respect to the working conditions and responsibilities.
- * Working out the working conditions policy in procedures and regulations.
- * Arranging sufficient training and instruction.

The policy also results in action points for supervisors:

- * Supervising compliance with procedures and regulations.
- * Carrying out periodical inspections.
- * Giving instructions and training regarding health risks in the workplace.
- * The systematic reporting of incidents.
- * Planned approach to tackling unsafe actions and situations.
- * Discussing problems and complaints related to the working conditions policy during the department and progress meetings.
- * Regular evaluation of the results of implemented improvement points resulting from unsafe actions and situations.



Use a push stick to safely push a small piece of work.

3.3 Accident prevention

Prevention is avoiding that accidents occur. Most important for accident prevention is and remains removing the direct causes. The risks can be caused by:

- * The human being.
- * The organisation.
- * The technology.
- * The surroundings.

The great majority of accidents (approximately 80%) is caused by human actions. Unsafe actions can be prevented in three ways:

- * Obtaining knowledge.
- * Possessing skills.
- * Behaviour, for instance being concentrated when working, tidy workplace.

Examples of preventive measures around the organization are:

- * Effective working methods.
- * Supervision by supervisors.

Examples of preventive measures around technique are:

- * Use of machines that have a safe technical design (CE marking).
- * Maintenance of tools.
- * Ergonomic design.

Many accidents can also be prevented by ensuring proper protection of the employee against dangers in the surroundings. This can involve work equipment suitable for the surroundings, for example suitable for extreme weather, and a safe layout of the work environment.



When performing work outside, the environment is a potential hazard.

3.4 Dealing with incidents

Especially in the case of accidents with serious injuries, fast action can prevent a lot of problems and possibly saves a life. What should you do when an accident occurs:

- * Act immediately to reduce the consequences of the accident. Ensure you do not become a victim yourself when undertaking a rescue attempt.
- * Report the company emergency response team and your supervisor or by severe accidents 112.
- * Record: Fill in the investigation form with the victim or witnesses.
- * Investigate the accident and make pictures.
- * Action plan (plan of approach): Draw up a final report with recommendations in order to prevent a subsequent accident.

In the case of an accident with serious injury you should always remember the following:

- * The accident must be reported to the immediate supervisor.
- * Accompany the victim to the medical department/ service.
- * Direct measures to avoid recurrence.
- * Follow the internal instructions.



Fast, but safe handling is important.



1-1-2: When every second counts.

With near-accidents or unsafe situations/actions it's a good thing to act immediately to prevent growing risks. Points of attention are:

- * Take direct measures (identify situations / halt or arrange to halt actions).
- * Report to the supervisor.

3.5 Recording incidents

Incidents, accidents, near-accidents, unsafe situations and operations have to be registered. The goal of this registration is:

- * Record details for the policy of the company.
- * Learn lessons from it and take measures to prevent repetition.
- * Comply with the statutory requirements in this area.

3.6 Accident investigations

All accidents, with or without serious consequences, should be reported to the management of the organisation. This also applies to unsafe situations, unsafe actions and near-accidents. The reason for this is to avoid the recurrence of similar accidents in the future.

An accident investigation consists of the following:

- * Investigate at the site of the accident.
- * Gather facts and data regarding what led to the accident.
- * Interview people involved and witnesses.
- * Analysis of investigation results.
- * Final report containing recommendations / action points to avoid recurrence.



The consequences of an explosion in a factory.

Points of attention for the end report of the accident investigation are:

- * Description, facts and details.
- * Analysis.
- * Recommendations in order to prevent a subsequent accident.

The hiring company is responsible for the accident investigations.

The temporary employment agency is responsible for the accident insurance.

The objective of the Wet Verbetering Poortwachter (Eligibility for Permanent Invalidity Benefit (Restrictions) Act) is to improve the opportunities for reintegration of sick employees within or outside of the company.

The obligations that the temporary employment agency has within the framework of the Eligibility for Permanent Invalidity Benefit (Restrictions) Act are:

- * To draw up and execute a plan of approach for the reintegration of the ill employee.
- * If the temporary employment agency is negligent with respect to the plan of approach, the obligation to pay wages can be extended by one year.

The hiring company is obliged to report occupational accidents involving temporary agency workers to the government's Health and Safety Inspection Service.

3.7 Action Plan (plan of action)

Following the results of Risk Inventory Evaluation or Job Hazard Analysis (JHA) and the proposed measures that the workplace is going to take, a written plan of approach is prepared.

The purpose of an plan of approach is:

- * To ensure that the results of the investigation lead to the implementation of effective measures.
- * To ensure that recurrence of incidents is avoided.

The action plan should have the following content:

- * The measures to be taken.
- * The required resources, people to employ and any costs.
- * Period for implementation.
- * Person responsible for the implementation.
- * Follow up.
- * Evaluation of measures.

Examination questions: Accidents: causes and prevention

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 What is the cause of most accidents?

☐ C Human error.
☐ C Poor working conditions.
☐ C Lack of proper tool maintenance.
- ☐ 2 According to Lateiner, which factors play a role in the occurrence of an accident?

☐ A Background, human error and unsafe situations and operations.
☐ B The surroundings, human error and unsafe situations an operations.
☐ C Environment, human error and unsafe situations and operations.
- ☐ 3 Drawing up procedures for taking action when accidents occur, is a measure towards:

☐ A Surroundings.
☐ B Organization.
☐ C Human beings.
- ☐ 4 Which parts, at least, have to be included in the final report of an accident investigation

☐ A Drawings of the accident, witness statements and measures taken.
☐ B Analysis, conclusions and recommendations.
☐ C Interview reports, relevant documents and photos of the location.
- ☐ 5 The two direct causes of an accident or incident are:

☐ A Unsafe actions or broken equipment.
☐ B Unsafe actions or unsafe situations.
☐ C Human error or broken equipment.
- ☐ 6 What is not an examples of person-related factors that can lead to an incident?

☐ A Insufficient planning.
☐ B Insufficient attention.
☐ C Insufficient experience.
- ☐ 7 To which person must an accident be reported to ?

☐ A The Labour Inspectorate.
☐ B Your immediate colleagues.
☐ C The immediate supervisor.
- ☐ 8 What must one do with all accidents or near accidents?

☐ A Provide more PPE's.
☐ B Record the happening.
☐ C Identify the factors that caused the accidents.
- ☐ 9 Who is responsible for the accident investigations?

☐ A The hiring company.
☐ B The temporary employment agency.
☐ C The Labour Inspectorate.

10 Who is responsible for the absence due to illness?

- A The hiring company.
- B The temporary employment agency.
- C The Labour Inspectorate.

11 What is the objective of the Wet Verbetering Poortwachter (Eligibility for Permanent Invalidity Benefit (Restrictions) Act)?

- A To improve the opportunities for reintegration of new employees within or outside of the company.
- B To improve the opportunities for reintegration of sick employees within or outside of the company
- C Both answers are correct.

12 What is an obligations that the temporary employment agency has within the framework of the Eligibility for Permanent Invalidity Benefit (Restrictions) Act?

- A Both answers are correct.
- B If the temporary employment agency is negligent with respect to the plan of approach, the obligation to pay wages can be extended by one year.
- C To draw up and execute a plan of approach for the reintegration of the ill employee.

4 Safety conduct

As a manager, you often hear from your employees: "If I need to work safely, activities take more time. But I don't have that time". It's difficult to adjust behaviour from employees, but it starts with you as a manager.

As a manager, you must lead by example. Both in safe working and in behaviour. Only then you can convince your employees of the necessity of safe working.

As a manager you know which employees always take the most risks or have problems with specific activities. You must slow down one employee, and the other you may need to train.



Before starting new work activities, a good instruction is important.

It's your role as manager to arrange this.

As a manager, take "Safe Working" with you in appraisal- and assessment interviews of employees. Do not wait with corrections, correct immediately when rules are violated.

4.1 Conduct and safety

The supervisor should influence the behaviour of the employee. This is a very important part of the HSE policy, because:

- * Behaviour or the acceptance of behaviour is an important cause of accidents.
- * After the policy focused on technology and organisation, the human factor is addressed through 'behaviour'.

The cause of unsafe behaviour can lie with the supervisor as well as the operational employee. Unsafe behaviour can, for example, be caused by tolerating violations:

- * Tolerating violations, lack of leadership.
- * Incorrectly assigning tasks (lack of skill, not being able to).
- * Lack of knowledge/ information (not knowing).
- * Unwillingness (not wanting to).
- * Conflicting objectives/ assignments.
- * Lack of example function.



Unsafe and dangerous behaviour.

In a company, some factors affect human behaviour so that work can be performed safely:

- * Policy focused on better safety performance.
- * Clearly formulated assignment.
- * Clarity regarding desired and undesired behaviour.
- * Clarity about the priority of safe working.
- * Sanction and reward policy.
- * Communication and awareness campaigns.
- * Company culture.

The role of the manager is important for the human behaviour because:

- * The supervisor determines the working method.
- * The supervisor supervises.
- * The supervisor has the required information about the employees, the work and the circumstances.

In a company there are several organizational elements which influence the human behaviour so that work is safe:

- * Clear meeting structure.
- * Safety behaviour observations.
- * The right man at the right place.
- * A proper introduction procedure.
- * Correctly dealing with questions and complaints.
- * Arranging the correct provisions, including sanitary facilities, canteen.

General rules of conduct to work safely within a company:

- * Employees comply with the safety regulations.
- * Employees call each other to account for unsafe actions.
- * Employees immediately intervene in unsafe situations.
- * Employees ask questions in the event of unclear situations.
- * Setting a good example.
- * Responding consistently.
- * Supervising and correcting.
- * Also being critical towards yourself.
- * Gaining advice in unclear situations.
- * Taking into account the opinions, proposals, wishes and advice of your employees.
- * Giving clear and convincing instructions to employees (how and why).

Safe working stimulate people to show the following behaviour:

- * Taking care of one's own health and safety and that of other people involved.
- * A positive attitude.
- * Adhering to the safety regulations and following up instructions.
- * Calling other employees to account for unsafe actions.
- * Reporting unsafe situations and incidents and intervening appropriately.
- * Attention to personal hygiene, order and tidiness.

Alcohol and drugs have a negative effect on our ability to react. Some effects of alcohol and drugs are:

- * Reduced alertness.
- * Reduced or problematic working.
- * Less effective assessment of situations.
- * Breaking down the barriers to displaying unwanted behaviour.
- * Overestimation of one's own capabilities.
- * Increased work load on colleagues and disruptions in the organisation of the work.

Working in a safe way also means working in a fun way.



As a manager you always need to have a positive attitude.

Working under the influence of medication which affects the reaction time can be dangerous. Discuss this with your manager!

In a working environment alcohol and drugs use is therefore not allowed. It causes unsafe situations and disrupts the work situation. Some rules:

- * Withhold from alcohol and drug use during work.
- * Inform the supervisor of problematic use and withhold from dangerous activities or work that could entail a hazard to others.
- * Search for help in the event of chronic, problematic use.
- * Do not work under the influence of alcohol and drugs.



Working under the influence of alcohol is forbidden.

4.2 Influencing behaviour

The consultant has in promoting safe work by the temporary following tasks:

- * The correct selection for the function with respect to: the request from the hiring company, the job content and the capabilities and experience of the temporary agency worker.
- * Verification of the basic knowledge regarding health and safety of the temporary agency worker (e.g. Basic safety VCA).
- * Evaluation of the temporary agency worker after they he/she has performed the assignment.
- * Information provided to the temporary agency worker concerning the necessity, the advantages and the importance of safe working.



Non-compliance with regulations can result in large fines.

In the following ways the mentality of the temporary agency worker can be improved:

- * Try to obtain as much insight as possible into the nature and the capabilities of the temporary agency worker.
- * Do not assign jobs to the temporary agency worker that require him/her working above or below his/her capabilities.
- * Take into account the opinions, proposals, wishes and advice of the temporary agency worker himself.
- * Inquire into any negative working experiences.
- * Inform the temporary agency worker about how you and the hiring company think about him, including his performance in the area of safety.
- * Evaluate the temporary agency worker as objectively as possible.
- * Ensure that when corrective action is taken, the self esteem of the temporary agency worker as a professional or as a human being is not affected.
- * Ensure that there is a healthy amount of self criticism.
- * Prevent emotional situations developing where possible.
- * Do not be 'bossy' as an intermediary.
- * Try to listen attentively to what someone else has to say to you.

4.3 Order and tidiness (good housekeeping)

Order and untidiness are one of the things that are not conducive to safe behaviour. Possible results of order and tidiness are:

- * Prevention of incidents.
- * Prevention of contamination, environmental damage.
- * Higher efficiency, time benefit.
- * More pleasant working environment, positive influence on motivation and mentality.
- * Prevention of loss of tools and materials.



A cluttered floor gives a higher risk of falling.

Examples of good housekeeping are:

- * Clearing up the area.
- * Removing/ storing residual material.
- * Storage system for tools and materials.
- * Hanging up cables (safety hooks) or working them away.
- * Good layout of the workshop/ yard.



Tripping hazard caused by an untidy environment.

4.4 Stumbling, slipping and tripping

Important causes of stumbling, slipping and tripping are:

- * An uneven or loose surface.
- * Small difference in height.
- * Larger height differences.
- * Causes related to the limitations of the human body, such as a step that is too high.
- * Loose tiles.
- * Running.
- * Slippery / Polished floors.
- * Unsuitable shoes.



A poor staircase with loose tiles.

Methods and measurements to reduce dangers with walking are:

- * Design: in the design and the preparations attention must be paid to preventing unsafe situations.
- * Good housekeeping: ensure an orderly and tidy workplace.
- * Correct unsafe situations as quickly as possible.
- * Pay adequate attention when moving by foot.
- * Do not carry objects that obstruct the sight.
- * Follow the planned routes and paths.
- * Wear suitable shoes.

Methods and measurements to control hazards while walking are:

- * By separating man and hazardous situations by using physical fencing.
- * Suitable lighting.

Examination questions: Safety conduct

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 Which of the examples below is a cause of an accident by falling and stumbling?
 - ☐ A A flat anti-slip floor.
 - ☐ B Loose cables and ropes.
 - ☐ C A tidied workplace.
- ☐ 2 What is not a cause of unsafe behaviour?
 - ☐ A Lack of example function.
 - ☐ B Unwillingness (not wanting to).
 - ☐ C Good formulated objectives.
- ☐ 3 Which policy element influences human behaviour so that work can be done safely?
 - ☐ A A sanction policy.
 - ☐ B Clarity about the priority of safe working.
 - ☐ C Policy focused on good working machines.
- ☐ 4 Why is the role of the supervisor important for human behaviour?
 - ☐ A The supervisor is a rolemodel for employees.
 - ☐ B The supervisor makes sure the working method is determined by the employees.
 - ☐ C The supervisor has the required information about the employees, the work and the circumstances
- ☐ 5 A general rule of behaviour for safe working is?
 - ☐ A Being critical towards yourself.
 - ☐ B Employees should not intervene in unsafe situations but report them to the supervisor.
 - ☐ C Record opinions, proposals, wished and advice of employees and store them.
- ☐ 6 Which of the statements below is not behaviour that promotes safe working?
 - ☐ A Attention to personal hygiene, order and tidiness.
 - ☐ B Only think about your own health and safety.
 - ☐ C Reporting unsafe situations and incidents and intervening appropriately.
- ☐ 7 The influence of alcohol and drugs on safe working causes:
 - ☐ A Reduced or problematic working.
 - ☐ B No changes in behaviour.
 - ☐ C Good assesment of situations.
- ☐ 8 What is expected with respect to alcohol and drug use in regards to working safely?
 - ☐ A Use of alcohol and drugs is allowed when supervisors know about it.
 - ☐ B Searching for help in the event of chronic, problematic use.
 - ☐ C The supervisor does not have to be informed about occaional problematic use of alcohol or drugs.

9 A task of the intermediary related to the promotion of safe working by the temporary agency workers is?

- A Both answers are correct.
- B Information provided to the temporary agency worker concerning the necessity, the advantages and the importance of safe working.
- C Verification of the basic knowledge regarding health and safety of the temporary agency worker (e.g. Basic safety VCA).

5 Tasks, rights, duties and discussions

The employer and employee both have (according to the Working Conditions Act) rights and duties. Important tasks of the employer are for example: making sure the working area is safe and healthy for the employee. According to the H&S legislation the employees are entitled to this.

The employee is obliged to participate in the education and training. Employees may not bridge or remove protections. The employee must report dangerous situations to his supervisor so that he can take actions.

According to the H&S regulation, the employer and employee have joint responsibility for safety and health. Collective agreement is very important.

This agreement is governed by the CA (Councils Act). A company can arrange consultation in a staff representation or a WC (Works Council).

For larger companies, a WC is mandatory. The staff representation and the manager discuss safety and health a few times a year.



Company rules and regulations indicate which rules apply in a company.

5.1 Tasks, rights and duties of the employer and employees

General obligations of the employer:

- * To promote the health, safety and welfare of the employees by adopting a prevention policy integrated in the company policy.
- * To plan the prevention and the implementation of the policy using a systematic approach (drawing up and implementing an overall prevention plan and annual action plan).
- * To execute a (written) risk analysis.
- * To consult experts (internal and external prevention service).
- * Consult with employees.
- * Provide effective protective equipment and monitor that it is used.
- * To abstain from any form of violence, harassment or undesired sexual behaviour in the workplace.
- * To investigate serious occupational accidents and report them to the government's Health and Safety Inspection Service.
- * Arranging a suitable introduction, information, training and instructions.
- * To monitor the compliance with instructions.
- * To take the necessary measures for the health and safety of third parties in the workplace.



The employer is obliged to arrange instructions about Health and safety.

Serious work accidents have to be reported to the government's Health and Safety Inspection Service.

The national phone number is 0800 2700000. The website of the Labour Inspectorate is: www.arbeidsinspectie.nl

The general law defines a number of general obligations of the employee in the workplace. The general obligations are:

- * To take care of one's own health and safety and that of other people involved.
- * To correctly use and take care of the personal protection equipment
- * To take care of one's own health and safety and that of other people involved.
- * To correctly use machines, appliances, tools, hazardous substances, means of transport and other equipment
- * To report incidents to the supervisor.
- * To report serious and immediate danger to health and safety to the supervisor.
- * To collaborate with the employer and the prevention service in the area of promoting welfare at work.
- * To positively contribute to the prevention policy.
- * To abstain from any form of violence, harassment or undesired sexual behaviour in the workplace.
- * To follow training
- * To follow information
- * To follow instructions
- * To not change or remove protective devices and to use them in the correct way

The obligations in relation with education and training for the employee are:

- * The employee has to take part in training.
- * The employee has to take part in general education.
- * The employee has to follow instructions.

The obligations in relation to working with machines for the employee are:

- * To correctly use machines, appliances, tools, hazardous substances, means of transport and other equipment.
- * To not change or remove protective devices and to use them in the correct way.

The general rights of an employee on the H&S (Health and Safety Act) basis are:

- * The right to information and education.
- * The right to interrupt the work in the event of a threatening or serious danger to people.
- * The right to a healthy and safe working environment.



Screensavers can also be inappropriate behavior.

When there is a serious or imminent danger to humans, this must be reported to the manager. The manager must take actions. The employee must stop his task and/or bring himself to safety. If the employee has sufficient technical knowledge and resources, he may take appropriate measures to avoid the impact of such a danger.

The statutory liabilities with respect to health and safety are:

- * Relationships between the hiring company, temporary agency worker and temporary employment agency.
- * The hiring company is responsible for enforcing the health and safety legislation.
- * The temporary employment agency has specific tasks: information to the temporary agency worker, medical supervision.
- * The temporary employment agency is responsible for the occupational accident insurance.

The difference between the formal and the actual employer is:

- * The formal employer actually employs the employee.
- * The actual employer is the one under whose authority the temporary agency worker works.

The actual employer, therefore the hiring company, is responsible for the working conditions.

The statutory liability with respect to health and safety is arranged as:

- * In the event of temporary work, there is a triangular relationship: the temporary employment agency, the hiring company and the temporary agency worker.
- * The temporary employment agency is liable with respect to the duty to pass on information.
- * The hiring company, as the actual employer, is liable for the working conditions of the temporary agency worker.

The Health and Safety aspects that are important when assigning a temporary worker are:

- * Health and Safety risks and management measures for the temporary jobs".
- * The procedure to be followed for applications for temporary workers and for recording agreements on Health and Safety aspects (instruction, information, personal protective equipment, training, experience).
- * Requesting the requisite Health and Safety documents and being able to interpret them properly - for example, the results of a risk analysis, Health and Safety rules and regulations, professional competence.
- * Carefully selecting temporary workers.
- * Providing information to temporary worker(s) about Health and Safety aspects and agreements made with the user enterprise.
- * The procedure to be followed in the case of a serious accident with temporary workers.

The elements that the temporary worker's dossier must contain are:

- * Personal data, social security data, work permits.
- * Copy of an identification document.
- * Copies of relevant qualifications related to professional training, follow-up training courses, training courses for hazardous activities.
- * Work experience.
- * Copies or evidence of requisite valid certificates/credentials/diplomas.
- * Evaluations, memorandums with respect to temporary work, incidents, etc..
- * Job-oriented limitations.
- * Linguistic competence.
- * Evidence of medical suitability (where applicable).



Keeping track on all dossiers is very important with temporary agency workers.

The necessary points of focus to be recorded when the user enterprise submits an application are:

- * Job name.
- * Place, department and working environment.
- * Work and activities to be carried out.
- * Hazardous jobs and management measures for the task and workplace.
- * Required medical suitability.
- * Risks present (hazardous substances, machines, situations, processes).
- * Which personal protective equipment will be used to control these risks.
- * Who will provide the personal protective equipment (employment agency or user enterprise).
- * Who will provide the instructions for use for the personal protective equipment (employment agency or user enterprise).
- * Professional training/level of education.
- * Experience.
- * Linguistic competence.
- * The possession of a valid certificate/credentials/diploma Basic Safety VCA (Basisveiligheid VCA), unless it is stated in the application and substantiated by the user enterprise why Basic Safety VCA is not obligatory.
- * Requirements for Hazardous jobs.
- * Other relevant training courses.



Who purchases PPE?

The points of focus that are important during selection are:

- * The selection procedure for temporary worker(s) who satisfy the requirements specified in the application.
- * Any deviation from or exception to the requirements specified in the application will be presented to the user enterprise before definitive selection takes place.
- * Additional preventive measures or agreements are recorded on the selection or application document(s).
- * The user enterprise must approve the selection that deviates from the requirements. If there is a deviation between the application and the selection, a memorandum will drawn up by the employment agency (date, name, job).

The temporary worker must know the following things about the specific Health and Safety risks and the Health, Safety and Environmental Quality rules and regulations at the location where he will be employed

- * The assignment for temporary work, the risks and management measures of the job in which the candidate will be placed.
- * Sector-specific Health and Safety rules and regulations.
- * The personal protective equipment stipulated at the user enterprise and instruction on how to use it properly.
- * Medical examination, if necessary.
- * Reporting procedure for accidents involving absence from work/unavailability for work.

This must be done in relation to verifying the agreements made with the user enterprise:

- * Reception at the company; meeting the supervisor.
- * Person to whom the worker must report for instructions about the workplace and task.
- * Provision and use of personal protective equipment.
- * The task to be carried out by the temporary worker; the work location and the conditions.
- * Competence and medical suitability for carrying out hazardous jobs.

The following items must be focused upon when evaluating the temporary work with the user enterprise:

- * Professional competence; theoretical knowledge; self-sufficiency; safe execution of tasks.
- * Motivation; attitude; communication; understanding instructions (language knowledge).
- * Compliance with Health, Safety and Environmental Quality rules and regulations; complying with work permits; keeping to agreements.
- * Proper use of tools; looking after tools; cleaning up.
- * Correct use of personal protective equipment; care of personal protective equipment.
- * Asking questions if something is unclear; reporting deviations; reporting unsafe situations.
- * Communication and contact with the employment agency.

The aspects that must be focused upon when evaluating the temporary work with the temporary worker are:

- * Introduction and instructions on rules and regulations (Health, Safety and Environmental Quality).
- * Issuing personal protective equipment and work clothing.
- * Communication with contact person and/or direct supervisor.
- * Work situation and working conditions with regard to health and safety.
- * Facilities such as changing room, canteen, sanitary amenities.
- * Focus on and supervision of safe working conditions.
- * Involvement in Health, Safety and Environmental Quality instruction sessions / meetings (toolbox meetings; instruction when starting work).

The RI&E of the client gives a lot of information about this.



During check-ups, checklists can be a good tool.



Respiratory protection must be kept clean.

The following must be done in the case of accidents involving absence from work/unavailability for work:

- * Procedure for reporting and registration
- * Communication of the procedure to temporary workers.
- * Correct and full completion of accident form.
- * Registration of all accidents.
- * Annually following up accident statistics (IF).

IF = Incident frequency.

What must be discussed with the user enterprise regarding medical examination is:

- * For which jobs or specific work locations at hiring companies is a medical examination is required.
- * Medical examination by authorised persons or certified medical services for company medical care.
- * Providing information to temporary workers about required medical suitability and medical examinations and re-examinations to be conducted.
- * Informatie naar uitzendkrachten over Verifying and keeping up-to-date the medical suitability of temporary workers for the work they are actually doing.
- * Check out which medical examinations the customer will ask to be performed himself and (if necessary) extend cooperation.
- * Awareness that the medical examination must not be used inappropriately to select the healthiest candidate.

The conditions in which it is allowed to deviate from the tasks originally agreed for the temporary agency worker are:

- * Information in advance to the temporary employment agency and the temporary agency worker.
- * Description of specific additional preventive measures.
- * Agreements about who informs and instructs the temporary agency worker and who issues protection equipment.



Medical suitability for the position must be checked.

5.2 Tasks, rights and duties of those in operational leading roles

HSE tasks of operational supervisors are:

- * To formulate proposals and advice to the employer.
- * To investigate accidents and incidents and to propose measures.
- * To check working equipment and personal protection equipment.
- * To obtain advice from prevention services.
- * To verify whether employees have the required skills and information.
- * To monitor compliance with instructions.
- * To ensure that the employees understand the information and actually use it.
- * To organise the introduction of the starting employee and to assign an experienced employee to supervise him.
- * To cooperate in risk management (RIE and TRA).
- * To execute safety inspections.
- * To organise HSE meetings (toolbox meetings).

An employee must follow training and receive instructions in the following situations:

- * Prior to commencing the activities (during recruitment, including temporary employees).
- * When being transferred or when changing job/ function.
- * When a new piece of equipment is introduced or equipment is changed.
- * When implementing new working procedures, new technology.

The characteristics of good information and training are:

- * Tailored to the nature, capacity and experience of the employee.
- * Tailored to the specific activities.
- * Understandable.
- * Practical, not only in writing.

The obligatory content of information and training is: is:

- * Health, safety and welfare in general.
- * Hazards at the workplace.
- * Measures taken.
- * Regulations and rules drawn up by the employer.

The manager has the following measures to increase his employees motivation:

- * Try to obtain as much insight as possible into the personality and the capabilities of your employees.
- * Do not assign jobs to employees that require him/her working above or below his/her capabilities
- * Take into account the opinions, proposals, wishes and advice of your employees
- * Tell the employees what you think about them, including their performance in the area of health, safety and the environment
- * Evaluate your employees as objectively as possible
- * Ensure that when corrective action is taken, the self esteem of the employee as a professional or as a human being is not affected
- * Ensure that there is a healthy amount of self criticism
- * Try to listen attentively to what someone else has to say to you
- * Prevent emotional situations
- * Tell the employees not only how to do something but also why
- * Do not be 'bossy' as a boss
- * Give a good example



Zou ik zo'n leidinggever willen hebben?

5.3 Workplace inspections

Benefits of workplace inspections are:

- * Increase safety awareness.
- * Provide good insight into the safety situation at the workplace.

5.4 HSE meeting (toolbox meeting)

HSE consultation is the discussion on safety, health and environment within the company between the employer and employees.

HSE consultation is the consultation within the company about safety, health and environment between employer and employee.

Characteristics of such consultations:

- * Takes place regularly.
- * It takes place structured (agenda and report).
- * Runs interactively from above and from below.

Some forms of H&S meetings are:

- * A meeting between employer and the employee representation.
- * Work meeting with the employees in an HSE meeting (toolbox meeting).

Participants in the H&S meeting are:

- * Employer (chairperson).
- * Employers (representation).
- * Employees (representation).
- * Experts (e.g. prevention consultant).



Discuss together what PPE must be worn.

In the meeting with respect to health and safety within the temporary employment agency involvement of the management and the supervisors is important.

The temporary agency worker is involved in the HSE meetings (toolbox meetings) by:

- * The intermediary informs the temporary agency worker about the existence and purpose of the HSE meetings (toolbox meetings)
- * The temporary worker must take part in the Health, Safety and Environmental Quality meetings (toolbox meetings) that are organised at the user enterprise

5.5 Certification, safety passport and training for high-risk tasks

The acronym VCA stands for: VGM Checklist Aannemers (Safety Checklist Contractors - SCC). The meaning:

- * VGM stands for: Health, safety and environment (HSE).
- * Checklist: questionnaire in the form of an analysis and screening system.
- * Contractors: companies that carry out work for and on the premises of others (customers).



The logo of VCA.

Purpose of VCA certification:

- * To control the activities on the work floor with respect to HSE.
- * Objective: prevention of incidents (HSE).

VCA certification is aimed at companies that make employees do high-risk activities or activities carried out in dangerous environments such as factories, installations and construction.

The VCA certificate is a mark to show that a company, according to the VCA rules (SCC), works on HSE management.

The different levels of SCC certification are:

- * SCC* (one star): This certification level focusses on the direct HSE care for the activities on the work floor.
- * SCC ** (two stars): In addition to the elements from SCC*, here the safety structures and systems in the contractor's company are also assessed.
- * SCC Petrochemistry: Companies that carry out high-risk activities in the petrochemistry/petrochemical industry are subject to additional requirements supplementary to SCC**.

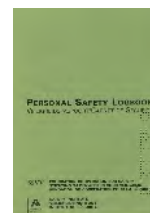
VCU stands for "Veiligheid en gezondheid Checklist Uitzendorganisaties" (health and safety checklist for temporary employment agencies).

The goal of VCU-certification is: allow temporary employment agencies to control the process of making temporary agency workers available with respect to health and safety at the workplace.

VCO stands for Veiligheid, gezondheid en milieu Checklist Opdrachtgevers (health, safety and environment checklist for customers).

The goal of a VCO-certificate is: Customers ensure that the conditions and circumstances are such that they allow VCA certified contractors and temporary agency workers, hired out via VCU certified agencies, and who come to carry out high-risk work on their premises, to work in a safe way.

The safety passport is a personal document (book) stating important details regarding education, training courses, medical details and vaccinations.



A safety passport.

The employer has drawn up a procedure to ensure that the safety passport is filled in correctly and in time.

In the petrochemical industry, it is mandatory that for a number of activities/functions the operational employees have passed an exam at a recognised centre. These courses are stated in the SSVV Opleidingen Gids.

For various activities in the petrochemical industry, an exam by an accredited examination center is required. These courses are stated in the Training Guide Risky Work. Some of the recognised trainings are:

- * Working with a fork-lift truck.
- * Working with a reach truck.
- * Working with a hydraulic platform.
- * Working on low-pressure flanges.
- * Working as an authorised attendant (confined space).
- * Working with self-contained breathing protection.
- * Moving loads.
- * Measuring gas: explosion hazardous and toxic substances and oxygen.

The employees of a VCA certified company need to be trained within 3 months after recruitment according to the requirements of VCA(SCC). For the various positions in a company there are several training requirements.

- * For operational employees: certificate, attest, diploma Basic safety VCA (B-VCA).
- * For operational supervisors of operational employees: certificate "Veiligheid Operationeel Leidinggevenden VCA" (VOL-VCA - Safety for Operational Supervisors VCA).
- * For high risk tasks: certificate of the associated test.

SSVV Opleiding

Look at: www.gorwvca.nl



Employees in a VCA certified company must be demonstrably trained.

The training requirements imposed on employees are:

- * Training Certificate Basic Safety VCA: (temporary) employees for working at a hirer with VCA certificate.
- * Safety Training Certificate operational managers (VOL): to operational management within a company.
- * Safety Training Certificate consultants and managers for a temporary employment agency (VIL).

The training requirements that are imposed on the temporary agency workers and on the employees of the temporary employment agency are:

- * The temporary agency workers who are posted to hiring companies that have a VCA certificate: certificate, attest or "Basic safety VCA" diploma.
- * The temporary agency worker, who is posted as an operational supervisor to a hiring company: certificate, attest or "Veiligheid voor Operationeel Leidinggevende VCA" or VOL-VCA (safety for operational supervisor VCA) diploma.
- * The intermediary and supervisors in the temporary employment agency with an VCU certificate: certificate, attest or "Veiligheid voor Intercedenten en Leidinggevenden VCU" or VIL-VCU (safety for intermediaries and supervisors VCU) diploma.

Examination questions: Tasks, rights, duties and discussions

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 What are the obligations of employers according to the Health and Safety legislation?
 - ☐ A To provide information and instruction about the hazards at work.
 - ☐ B To report hazardous situations or risks to health and safety to the manager/supervisor.
 - ☐ C Call in the government's Health and Safety Inspection Service during the drawing up of a risk assessment.
- ☐ 2 What are the obligations of employees according to the working conditions act? They are obliged to
 - ☐ A Call in the labour inspectorate during the drawing up of a risk assessment.
 - ☐ B To report hazardous situations or risks to health and safety to the manager/supervisor.
 - ☐ C To provide information and instruction about the hazards at work.
- ☐ 3 When do employees have the right to interrupt their work?
 - ☐ A If personal protective equipment is not available.
 - ☐ B If the Labour Inspectorate comes to carry out an investigation.
 - ☐ C If immediate and serious danger for human beings occurs.
- ☐ 4 Who participates in the works council consultation?
 - ☐ A Departmental management and employees.
 - ☐ B All employees, including temporary personnel.
 - ☐ C Employer and employees representatives.
- ☐ 5 What is the difference between SCC* and SCC** certification?
 - ☐ A SCC* focuses on the safety structures, SCC** also includes the environmental care system.
 - ☐ B SCC* focuses on the direct safety control, SCC** also includes the safety structures.
 - ☐ C SCC* focuses on the quality care system, for SCC** the working conditions care system is also included.
- ☐ 6 What must one do in the event of immediate danger?
 - ☐ A Fix the risks which caused to danger.
 - ☐ B Keep working until someone tells u to stop.
 - ☐ C To stop work and/ or ensure your own safety.
- ☐ 7 What is a benefit of work place inspections and observation rounds?
 - ☐ A Provide good insight into the safety situation at the workplace.
 - ☐ B Increase safety awareness.
 - ☐ C Both answers are correct.
- ☐ 8 A statutory liability with respect to health and safety is:
 - ☐ A The temporary employment agency is responsible for the occupational accident insurance
 - ☐ B The hiring company has specific tasks: information to the temporary agency worker, medical supervision
 - ☐ C Relationships between the hiring company, temporary agency worker and Labour Inspectorate.

9 The difference between the formal and the actual employer is:

- A The actual employer is the one under whose authority the temporary agency worker works.
- B Both answers are correct.
- B The formal employer actually employs the employee.

10 Who is responsible for the working conditions in the event of temporary work?

- A The temporary employer has to deliver his set of personal working conditions to his hiring company.
- B The temporary employment agency is responsible for the working conditions
- C The actual employer, therefore the hiring company, is responsible for the working conditions.

11 A statutory liability with respect to health and safety is arranged is:

- A In the event of temporary work, there is a special relationship: the temporary employment agency, the hiring company, the Labour Inspectorate and the temporary agency worker.
- B In the event of temporary work, there is a triangular relationship: the temporary employment agency, the hiring company and the temporary agency worker.
- C The hiring company is liable with respect to the duty to pass on information.

12 A Health and Safety aspect that is important when assigning a temporary worker is:

- A Requesting the requisite Health and Safety documents and being able to interpret them properly - for example, the results of a risk analysis, Health and Safety rules and regulations, professional competence
- B The procedure to be followed for applications for temporary workers and for recording agreements on Health and Safety aspects (instruction, information, personal protective equipment, training, experience).
- C Both answers are correct.

13 What must be discussed with the user enterprise regarding medical examination?

- A For which jobs or specific work locations at hiring companies a medical examination is required.
- B Medical examination by the supervisor of the company.
- C Providing information to temporary employment agencies about required medical suitability and medical examinations and re-examinations to be conducted.

14 Elements that the temporary worker's dossier must contain are:

- A Job-oriented limitations.
- B Copy of an identification document.
- C Both answers are correct.

15 What is not a necessary point of focus to be recorded when the user enterprise submits an application?

- A Required medical suitability
- B Who will provide the instructions for use for the personal protective equipment (employment agency or user enterprise)
- C The possession of a valid certificate/credentials/diploma VOL SCC (VOL VCA), unless it is stated in the application and substantiated by the user enterprise why VOL SCC is not obligatory.

16 What must be done in relation to verifying the agreements made with the user enterprise?

- A A formal introduction to co-workers.
- B Reception at the company; meeting the supervisor
- C The temporary employee has to bring his own PPE.

17 An aspect that must be focused upon when evaluating the temporary work with the temporary worker is:

- A Issuing personal protective equipment and work clothing.
- B Facilities such as changing room, canteen, sanitary amenities.
- C Both answers are correct.

18 What is important in the meeting with respect to health and safety within the temporary employment agency?

- A Involvement of the Labour Inspectorate.
- B Involvement of the co-workers.
- C Involvement of the management and the supervisors

19 What is the involvement of the temporary agency worker in the HSE meetings (toolbox meetings)?

- A The temporary worker must take part in the Health, Safety and Environmental Quality meetings (toolbox meetings) that are organised at the user enterprise.
- B The intermediary informs the temporary agency worker about the existence and purpose of the HSE meetings (toolbox meetings).
- C Both answers are correct.

6 Procedures, instructions and signs

High risk activities have to be well prepared and coordinated to keep the chance of accidents as small as possible. As part of this preparation, one identifies which tasks will be performed by employees. After this identification, the risks of these tasks will be determined and the preventive measures will be set. All these steps are defined in a Task Risk Analysis (TRA).

In order to transfer the measures correctly to the executer, these measures are documented in procedures and work permits.



Working at height requires a thorough preparation.

Thanks to this documentation, the chance on any mistakes and accidents will be decreased.

The measures taken are established in a work permit. This work permits defines the work content and how and where to work. A work permit is required when working in the petrochemical industry.

6.1 Safety procedures and instructions

Every organization has rules and procedures for working well and safely. Informal rules and procedures are risky because they are not known to everyone, but also because they may not comply with applicable laws and regulations.

Formal rules and procedures ensure unambiguous commitments for working well and safely and therefore should always be documented.

Formal rules and formal procedures ensure unambiguous agreements to operate safely and properly and should therefore always be documented. General safety rules are meant for:

- * The entire organization.
- * Company personnel.
- * Temporary agency workers.
- * Other people at location.
- * Personnel of contractors and subcontractors.

Guidelines for the use of general safety rules are:

- * Must be recorded in writing.
- * Must be notified to every person who enters the location for the first time.
- * Must be sufficiently clear and unambiguous.
- * Must also be able to be understood by people how only speak other languages.

General safety rules may include:

- * Notifying presence and absence.
- * Traffic rules on the site.
- * How to act in the event of incidents.
- * The sorting of waste.
- * The reporting of accidents, fire and incidents.



A sign with general safety rules.

Specific safety rules are meant for activities, functions and operations with high risk.

Specific safety rules may include:

- * Entering confined spaces.
- * Working at height.
- * Warm/ hot work.
- * Working in explosion hazardous surroundings.
- * Use of the required personal protection equipment.
- * Excavation activities.
- * Use of specific tools, devices and equipment.
- * Screening.
- * Ventilation.
- * Signalling.
- * Marking.

On specific safety rules must be no doubt because they are used at high risk activities. Therefore, there are guidelines regarding the usage. Specific safety rules:

- * Must be recorded in writing.
- * Must be sufficiently clear and unambiguous.
- * Must be available to the employee concerned/ involved.
- * Must be explained to the employees involved.
- * A check must establish whether the employees involved have understood the rules.
- * Should also be understood by native speakers.

General education to employees includes:

- * General health and safety rules.
- * General instructions for use in the event of fire and alarm.
- * Reporting procedure for (near) accidents and unsafe situations.
- * The guidelines to adhere to in the event of an accident during or on the way to or from the workplace.
- * Necessity and use of personal protection equipment.
- * Information about the workfloor (job requirements, hazards, control measures) based on information from the manager.

Specific education from the manager to the new employees is education about:

- * Hazards on the workfloor.
- * Measures taken.
- * Specific workplace-bound health and safety rules.
- * Specific health and safety rules related to the installations, machines, equipment.
- * Specific company-related details regarding danger zones, evacuation routes, canteen, sanitary facilities, etc.
- * Information about the workplace(job requirements, hazards, control measures) based on information from the client.

The duty to pass on information entails:

- * The hiring company and the temporary employment agency have a shared duty with respect to providing the temporary agency worker with information.
- * The temporary employment agency must pass on information to the temporary agency worker regarding the workplace and the requirements placed on the job and the activities.
- * This information is provided by the hiring company.



A specific safety rule for entering a confined space.

The temporary agency worker must receive the following instructions from the temporary employment agency within the framework of VCU:

- * General health and safety rules.
- * General instructions for use in the event of fire and alarm.
- * Reporting of unsafe actions and situations.
- * What to do in the event of accidents during the work and in the event of accidents on the way to and from work.
- * Details concerning the temporary employment agency (office) and of the contact person.
- * Policy declaration of the temporary employment agency.
- * Necessity and rules governing the use of personal protection equipment.
- * Sector specific information (e.g. pictograms).
- * Information regarding the workplace: job requirements, hazards and control measures.
- * Specific information supplied by the hiring company.
- * Specific rules and regulations of the hiring company.
- * Explanation of the permit system of the hiring company.
- * Handing over documentation.

6.2 Making the workplace and installation safe

Working in a confined space is risky, therefore there are special measures for entering a confined space. These are preventive measures.

These preventive measures reduce the risk of entering a confined space. Securing the installation is an important preventive measure.

By securing, you are certain that you can never get hurt while working in or near the installation.

The function of a connecting flange is:

- * To separate sections of pipe.
- * To shut off supply lines to vessels, tanks or installations.

Situations in which a connecting flange is used are:

- * When cleaning, repairing, inspecting, flushing, degassing or steaming out pipelines, vessels/ tanks and/or installations.
- * When replacing shut-off valves and devices on pipes that transport products.
- * When working in a confined space (supply pipes must be shut off).



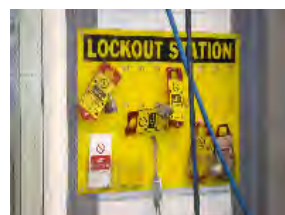
A connection flange.

Securing installations and equipment is important to prevent accidents. With securing you prevent:

- * To prevent equipment and system components starting to move unintentionally.
- * To prevent equipment, machines and system components becoming live unintentionally.
- * Check that the installations and/or equipment are not pressurised.

Examples of actions for making installations and equipment safe are:

- * Only to be secured by authorised persons
- * Decommission.
- * Secure against accidentally switching on again and mark.
- * Check whether installation or equipment has been secured.



Locks for lockout situations for locking out switches.

6.3 Safe working permits (work permit)

A working permit is not a piece of paper that you just have to sign. A working permit gives you formally permission to perform certain risky activities. All possible working measures are included in this working permit so that all work activities can be performed safely.

Carefully read a working permit because it is about your own safety!

A work permit is not just a piece of paper which u need to sign. A work permit give u a formal permission to perform certain risky activities. The work permit includes all possible measures so that the work can be performed safely.

The purpose of supplementary work permits are:

- * The work is associated with specific risks.
- * The work requires specific training or explanation.
- * A condition for carrying out the work: it is written down.
- * A condition for carrying out the work: it has been explained in detail.
- * A condition for carrying out the work: it has been signed.

The purpose of specific permits are:

- * Ensuring that consultation takes place, and that formal approval is given before starting the work
- * Specific training or instructions

In the following activities u need an additional permit:

- * Entering a confined space.
- * Work that poses the risk of fire, also called a fire permit or hotwork permit, e.g.: welding, grinding.
- * Digging in contaminated soil (soil that contains pipes, opening up floors, streets).
- * Working on or near hazardous radiation sources.
- * Lifting activities above or in the vicinity of important installations.
- * Working with or demolishing materials containing asbestos.
- * Working with hazardous substances.
- * Working at height.

The functions of permits are:

- * To consult with everyone involved in the work.
- * To make binding agreements with everyone involved in the work.
- * To record the conditions in which work must be done.
- * To grant permission for starting the activities.

A work permit describes, as good as possible, the activities that have to be performed. In a working permit the following is described:

- * Time of validity (usually 1 day).
- * The activities that are being requested.
- * Who will do which job.
- * Measures to take/ that have been taken by the issuer of the permit.
- * Measures to be taken by the operational employees, those who carry out the work.
- * Measures to ensure that the work area is left in a safe condition after the work has been done
- * Approval/ signature.
- * Conditions for extending the permit.



With risk of fire, one needs a working permit.

In the working permit there is a section "requested activities". This section describes:

- * This states what the activities entail.
- * How the activities will take place.
- * Where the activities will take place.

In the working permit there is a section "measures prescribed by the permit issuer". This section describes the measures to work safely.

Examples are:

- * The measures to be able to work safely.
- * Making safe: connecting flanges, electrical and/or mechanical locking.
- * Measuring and detecting.

The work permit contains a section "Measures taken by an operational employee". This section describes the measures that have to be taken by operational employees to work safely. All personal protection has to be noted in this section.

Signing a working permit means that those who sign agree with the contents of the permit. The permit is signed by the issuer, the holder(s) and the supervisor(s) of the executive(s) or the executive(s) if he is so entitled.

The work permit is signed by:

- * The issuer.
- * The holder.
- * The supervisor(s) of the executive(s).
- * The executive(s) if he is so entitled.

The obligations of a permit issuer are:

- * To check that the installation has been released for the activities before signing the permit.
- * To discuss with the permit holder the nature of the activities, the conditions and the measures that must be taken before signing the work permit.
- * To sign the permit.
- * If necessary, the issuer ensures that the necessary measurements have been taken.
- * If various activities are carried out concurrently, the issuer arranges the necessary coordination and agreements between the parties involved.

Usually the holder of the work permit is the supervisor of operational employees. The obligations of the holder of the work permit are:

- * The holder provides an explanation to the operational employees.
- * The holder signs the permit.
- * The holder monitors that the activities proceed in accordance with the conditions on the permit.
- * The holder ensures that the permit is on site.



Monitoring of the CO2 level may be prescribed in the working permit.



Carefully read the working permit. If something is not clear, ask your supervisor.

The obligations of operational employees while working with work permits are:

- * To become acquainted with the content of the permit.
- * To comply with the conditions for the activities as described in the permit.
- * To comply with the measures as stated in the permit.
- * To only work with a valid permit and to take into account the duration of validity.

Duties of the supervisor of operational employees are:

- * The supervisor must discuss all measures thoroughly in advance with the operational employees
- * The supervisor must ensure that the permit is present at the workplace
- * The supervisor must check the measures to be taken by the issuer and the holder

Examination questions: Procedures, instructions and signs

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 The procedure for reporting a fire falls under the.

- A SCC procedures.
 - B General safety rules.
 - C Specific safety rules.
- ☐ 2 What must be described AMONGST OTHER THINGS in a work permit?

- A The relevant control points from the SCC.
 - B The number of employees with an IER (BHV) diploma.
 - C The measures to be taken by the issuer of the permit and the permit holder.
- ☐ 3 General safety rules are intended for:

- A One department of the company.
 - B Temporary agency workers.
 - C Contractors and subcontractors
- ☐ 4 General safety rules:

- A Must be recorded in writing.
 - B Are tool specific rules on the usage.
 - C Must be written in one language.
- ☐ 5 Specific safety rules do not relate to:

- A Lunch breaks.
 - B Ventilation.
 - C Entering confined spaces.
- ☐ 6 A guideline concerning the use of specific safety rules is:

- A Should also be understood by native speakers.
 - B Must be released to all employees.
 - C When the rules are explained, one can conclude everybody knows what they mean.
- ☐ 7 A function a connection flange is:

- A To shut off supply lines to vessels, tanks or installations.
 - B To separate sections of pipe.
 - C Both answers are correct.
- ☐ 8 What is not a section of the work permit concerning the requested activities?

- A How the activities will take place.
 - B Where the activities will take place.
 - C Why the activities will take place.
- ☐ 9 What is a section in the work permit concerning the measures to be taken by the operational employee.

- A All items of personal protection equipment must be stated here
 - B The measures that the operational employees must take to work safely.
 - C Both answers are correct.

10 What is the duty to pass on information entails?

- A Both answer are correct.
- B The temporary employment agency must pass on information to the temporary agency worker regarding the workplace and the requirements placed on the job and the activities.
- C The hiring company and the temporary employment agency have a shared duty with respect to providing the temporary agency worker with information.

7 Preparations for emergencies

Despite all precautionary measures accidents with injury or damage may still happen. In those cases the company has to respond well to limit the consequential damage or injury. Therefore the Emergency Response Team (BHV) has been made.

There has to be an Emergency Response Team in each organization with employees. The Emergency Response Team members must be trained on the risks within the company.

A company must have an Emergency Response instruction and sufficient Emergency Response resources such as fire extinguishers and first aid-equipment.



The Emergency Response Team must bridge the time until external support is present. They must act.

From the RI&E, the size and nature of the Emergency Response organization has to be established. Also the RI&E describes the possible incidents that may occur in the organization. The Emergency Response team exercises should be based on this.

7.1 Concepts

Emergency situations can be caused by:

- * Emergency situations that are caused by accidents, fire, explosion, uncontrolled escape of liquids, gas, biological agents or radioactive radiation.
- * Threat of an emergency situation due to excessively bad weather, natural disasters, social unrest, (threat of) terrorist attack, loss of infrastructure.

7.2 Objectives and components of the (internal) company emergency plan

A company emergency plan is a description of the measures and facilities that a company has prepared in case of an emergency.

The purpose of a company emergency plan is to reduce the effects of possible emergencies.

While entering company premises the (temporary) employees and operational employees must receive information about the company emergency plan. Important sections of this information are:

- * Map of the company site (the evacuation plan).
- * Warning signs (the evacuation plan).
- * Alarm (the evacuation plan).
- * Drills (the evacuation plan).
- * Location of available aids (the intervention).
- * First Aid (the intervention).
- * Removing the cause (the intervention).
- * Reporting accidents, emergency situations (the reporting).
- * Warn the emergency services (the reporting).



An instruction sign.

7.3 Controlling and tackling emergency situations

The important phases while controlling and tackling emergencies are:

- * First alarm.
- * Actions and measures.
- * Termination.

During emergency situations the concept "the first report" is the description in which ways an emergency can be reported, the information needed and the following actions.

In a company emergency plan there is a description of "actions and measures". This section describes what has to be done when there is a report of an emergency situation. Possible actions or measures

- * Evacuation.
- * Intervention: extinguishing, securing and cleaning up.
- * Rescue operations.
- * Assistance.

A company emergency plan contains a chapter "termination of the emergency situation". This chapter describes:

- * Who is/are authorised to declare the emergency situation terminated.
- * The way in which the termination is notified.

Education and training of the emergency response team is very important. The goals of training and practice regarding company emergency plans are:

- * Preparing staff for emergencies.
- * Testing the company emergency plan.
- * Testing whether the personnel is prepared for emergencies.

7.4 Company Health and Safety

The obligations with respect to company emergency response are:

- * Must be organised in every company/ every organisation.
- * Company emergency response must be guaranteed for the company concerned.
- * Certain training requirements must be met.
- * It concerns an established set of tasks.

The statutory duties of emergency response are:

- * To provide first aid.
- * To fight a starting fire.
- * In emergency situations, to evacuate employees and other people present on the company premises.



This car was parked too close to the burned out building.



The emergency response team needs to practice their duties on a regular basis.



When the fire brigade arrives, they take over the tasks of the emergency response team.

Every company must at least have First Aid equipment available and needs educated (in-company) emergency response team members.

7.5 Evacuation

The employees should be instructed about a possible evacuation. Important subjects in this instruction are:

- * Where are the assembly point(s)?
- * What are the evacuation possibilities?
- * What are the customers instructions?

At least one time a year there needs to be an evacuation drill. Employees need to attend in an evacuation drill.

During an evacuation signal employees need to take the following actions:

- * Immediately stop work, stop external communication.
- * Follow the instructions of the supervisor.
- * Go to the assembly point, in accordance with the evacuation plan.
- * Do not use any lifts.
- * Evacuate at right angles to the wind direction.
- * On arrival at the assembly point report presence.



The emergency response member at the gate has an important job: he must collect the assistance.

Upon the introduction of a temporary agency worker, he must receive the following items regarding the evacuation plan and the first aid intervention:

- * Routes.
- * Warning equipment.
- * Alarm: types.
- * Exercises and emergency situations.
- * Raising the alarm, reporting an emergency situation and reporting when an emergency situation is no longer present.
- * Warn internal and external emergency services.
- * First Aid equipment.

The temporary employee has to receive information about the first intervention during emergencies. This information may contain:

- * Raising the alarm, reporting an emergency situation and reporting when an emergency situation is no longer present
- * Warn internal and external emergency services
- * First Aid equipment

Examination questions: Preparations for emergencies

The correct answers of these questions can be found in annex 1 of this book.

1 Which of the tasks below belongs to the role of the emergency response officers?

- A Extinguishing a starting fire and restricting damage and injury.
- B Participating in the accident investigation team.
- C Carrying out gas detection measurements.

2 What is a concept of an emergency situation?

- A Emergency situations that are caused by accidents, fire, explosion, uncontrolled escape of liquids, gas, biological agents or radioactive radiation.
- B Threat of an emergency situation due to excessively bad weather, natural disasters, social unrest, (threat of) terrorist attack, loss of infrastructure.
- C Both answers are correct.

3 A company emergency plan is:

- A A description of the measures and provisions that a company has prepared in case emergency situations arise.
- B A list of measures and provisions that a company has prepared to make sure working conditions are safe.
- C A list of measures which employees in a company have to prepare before starting work activities.

4 Limit effects of possible emergency situations is the purpose of:

- A The set of working conditions.
- B An (internal) company emergency plan.
- C The emergency response team.

5 What is not a phase in controlling emergency situations?

- A First alarm.
- B Second alarm.
- C Termination.

6 The first alarm in emergency situations is:

- A A list of ways to alarm people in emergency situations.
- B A description of the ways an emergency situation can be reported and what information must be communicated.
- C A loud bell which signals there is an emergency.

7 A component of the termination of the emergency situation is:

- A The way in which the termination is notified
- B Who is/are authorised to declare the emergency situation terminated
- C Both answers are correct.

8 One of the minimum requirements with respect to first aid is:

- A Trained emergency response team members
- B First Aid equipment always available
- C Both answers are correct.

9 Which sequence is correct in the event of an evacuation signal?

- A 1: Immediately stop work, stop external communication,
2: Do not use any lifts,
3 :Go to the assembly point, in accordance with the evacuation plan.
- B 1: Immediately stop work, stop external communication,
2: Do not use any lifts,
3: On arrival at the assembly point report presence.
- C 1: Immediately stop work, stop external communication,
2: On arrival at the assembly point report presence,
3: Evacuate at right angles to the wind direction.

8 Hazardous substances

Who isn't dealing with hazardous substances? These substances are everywhere; at home (for example cleaning products) and at work (for example quartz, lead, printer toner, cleaning products and paint).

Actually, all substances are dangerous. If u drink a lot of water over a short period of time, even water can be lethal! Make sure u understand the risks that come with hazardous substances.

The law Milieugevaarlijke Stoffen (WMS) makes sure hazardous substances are recognizable. The producer of hazardous materials is required to put important information in Material Safety Data Sheets (MSDS) and provide them to users.

Before using hazardous substances, always read the label and the MSDS. If u follow these instructions, you are able to work safely.



Ensure proper storage of hazardous substances. Do not store sensitive resources near hazardous substances.

8.1 Classification and dangers of hazardous materials

Hazardous substances can be divided in the following categories:

- * Explosive substances.
- * Oxidising substances.
- * Extremely flammable, highly flammable, flammable substances.
- * Toxic and very toxic substances.
- * Harmful substances.
- * Irritating substances.
- * Caustic or corrosive substances.
- * Carcinogenic substances.
- * Environmentally hazardous substances.
- * Sensitizing substances.

Hazardous substances are materials that pose a danger to health and/or to the environment

Working with hazardous substances comes with many risks. Therefore it is important to recognize hazardous substances and know the risks that come with them. The most important risks that come with flammable substances are:

- * Explosive materials explode easily, also without the action of oxygen, at a certain temperature, in contact with other substances, in the event of shock or friction, e.g. ammunition, TNT, gunpowder.
- * Extremely and highly flammable materials ignite very easily, even at normal ambient temperatures (21°C), e.g. Petrol, acetone, white spirit.
- * Oxidising materials or those that promote fire release oxygen and therefore react fiercely with other (flammable) materials and promote fire, e.g. oxygenated water/hydrogen peroxide, ozone, oxygen.
- * Very toxic and toxic substances pose a substantial risk of serious problems if breathed in, swallowed or when absorbed through the skin, e.g. carbon monoxide, H₂S, methanol, benzene.
- * Harmful substances pose fewer problems than toxic substances when breathed in, swallowed or when absorbed through the skin, e.g. paint, lacquer, wood protection products,
- * Corrosive substances affect the skin, eyes, lungs and mucous membranes in the event of contact, e.g. acids and bases.
- * Irritating substances have less impact than corrosive products, they can cause infections in the event of contact with the skin, eyes, lungs and the mucous membranes, e.g. diluted or weak acids and bases, many solvents, polyester mastic. Carcinogenic substances can lead to cancer, e.g. asbestos, benzene, vinyl chloride, diesel fumes.
- * Environmentally hazardous substances pose a danger to the environment and animals, e.g. CFCs, certain pesticides.
- * Sensitizing materials can cause allergies, e.g. certain resins and colorants, some paints, metal treatment products, hair-dyes.

Using hazardous substances at your workplace and transport these substances is very risky. The danger of hazardous substances is related to the danger that can happen during use and transportation.

When transported around the workplace, the danger posed by the transport is also taken into account.

- * Hazardous substances at the workplace are indicated with a square danger symbol, black on an orange background.
- * Hazardous substances that are transported by road are indicated with a diamond-shaped danger symbol with the point upwards and the colour dependent on the category.



A tanker with danger signs.

8.2 Exposure to, ingestion of and mode of action of hazardous materials

Measures that have to be taken in succession against exposure to hazardous substances, must be performed according to the workplace strategy:

- * Firstly, measures focused on the source (use other products, use tablets instead of loose material).
- * Secondly, technical, collective and organisational measures (local extraction, ventilation).
- * Thirdly, personal protection equipment.

Measures at the source prevent u from having to work with hazardous substances. This can be done in various ways:

- * Elimination: by not using a hazardous substance.
- * Replacement: replacing the substance with a less hazardous substance (e.g. paint containing solvents by water-based paint).
- * Adaptation (e.g. not using the raw material as a powder but in tablet form).

Local extraction of hazardous fumes, gas or dustparticals can reduce the duration of the exposure. For example: in welding, u use spot extraction.



Local extraction of fumes.

When separating people from the source, you ensure that employees cannot come into contact with the hazardous substance by completely screening the place where the substance is emitted. This can be achieved by applying walls, panels or a hood.



A welding screen is an example of separating people and source.

During ventilation at working with hazardous substances, the contaminated air gets sucked out and clean air is brought in.

Factors that are important in using a personal monitor are:

- * Wear at chest level and not underneath clothing
- * Test before use, or must have been tested before use
- * Actions to be taken in case of an alarm

Hazardous substances can be absorbed by your body. Possible absorption ways are:

- * Through inhalation.
- * Through the skin.
- * Via respiratory organs.
- * Through direct absorption in the bloodstream.

The manner in which hazardous substances are absorbed are:

- * Through the mouth: Eating and drinking with dirty hands.
- * Through the skin: Touching solvents or solvents in paint.
- * Through the lungs: Inhaling powder, gas, haze and vapour.
- * Into the bloodstream: Through open wounds.

The absorption of hazardous substances can be prevented:

- * Eating and drinking in a special room, taking off dirty work clothing (dirt and dust collect in the clothing), washing hands and face, cleaning, treating and covering a wound as quickly as possible.
- * Use of effective personal protection equipment.

The absorption of hazardous substances into the human body also depends on the following environmental- and product factors:

- * State or occurrence of the substance (dust, mist, gas, vapour, solid, liquid).
- * The toxicity of the substance.
- * The concentration of the substance in the oxygen.
- * The duration of the exposure.



A snake bite can be very toxic.

The extent of the injury and poisoning by hazardous substances is subject to the following human factors:

- * Physical exertion: when performing strenuous work, you breath deeper and faster and therefore will inhale more of the hazardous substance.
- * The fitness of the victim: fitter people have more resistance.
- * The weight of the victim: the heavier the person the less influence the substance normally has.
- * The candidate can state the human factors that influence the degree of poisoning and damage due to hazardous substances.
- The age of the victim: young and old people are usually affected to a greater degree.
- * The size of the exposed skin area: the larger the skin area that has been in contact with the substance, the greater the impact.
- * The thickness of the skin: the thinner the skin that has been in contact with the substance, the greater the impact.
- * Sensitivity: some people are more sensitive to certain substances than others.



A cup of coffee is tasty, but too much coffee can be harmful for your health.

Hazardous substances can have several effects on the human body:

- * Headache.
- * Dizziness.
- * Balance disorders.
- * Stomach cramps.
- * Nausea.
- * Vomiting.
- * Palpitations.
- * Distress.
- * Blurred or double vision.
- * Allergy.

When dealing with hazardous substances, there are two possible types of poisoning:

- * Chronic poisoning: poisoning due to long-term or repeated exposure to the hazardous substance (e.g. solvents, heavy metals in low doses).
- * Acute poisoning: poisoning occurs rather rapidly during or directly after the exposure to the hazardous substance (e.g. CO poisoning).

Mercury poisoning is a chronic poisoning.

8.3 Meaning of threshold values and perception of odours

In work related exposure to hazardous substances, the concentration of these substances may not exceed the threshold. The threshold is:

- * The maximum concentration of a hazardous substance as time-weighted average over a reference period, above which no employee must be exposed.

The use of limit values for exposure to hazardous substances subject to conditions. These are:

- * For normal, healthy people.
- * At a normal day's work.
- * At a normal working week.
- * At normal working conditions.
- * During normal physical exertion.

The sense of smell is a bad indicator of the presence of hazardous gases. One should never rely on his perception. The reasons are:

- * Many hazardous vapours or gases have no odour or colour.
- * The smell limit can be above the accepted threshold limit value.
- * The smell of the hazardous substance can be masked by other non-hazardous substances.
- * Smell is subjective or dependent on the person.



Even your nose can malfunction sometimes.

8.4 Labelling, hazard pictograms and product information

An old label on the packaging of hazardous materials needs to contain important information about the substance. A label must include the following information:

- * Name(s) of the product(s).
- * Hazard pictograms.
- * The R-phrases (risk phrases).
- * The S-phrases (safety measures).
- * Details of the supplier/ importer



Small packages must comply with the requirements as well.

The following elements have to be listed on the new labels of dangerous substances:

- * Name(s) of the product(s).
- * Hazard pictograms.
- * Risks and hazards (H-phrases).
- * Precautionary measures and safety recommendations (P-phrases).
- * Details of the supplier or importer.

Hazardous substances are marked with hazard symbols. For example:

- * Explosive substances.
- * Highly flammable and extremely flammable substances.
- * Oxidizing substances.
- * Toxic and very toxic substances.
- * Harmful and irritating substances.
- * Corrosive substances.
- * Environmentally hazardous substances.
- * Long term health danger.
- * Pressured gases.

You can recognise the asbestos pictogram with this symbol:



R-phrases describe the dangers (risks) of that product (R = Risk).

Some examples of R-phrases are:

- R 23 Poisoning by inhalation;
- R 24 Poisoning in contact with skin.

S-phrases describe the security measures needed to be taken to avoid risks when working with the product. (S = Safety).

Some examples of S-phrases are:

- S 22 Do not breath in this substance;
- S 23 Do not breath in gas/ fumes / vapour / spray mist.

The purpose of the material safety data sheet is to provide information about the hazards of the substances and information regarding the safety measures to take.

The key points on a material safety data sheet are:

- * Maximum Acceptable Concentration (MAC).
- * Composition.
- * Manner of intake.
- * Manner of storage.
- * Direct hazards.
- * Labelling.
- * First aid.

The meaning of the four sections of the hazard diamond are:

- * Red section (top): Flammability.
- * Blue section (left): Health.
- * Yellow section (right): Reactivity.
- * White section (bottom): reaction with fire water and possibly specific hazards.

The numbers in the hazard diamond include information about the hazard. The diamond is numbered from 0 to 4; the higher the number the greater the hazard.

The hazard phrases on the new labels can be associated with a substance or product.

The meaning of the safety recommendations on the new labels is to describe the measures to be taken to work safely with the product.

8.5 Monitoring and medical examinations

A medical examination is required when a person works with hazardous substances; based on the exposure and the state of the worker's health an investigation determines whether he is and will remain suitable.

A medical examination determines whether the employee will remain suitable to continue working with the hazardous substance. Periodicity depends on the product and the exposure, for most materials it is once a year.

8.6 Specific hazardous materials

When working with asbestos, the following statutory provisions apply:

- * New use and processing of asbestos is prohibited.
- * Every company must make an inventory of the asbestos present in the company; if there is asbestos present, the employer draws up a control programme to keep the exposure to asbestos as low as possible.
- * Waste containing asbestos must be collected and disposed of separately.
- * A specific label is displayed where asbestos is present, (this applies to everything included in the inventory as well as for the recipient of asbestos waste).
- * The use of high-speed mechanical equipment (high pressure cleaners, air compressors, dry blasting equipment, sanding discs and grinders) to process, cut or clean asbestos (-containing) material or to remove asbestos is prohibited.



Always be careful with (suspicion of) asbestos.

Gaskets that can be removed without breaking/chipping may be removed by anyone.

Fixed (bonded) asbestos products that can be removed without breaking.

These works may only be carried out by appropriately trained people (experts in removal asbestos).

While being exposed to asbestos, an employee will be exposed to the following dangers:

- * Chance of Asbestosis. This is a disorder that belongs to black lung. Tissue will connect which lead to lesser lung capacity. This way the heart can get overloaded.
- * Chance of Asbestos lung cancer. Lung cancer caused by asbestos.
- * Chance of Mesothelioma. This is a tumor of the lung membranes.



Een An asbestos fiber.

Possible asbestos locations are:

- * Heat insulation on a range of appliances and equipment.
- * Flanges and bearing seals.
- * The materials used to make steel structures and above-ground tank structures fire-resistant.
- * In and behind the brickwork of furnaces and boilers and structures of above-ground tanks.
- * Insulation in electrical equipment.
- * In fire-resistant blankets.
- * Brake linings.
- * Sewage pipes.
- * Sealing material.
- * Roofing material and wall cladding.
- * Floor coverings.

If asbestos may be present in a working location, this working order should be followed:

- * A risk assessment based on an asbestos inventory must have been carried out.
- * Any exclusions in the asbestos inventory do not concern the location where the work is to be done.

Materials that are suspected to contain asbestos should be handled as follows:

- * Before any work is carried out in locations where asbestos may be present, an assessment is to take place of the risk of exposure to asbestos. This assessment is to be based on an asbestos inventory.
- * If there is a suspicion that asbestos is present, the work must be immediately halted.
- * Alert the operational supervisor and the client.
- * The client has a sample taken by an independent laboratory to determine whether asbestos is concerned.
- * If asbestos is present, remove in accordance with statutory instructions.

A lot of products contain hazardous substances. For example:

- * Organic solvents: often made from crude oil, for instance white spirit.
- * Cyclic compounds: are often used as solvents, for instance benzene, phenol, toluene and xylene.
- * Heavy metals: for instance lead, mercury and zinc.

The harmful effects of hazardous substances depend on the nature of the substance. As an example, the effects of some substances are stated down below.

- * Organic solvents: through inhalation, headache in the short term, brain damage in the long term.
- * Cyclic compounds: benzene and phenol are toxic, benzene is also carcinogenic.
- * Heavy Metals: very toxic.
- * Carbon monoxide: very toxic by inhalation, prevents oxygen intake into the blood and explosion hazard.
- * Cement: irritating to the respiratory system and the skin, risk of eye injury, in wet condition chemical burns in the event of long-term contact.
- * Quartz dust: prolonged inhalation may lead to pneumoconiosis

At home, there are also many hazardous substances. The dangers of some products are:

- * Agents for unblocking drains: (caustic soda) corrosive.
- * Paints, varnishes: possibly containing organic solvents.
- * Organic solvents: flammable or highly flammable, in the event of long-term use brain damage, degreases the skin (e.g. white spirit).
- * Detergents for dishwasher: irritating, chance of eye injury.

8.7 Leaks and biological materials

Consequences of leaks are:

- * Spread of product in the air breathed in.
- * Fire hazard.
- * Environmental contamination.
- * Slipping hazard.



A concrete depositor may have prolonged contact with concrete.

Possible causes of leaks:

- * Poor maintenance of the facilities.
- * Incorrect installation of flanges.
- * Erroneous procedure when transferring from one container to another.
- * Leaking taps/seals.
- * Damages (e.g. Fork-lift truck collision).

Measures to be taken against leaks and corrective measures to be taken after leaks have occurred are:

- * Regularly check whether the system is in order.
- * Only allow trained personnel to perform repairs and activities.
- * Every starting leak must be reported.
- * Sufficiently large leak trays under tanks.
- * Professional repair of leaks.
- * Professional removal of spilt product.

A preventive measure is a measure to prevent incidents. A corrective measure is a measure to mitigate or fix the consequences from an incident.

One could be exposed to biological agents that could cause damage to health:

- * Waste processing.
- * Health care.
- * Agriculture.
- * Food and beverage industry.
- * Contact with animals.
- * Work in sewage pipes.
- * Working in water treatment plants.
- * Working in contaminated soil.
- * Working in the pharmaceutical industry.
- * Biotechnology industry.

Some types of health damage caused by biological agents:

- * Infections / diseases.
- * Poisoning.
- * Allergy.
- * Fungal infections.
- * Parasites.

Some prevention and hygiene measures while working with biological substances are:

- * Protection of the skin: - Protective clothing. - Gloves. - Protection cream.
- * Washing hands.
- * Vaccinations.



The usage of PPE in the horticultural sector.



Chance of health damage cause by spoiled food.

8.8 Industrial gas cylinders

The principle used to distinguish gases in cylinders from each other is by colour coding. Colour coding can be done in accordance with the old and new code.



Always be cautioned when working with gas cylinders.

Colour codes on several gas cylinders are:

- * Oxygen (white shoulder).
- * Nitrogen (black shoulder).
- * Carbon dioxide (grey shoulder).
- * Air (light green shoulder).
- * Acetylene (chestnut brown shoulder).

The colour code is only required for the shoulder of the cylinder. The capital letter 'N' indicates that the colour code is according to the new coding.

When choosing a gas cylinder you should pay attention to the color code on the cylinder. Based on the description on the label you must check again whether u picked the right cylinder.

Safe storage of gas cylinders is very important to prevent accidents. Considerations for storing gas cylinders are:

- * Secure the cylinders correctly.
- * Protect against negative weather influences.
- * Screen from sun and other radiated heat sources.
- * Do not set up series of gas cylinders at the workplace.
- * Sufficient ventilation of the storage location.
- * Do not store in/near cellars and excavations/wells (many gases are heavier than air and can remain 'hanging').
- * Keep oxygen cylinders separate from cylinders containing flammable gases.
- * Ensure there are the correct fire-fighting agents and water for use as a coolant near the storage location.



A well organized storage.

8.9 The dangers of excessively low or high oxygen concentrations in the air for humans and the surroundings

The concentration of oxygen in the air under normal conditions is 21%.

The minimal concentration of oxygen in the air during work should be 19%.

The effects of deviation from a normal concentration of oxygen depend on the oxygen content:

- * Higher than 21%: increased fire and explosion hazard.
- * Lower than 21%: lack of oxygen, effects depend on the percentage: dazedness, unconsciousness and death.

A low oxygen concentration can be caused by various conditions.

Possible causes are:

- * Lack of ventilation.
- * Corrosion.
- * Other reactions in which oxygen is used: e.g. bacteriological and biological reactions, fire.
- * Substances that are released into a room and as a result displace oxygen (e.g. nitrogen, other inert gases).

A too high oxygen concentration can be caused by oxygen leaking from pipes or oxygen cylinders.

Before working in a room with a low oxygen concentration, employees have to take at least one of the following actions:

- * Eliminate the lack of oxygen by mechanically ventilating the room.
- * Use self-contained breathing apparatus.

Measures against a too high oxygen concentration in a room:

- * Use hose break protection.
- * No oxygen cylinders in the area (e.g. confined space).

Examination questions: Hazardous substances

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 You are reading the label of a hazardous product. How can you tell which risks the product presents?
 - ☐ C From the S phrases.
 - ☐ C From the M phrases.
 - ☐ C From the R phrases.
- ☐ 2 What does the yellow surface of the diamond hazard warning sign indicate?
 - ☐ C Reactivity hazard.
 - ☐ C Health hazard.
 - ☐ C Fire hazard.
- ☐ 3 Which poisonous effect does carbon monoxide have?
 - ☐ A Paralysis of your sense of smell.
 - ☐ B Oxygen is no longer absorbed into the blood.
 - ☐ C An irritating effect on the airways.
- ☐ 4 What information is contained in a material safety data sheet?
 - ☐ A Information about the steps which must be taken to be able to work with this product in a confined space.
 - ☐ B Information about the permits which are necessary to work with the product.
 - ☐ C Information about the hazards of the substances of which a product consists and about the safety measures which you must take.
- ☐ 5 Under normal conditions, the oxygen in the air is:
 - ☐ A 21%.
 - ☐ B 19%.
 - ☐ C 25%.
- ☐ 6 Lack of ventilation:
 - ☐ A Eliminates the lack of oxygen.
 - ☐ B Can cause an oxygen concentration which is too low.
 - ☐ C Causes problems with breathing.
- ☐ 7 A white shoulder on a cylinder means:
 - ☐ A Carbon dioxide.
 - ☐ B Air.
 - ☐ C Oxygen.
- ☐ 8 When storing gas cylinders:
 - ☐ A One has to secure them properly.
 - ☐ B One should put them in special cellars.
 - ☐ C They should be placed near the workplace for easy acces.
- ☐ 9 What is the meaning of 'hazardous substances'?
 - ☐ A Materials which explode easily if they get mixed with a high concentration of oxygen.
 - ☐ B Materials that pose a danger to health and/or to the environment.
 - ☐ C Materials that require special training before use.

- 10 Hazardous substances at the workplace are indicated with:
- A A square danger symbol, black on a red background.
 - B A square danger symbol, black on an orange background.
 - C A square danger symbol, black on a yellow background.
- 11 Vaccinations is a precautionary measure for working with:
- A Biological substances.
 - B Flammable substances.
 - C Irritating substances.
- 12 Biological substances can be present in:
- A Agriculture
 - B Contaminated soil
 - C Both answers are correct.
- 13 Professional removal of spilt product is a preventive measure to take against:
- A Fire.
 - B Falling objects.
 - C Leaks.
- 14 What is not a cause of leaks?
- A Damage (e.g. fork-lift truck collision).
 - B Erroneous procedure when transferring from one container to another.
 - C Correct installation of flanges.
- 15 Slipping is a consequence of:
- A Environmental contamination.
 - B Leaks.
 - C Using hazardous substances.
- 16 The hazard of organic solvents is:
- A Flammable or highly flammable.
 - B Corrosive.
 - C Irritating.
- 17 An example of cyclic compounds is:
- A Zinc.
 - B Phenol.
 - C White spirit.
- 18 The chance of mesothelioma is a health effect related to:
- A Benzene.
 - B White spirit.
 - C Asbestos.
- 19 What is not a possible location of asbestos?
- A In newly build houses.
 - B Roofing material and wall cladding.
 - C In and behind the brickwork of furnaces and boilers and structures of above-ground tanks.

20 What is not a legal stipulation with respect to asbestos?

- A New use and processing of asbestos is allowed.
- B Waste containing asbestos must be collected and disposed of separately
- C Where there is a risk of contact with asbestos, a specific label is applied.

21 A concept and objective of the periodic medical examination is:

- A Regular medical examinations must determine whether the candidate remains suitable to continue to work with substances
- B Regular medical examinations must determine whether the candidate remains suitable to continue to work with the hazardous substance
- C Periodicity depends on the worker.

22 Safety recommendations on the new labels describe:

- A The measures to take to avoid change in the working conditions.
- B The measures to take to avoid risks when working with the product.
- C The measures to take to avoid fire.

23 The numbers on a hazard diamond go from:

- A 1 to 5.
- B 0 to 4.
- C 0 to 5.

24 The measures to take to avoid risks when working with the product are described in:

- A The H phrases.
- B The R phrases.
- C The S phrases.

9 Fire and explosions

In the Netherlands there are dozens of deaths caused by fire yearly. The damages caused by fire are tens of millions euros.

The dangers of fire are often underestimated (for example repairing a roof where fire is used). This underestimation can have major consequences.

An explosion is a very fast combustion of substances which generates very high energy in a short period of time. Because of the heat generation the temperature of the air rises very quickly and expands with great force.



Burning gas.

Preventing fire and explosions is very important. During fire hazardous activities, prevention always comes first!

Make sure fire extinguishers are present during fire hazardous activities and check after half an hour after finishing the work if the fire actually is off.

9.1 Dangers of fire and explosions

For the occurrence of fire or explosion, the following factors are necessary. These factors are also called Fire triangle.

- * Flammable substance.
- * Ignition energy.
- * Oxygen.

The flash point (ignition temperature) is:

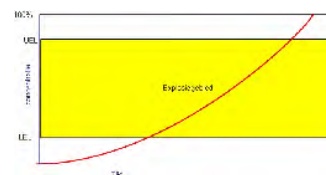
- * The lowest temperature at which a liquid develops so much flammable vapour that it can be ignited in the air that is present, under normal air conditions.

Explosion limits (explosive limits) are the boundaries of the explosion area. The definition of explosion limits is:

- * Minimum or maximum amount of gas/ vapour in the air which makes an explosion possible.

The ignition temperature is the lowest temperature at which a liquid develops so much flammable vapor that it can be ignited in the air that is present, under normal air conditions.

The explosion area is the area between the lower and upper explosion limit. In this area, the mixing ratio between air and gas/vapor where an explosion can occur.



The yellow area is the explosion area between the LEL and the UEL.

The lower explosion limit (LEL) is the minimum amount of gas/vapor that must be present in the air for an explosion.

The upper explosion limit (UEL) is the maximum amount of gas/vapor in the air which can cause an explosion.

An ignition source provides the energy needed for a substance to burn or explode. Examples of ignition sources are: naked flames and hot gases, mechanical sparks or welding sparks, electrical installations and equipment, hot surfaces, static electricity or lightning.

An explosion is a very rapidly developing fire that releases a pressure wave. A pressure wave is a very fast increase of air pressure followed by a sudden drop in air pressure caused by an explosion.



An exploded ink factory in Denver.

Certain products can influence the fire (reaction of fuel and oxygen):

- * A positive catalyst promotes the reaction speed, therefore this will fan the fire.
- * A negative catalyst slows down the reaction speed, therefore this will slow down the fire.

The combustion gasses in a fire are lighter than air. The occurrence of smoke and combustion gasses in a fire have the following impact on the surroundings:

- * Limited visibility.
- * Possibly toxic or hazardous.



A raging fire with big smoke.

During a raging fire the heat has a strong influence on the surroundings because:

- * Fire heats up products in the vicinity, as a result of which they can reach their spontaneous ignition temperature and ignite spontaneously.
- * Radiated heat heats up gas cylinders and reservoirs in the vicinity so that they can crack and explode.
- * Due to heat it is difficult to get closer to the fire.

9.2 Classification of fires and the associated extinguishing equipment

The fire classes can be stated as:

- * Class A: Fire involving solid materials.
- * Class B: Fire involving liquids.
- * Class C: Fire involving gases.
- * Class D: Fire involving combustible metals.
- * Non-classified fires.

The different fire classes are:

- * Class A: fires involving solid substances, so called dry fires, for instance wood, paper, cotton, plastic, textile.
- * Class B: fires involving liquids or substances that become liquids, for instance petrol, oil, alcohols, paints, rubber, paraffin, solvents.
- * Class C: fires involving gases, including methane, propane, butane, acetylene.
- * Class D: fires involving combustible metals, including magnesium, aluminium, potassium, sodium, metal alloys.
- * Non-classified fires: fires such as fires in and on electrical appliances.

The extinguishing methods for a class A fire are:

- * Extinguish using water.
- * Extinguish using fire extinguishing powder (in practice ABC powder).
- * Extinguish using foam.
- * Fire blanket, can be used on burning products and persons and for fire on a flat surface.

The extinguishing methods for a class B fire are:

- * Extinguish using fire extinguishing powder (in practice ABC or BC powder).
- * Extinguish using foam.
- * Extinguish using sand.

The extinguishing methods for a Class C fire are:

- * Shut off gas supply.
- * Extinguish using fire extinguishing powder (in practice ABC or BC powder).

The extinguishing methods for a Class D fire are:

- * Extinguish using a specific extinguishing powder.

The extinguishing methods for a non-classified fire (for instance an electrical device) are:

- * Extinguish using carbon dioxide.
- * Extinguish using modified foam.

9.3 Extinguishing equipment

The most important extinguishing principles are based on several parts of the fire pentagon:

- * Switch off/ remove fuel.
- * Reduce ignition energy and remove heat by cooling.
- * Displace oxygen or oxygen occlusion.
- * Remove the ignition source.

Several hazards and disadvantages of extinguishing using water are:

- * Causes water damage.
- * Conducts electricity.
- * There are many chemical substances that react fiercely with water.
- * Hazardous if used on burning liquid: causes burning liquids to splatter resulting in jets of flame; many liquids float on water allowing the fire to spread.
- * Sensitive to freezing.
- * Environmental damage due to contaminated fire water.



Reservoirs crack due to the heat.



Fire reel with water and a small extinguisher.

The disadvantages of extinguishing using foam are:

- * Causes limited damage.
- * Sensitive to freezing.
- * Foam can conduct electricity.

The disadvantage of extinguishing using sand is: sand rapidly cakes on and hardens.

Extinguishing powder is an ideal agent to extinguish fire, but it has several disadvantages:

- * Has a limited cooling effect.
- * Reduces visibility in small areas.
- * Causes damage to electronics.
- * Contaminates.



Powder extinguisher.

Using carbon dioxide for extinguishing fires, there is a chance of frostbite because of the low temperature. This can occur when you hold the agent in the wrong way. A different risk is suffocation because oxygen is displaced.

Hazards and disadvantages of extinguishing using fire blankets are:

- * One has to get close to the seat of the fire.
- * If not applied well: danger for the person fighting the fire.
- * If incorrectly or incompletely covered, there is a greater chance of injury and damage.

Extinguishing characteristics of water are:

- * Limited oxygen displacement due to steam.
- * Cooling ability.

Extinguishing characteristics of foam are:

- * Excludes oxygen from the seat of the fire.
- * Foam cools.



Foam extinguisher

Sand has the characteristic that it excludes oxygen.

The extinguishing characteristics of extinguishing powder are:

- * Slows down the reaction between the flammable substance and oxygen, negative catalyst.
- * Excludes oxygen to a limited degree.

Carbon dioxide has the characteristics:

- * Displaces oxygen.
- * Limited cooling capacity.



CO2 extinguisher.

A fire blanket excludes oxygen. You need to use a fire blanket on burning products, persons and fire on flat surfaces.



Fire blanket.

9.4 How to act in the event of a fire

The actions to take when a fire is detected are:

- * Ensure your own safety.
- * Report the fire.
- * Warn the people in the vicinity of the fire.
- * Close doors and windows.
- * Guide people to safety.
- * Extinguish the fire, if you can.

The actions to take when extinguishing a fire are:

- * Ensure your own safety at all times.
- * Choose the correct extinguishing agent.
- * Extinguish the fire.
- * Remain alert even when the fire appears to be extinguished, it can still fan up again.
- * If you cannot extinguish the fire, stop trying to do so and evacuate.

When treating burns you should rinse with water as soon as possible for at least 15 min.

The tasks of the hot work watch are:

- * Preventive actions to prevent a fire.
- * Supervise the requirements of the hot work permit.
- * In the event of a fire, call in emergency services.
- * Take action to extinguish a starting fire until the intervention team arrives.
- * Provide first aid.



The fire watch must have had a decent education for extinguishing an outbreak of fire.

A fire watch has to follow a special training.

9.5 Explosive environments

An explosion risk environment is an area where there may be an explosive mixture. In these environment, special measures are needed to prevent the ignition of possible hazardous mixtures.

Branches/sectors where explosion risks may be present include: chemical industry, rubbish dumps, power plants, waste-processing companies, gas companies, timber-processing industry, agricultural businesses (biogas), metal-processing companies, food and feed industry, pharmaceutical industry, refineries, recycling companies.

- * Branches/sectors where explosion risks may be present include: chemical industry, rubbish dumps, power plants,

The warning sign with indication Ex indicates an explosion risk zone.



Warning for explosion zone

Factors that are important in using a personal explosion meter in explosion risk environments are:

- * The dust density (high concentration of the substance) is large (several metres sight), footprints stay in the dust. The dust must be composed of fine particles.
- * Wear at chest level and not underneath clothing.
- * The dust spreads all over the room caused by resuspension. Or when there are smolderings.
- * Test before use, or must have been tested before use.
- * Actions to be taken in case of an alarm.

Examination questions: Fire and explosions

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 What is a disadvantage of extinguishing a fire with water?

☐ A The burning material may react severely to water.
☐ B Operators are hindered by the smoke that is created.
☐ C Using water as an extinguishing agent is relatively expensive.
- ☐ 2 What must you know before you start extinguishing a fire?

☐ A How long it has already been burning.
☐ B How high is the temperature.
☐ C What substance is burning.
- ☐ 3 What must be done first when a fire is detected?

☐ A If possible extinguish it yourself.
☐ B Guide people to safety.
☐ C Report the fire.
- ☐ 4 What does an explosion limit signify?

☐ A The highest or lowest temperature of a solid substance at which an explosion can occur.
☐ B The highest or lowest concentration of a gas in the air at which an explosion can occur.
☐ C The highest or lowest concentration of metals at which an explosion can occur.
- ☐ 5 What is meant by the flash point?

☐ A The highest or lowest concentration of a mixture of gas and air at which an explosion can occur.
☐ B The highest or the lowest temperature of a solid substance at which an explosion can occur.
☐ C The lowest temperature of a liquid at which the vapour from that liquid can be ignited with an ignition.
- ☐ 6 What result can an excess of oxygen have during a fire?

☐ A Easier ignition of flammable substances.
☐ B Prolonged burning of the flammable substances.
☐ C Suffocation of the fire.
- ☐ 7 A flammable substance:

☐ A Is needed for a fire or explosion.
☐ B Always is a gas.
☐ C Always causes fire or explosion.
- ☐ 8 What is an explosion?

☐ A A fire caused by a flammable substance.
☐ B A very rapidly developing fire that releases a pressure wave.
☐ C A fire in class C.
- ☐ 9 What does a class A fire involve?

☐ A Gasses
☐ B Liquids.
☐ C Solid metals.

10 The sign that indicates an explosion risk zone:

- A Displays Ex.
- B Displays Er.
- C Displays a fire symbol.

11 An explosion risk environment is:

- A An environment where special measures must be taken to prevent the ignition of explosive mixtures that may be present.
- B An environment where it is not possible for an explosive mixture to arise.
- C An environment where special measures must not be taken to prevent the ignition of explosive mixtures that may be present.

12 A requirement for the hot work watch is:

- A Minimal five years of experience in working with fire.
- B A specific training.
- C Minimal a year of experience in working with fire.

13 When treating burns:

- A Flush with water for at least 5 minutes.
- B Flush with water for at least 15 minutes.
- C Flush with water for at least 10 minutes.

14 When extinguishing a fire one must not:

- A Remain alert even when the fire appears to be extinguished, it can still fan up again.
- B Ensure your own safety at all times.
- C Just walk away when the fire is extinguished.

15 Fire blankets:

- A Cools.
- B Displace oxygen.
- C Exclude oxygen.

16 A disadvantage of extinguishing with a fire blanket is:

- A Chance of freeze wounds due to low temperatures.
- B One has to get close to the seat of the fire.
- C Conatmination.

17 A disadvantage of extinguishing with carbon dioxide is:

- A Can create a suffocating atmosphere due to oxygen displacement.
- B Chance of freeze wounds due to low temperature.
- C Both answers are correct.

18 An extinguishing characteristic of sand is:

- B Sand displaces oxygen.
- B Sand excludes oxygen.
- C Sand has a cooling ability.

10 Equipment

"Good tools for the job". The advantages of good tools are numerous. They are safer, sustainable and more efficient.

The use of good tools is not sufficient. The tools have to be inspected periodically. Before every use the tools have to be visually inspected and operated according to the regulations.

Always read the regulations before usage and maintenance of your tools!



A grinding wheel with insufficient protective caps and a poor tool rest.

The H&S regulation states that the employer must provide good tools and maintain them.

Each type of tool has its own specific risks. If you are not authorized to use certain tools, then don't!

10.1 Machinery, powered tools and hand tools

General dangers while working with machines, powered tools and handtools are:

- * Being caught by moving parts.
- * Being hit by flying material/ particles.
- * Getting trapped.
- * Electrocution.
- * High or low temperature of the workpiece.
- * Hearing damage due to noise.
- * Physical complaints due to vibration.
- * Physical overloading due to incorrect posture.
- * Failure of the control system or energy source.
- * Distracting the operator.
- * Cuts or grazes.
- * Poor maintenance.
- * Dislocation of hand and wrist and bruises due to incorrect use.
- * Fire or explosion due to sparks.
- * Burns due to short circuit.
- * Injury caused by the tool snagging.
- * Injury due to an air hose coming loose.
- * Machines taking a long time to slow down.
- * Touching cutting parts.
- * Inhalation of hazardous substances.
- * Flying nails (nailer/ stapler).
- * Rebounding nail from hard material (nailer/ stapler).
- * Penetrating workpiece with nail (nailer/ stapler).

General dangers while working with a nailer or stapler are:

- * Flying nails.
- * Rebounding nail from hard material.
- * Penetrating workpiece with nail.



The requirements machinery and powered tools should meet are:

- * Periodic inspection.
- * Period of validity shown by label or colour coding.
- * User and maintenance instructions in the language of the country where it is used.
- * After 1995, having a CE mark.
- * Moving parts are guarded.
- * Being in a good state of repair and being suitable for the work to be carried out.



On all risky work equipment, there has to be a date of inspection.

General safety measures while working with machinery/power tools are:

- * Switched off during maintenance activities.
- * Never operate with drive guards open.
- * The floor around machines and permanently installed tools must be cleared up, clean, level, dry and not slippery.
- * There must be sufficient space to walk and manoeuvre around the machine to be able to operate it.
- * Personal protection equipment must be present and correctly used.
- * Operated in the correct manner.
- * Screening off of danger zone.
- * Brake to ensure that the machine does not take too long to slow down.
- * Stationary machines are fitted with an emergency stop.
- * Operating instruction card (machines) present.
- * Effective dust extraction (if applicable).
- * In areas where there is a risk of explosion, only use explosion safe machines and power tools.
- * Powered manual tools are fitted with a dead man's switch.

A deadman's switch on a machines makes sure that if you release the operating handle, the machine or tool stops immediately.



A grinding wheel with deadmans switch.

An emergency stop on a machine makes sure that the event of emergency, used to stop the machine as quickly as possible.

The requirements for an emergency stop on a machine are:

- * Easily accessible.
- * Clearly visible and recognisable.
- * After using the emergency stop, the system can only be restarted by using the normal start-up procedure.



An emergency stop device.

A no-load switch ensures that the machine does not start automatically after the power supply has failed and then returned.

During the operating of machinery, powered tools the operator of the machine has to meet the following requirements:

- * Sufficient training and experience.
- * Older than 18 years of age (for operating hazardous machines).
- * Must not wear loose-fitting clothing, loosely-hanging jewellery or have loose hair.
- * Must not wear gloves if contact with rotating parts is possible and/or if there is a risk of becoming caught.
- * Must not bypass, take out of operation or remove safety devices.
- * Must not leave running machines unattended.
- * Must know the location and function of the emergency stop.



Protections should not be disabled.

10.2 Hand tools

A hammer must meet the following requirements:

- * The head of the hammer must be firmly fixed on the handle.
- * The handle must be undamaged.
- * The head of the hammer must be intact.

Spanners must comply with:

- * They must be undamaged.
- * The jaws of the spanner must fit the nut.

When working with spanners, the following applies:

- * Only extend the length using the correct accessories.
- * Preferably use a ring spanner instead of an open-ended spanner.
- * Slogging spanner has a cord to prevent it from falling/ flying away if it is knocked off.



An impact wrench.

Screwdrivers must be exactly tailored to the screw slot and the blade should not be too sharp.

When working with small screwdrivers, one must clamp the workpiece or secure it.

Requirements files have to meet are:

- * The handle should be in good condition and undamaged.
- * The handle is firmly attached to the file.

Requirements for chisels are:

- * No burrs on the tool.
- * Tool must be equipped with hand protection.

Pliers must have jaws and a hinges which are clean and intact.

Knives should be sharp and be adapted to the work they are used for.

Considerations when working with knives:

- * Cut away from your body.
- * Do not slide an extendable blade out too far.
- * Break off the worn part of the blade using a blade breaker or pliers.

When picking a saw, one should pay attention to which kind of work has to be performed. The saw needs to be tailored to the work you are performing.

Safety measures when working with saws are:

- * Only use well sharpened, sharp saw blades.
- * A good setting of the saw.
- * Blade well tensioned.
- * The teeth should be facing forward.

10.3 Fixed machines

The dangers while working with (pillar) drills are:

- * The workpiece becoming unclamped.
- * Injury due to the drill breaking.
- * Injury due to wiping away swarf with the hand.
- * Injury due to swarf.
- * Splashes of cooling oil or cutting oil.



A pillar drill.

While working with (pillar) drills there has to be a transparent screen between the operator and the drill.

While working with (pillar) drills, the workpiece has to be clamped securely and swarf has to be brushed away using a swarf brush/hook.

The dangers while working with bench grinders are:

- * Flying particles: eye injury.
- * Exploding grinding wheel.
- * The workpiece becoming trapped against the grinding wheel.
- * Injury due to touching a rotating grinding wheel.
- * Inhaling grinding dust.



A fixed grinder.

Requirements that bench grinders need to meet are:

- * Grinding wheels are sufficiently round.
- * The grinding surface of the wheel is flat.
- * Two grinding wheels on one machine should not differ too much in size.
- * The distance between the rest and the grinding wheel is a maximum of 3 mm.
- * The sides of the wheels are guarded.
- * Protective shield present.



The distance between the tool rest and the grinding wheel is too big.

Safety measures while working with bench grinders are:

- * Only professional personnel are allowed to install or replace the grinding wheels.
- * The rest must be regularly adjusted.
- * The rest must only be adjusted when the machine is stopped.
- * Protective shield must always be used.

The dangers for employees while working with table circular saws are:

- * Getting hit or caught by the saw or other moving parts.
- * Getting hit by sawn off, flying parts of the product.
- * Health complaints due to inhaling hazardous dust.



A fixed circular saw.

Requirements that table circular saws need to meet are:

- * The machine must have a suitable guard, attached to a robust stand.
- * Riving knife and support present suitable for the saw blade.
- * Easily adjustable, suitable guide present.
- * One or more connections for dust extraction present.



The riving knife behind the blade.

Safety measures while working with saw benches are:

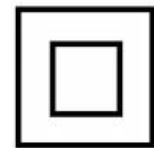
- * Use a push stick to safely push/ guide small or narrow pieces of work.
- * Correctly adjust the saw blade.
- * For large pieces of work, use a second person or a roller conveyor.



A push stick.

10.4 Powered hand-held tools

Electric power tools must be double insulated and must not be grounded. The double insulation is no protection against water or humidity.



The symbol for double insulated tools.

Pneumatically powered hand tools are double insulated. This is no protection against water or humidity.

Safety measures while working with pneumatically powered hand tools are:

- * Take regular breaks when using vibrating tools.
- * Shut off air supply after use.

Requirements for hand-held grinders are:

- * Must have a handle on the side (dependent on the power of the hand-held grinder).
- * Protective cover over the grinding disc.
- * Hand powered tools must have a dead man's switch.



An angle grinder.

On the grinder you can read the following information:

- * The manufacturer's name.
- * Maximum allowable rotational speed.
- * Disc dimensions.
- * Application, year, specification of material for which the disc is intended.

Safety measures for working with hand-held grinders are:

- * The workpiece must be clamped.
- * The maximum rpm of the disc must never be exceeded.
- * The hand-held grinder must only be put down when the disc has stopped rotating.
- * Do not use the side of a cutting disc (for instance for deburring).
- * Use grinding discs for deburring.
- * Only use cutting discs for cutting, not for deburring.

A grinding wheel is calculated on its lateral load and a cutting wheel is not.

Requirements for a hand-held circular saw are:

- * Protective guard completely covers the part of the saw that does not cut.
- * Automatic guard over the part of the saw that cuts.
- * Riving knife: suitable for the diameter and thickness of the saw.



A hand-held circular saw.

Safety measures while working with a hand held circular saw are:

- * Adjust the saw blade and the guide so that the saw blade protrudes as little as possible under the workpiece to be cut.
- * If necessary, obtain assistance from a second person.
- * Do not allow the saw to jam.
- * Always keep the power cord behind the saw.

Nailers and staplers must have a protection against undesired firing.

Safety measures for working with nailers/staplers are:

- * Check the maximum working pressure.
- * Guide empty when recharging.
- * Use staples or nails suitable for the machine, material and shape of the workpiece.
- * Keep the free hand far away from the machine.
- * Force the head onto the workpiece.



A nailer.

Requirements a chainsaw must meet:

- * Handles: vibration free with anti-slip.
- * Internal vibration damping.
- * Rear handle: protection against broken chain.
- * Protective bracket.
- * Chain brake with anti-kickback device.
- * Chain catch mechanism.
- * Double handed operation.
- * Protective guard (for the chain).
- * Balanced.
- * Locking the operating controls with automatic decoupling of the operating button.



A chainsaw.

While working with chainsaws the following counts:

- * The operator of the chainsaw needs to follow appropriate training with exam.
- * Trousers and gloves made of fibre that is difficult to cut and that will block the saw.
- * Stand so that if the saw recoils the chain does not hit the body.
- * Use the correct saw blade / chain combination.
- * Always use two hands to operate the chain saw.

10.5 Internal transportation

A pallet trolley is a vehicle with a lifting height of 20 cm that is suitable for manually or electrically moving pallets.

Possible dangers while working with pallet trolleys are:

- * Back complaints due to incorrect working posture.
- * Painful shoulder and arms due to pulling a pallet trolley that is carrying too heavy loads.
- * Trapping of fingers, ankles, feet and toes.
- * The load falling.
- * Collision with people, goods, buildings.
- * Damage to goods and equipment due to incorrect use.



A pallet trolley.

The precautionary measures to take when working with pallet trolley are:

- * Ensure there is sufficient room to manoeuvre.
- * Travel over an even floor.
- * Load in a stable way, distribute the load over the two forks.

The operator of the pallet trolley should not push the pallet trolley but pull it. You need to think of your posture when you are pulling the pallet truck (straight back) to prevent physical injury.

A forklift truck is a vehicle used to move loads on forks or using special accessories.

Possible dangers while working with forklift trucks are:

- * The load can fall.
- * The load can tilt.
- * The vehicle can tilt.
- * Collision with people, goods, buildings.
- * Damage to goods and equipment due to incorrect use.
- * Inhaling diesel engine emission when a diesel-driven forklift truck is used in a confined space.



A forklift truck

Working with forklift trucks is very risky. The following measures have to be taken while working with these trucks:

- * Never increase the contra weight.
- * Load in a stable way, distribute the load over the two forks.
- * Do not lift people without a special safety cage.
- * The driver's visibility must be optimal.
- * Do not hoist without a special accessory.
- * Do not take a passenger if no special seat is available.
- * Take account of people in the vicinity.



This is forbidden and dangerous!

A forklift driver has to wear his safety belt at all times. The forklift driver has to possess the tested expertise.

Examination questions: Equipment

The correct answers of these questions can be found in annex 1 of this book.

- 1 What must you look out for when using chisels?
 - A Only use chisels that have no burrs on the cutting edge.
 - B Only use chisels that have been tested.
 - C Only use chisels of the right strength.
- 2 Why may you never wear gloves when working with machines that have rotating parts?
 - A Because your hands do not come into contact with the rotating parts.
 - B Because the machine can be damaged if the glove is dragged into it.
 - C Because the glove and therefore also your hand can be dragged into the machine.
- 3 What does working safely with pillar drills mean?
 - A Making use of the riving knife on the drill column.
 - B Clamp the workpiece in a secure way.
 - C Making use of counter pressure when drilling.
- 4 What must hammers be provided with to prevent accidents? They must be provided with
 - A A plastic handle.
 - B A metal handle.
 - C A handle which is secured in the hammer-head.
- 5 What is the purpose of a dead man's switch on a pneumatic tool?
 - A This allows the operator to take breaks.
 - B To put the tool in the warm up position.
 - C To immediately stop the machine when the operating handle is released.
- 6 What must be present on a table circular saw?
 - A A glass shield.
 - B A blade guard attached to a robust stand.
 - C A recoil protection.
- 7 Electrical hand tools for a voltage of 230 volts
 - A Must be double insulated.
 - B Must be provided with safety earthing.
 - C May only be used by a sufficiently instructed person.
- 8 A periodic inspection has to be done:
 - A On machines and power tools.
 - B Every week.
 - C To make sure every employee likes his job.
- 9 A machine will stop as quickly as possible when:
 - A The emergency stop is used.
 - B The operating handle is released.
 - C If the rotating parts are blocked.

10 What is a pallet trolley?

- A A vehicle to transport people in the workplace.
- B A vehicle used to move loads over different heights.
- C A vehicle with a lifting height of 20 cm.

11 A requirement that is placed on a forklift truck driver is:

- A Use of the safety belt.
- B Possessing the tested expertise.
- C Both answers are correct.

12 What is not a safety measure to take when working with forklift trucks?

- A Do not take a passenger if no special seat is available.
- B One can increase the contra weight when needed.
- C Both answers are no safety measure.

13 When working with saws:

- A The blade must be well tensioned.
- B The blade must not be too sharp.
- C The operator must have followed the suitable training.

14 When working with knives:

- A Teeth must face forward.
- B Do not slide an extendable blade too far out.
- C The blade must be well tensioned.

15 When working with small work pieces:

- A One must hold them with one hand.
- B One must clamp the workpiece.
- C Both answers are correct.

16 Trousers and gloves made of special fibre are used when:

- A Working with a chain saw.
- B Working with screwdrivers.
- C Working with hammers.

17 A nailer/stapler must have:

- A An extra handle on the side of the tool .
- B A protection against undesired firing.
- C A transparent screen.

18 A riving knife is used with:

- C
- A A hand-held circular saw.
- B A chain saw.
- C A screwdriver.

19 Which information has to be present on a grinding disc?

- A The minimum allowable rotational speed.
- B The manufacturer's name.
- C The user of the disc.

20 When working with pneumatically powered hand tools one must:

- A Shut off air supply after use.
- B Take regular breaks when using vibrating tools.
- C Both answers are correct.

21 When working with bench grinders, what is not a possible hazard?

- A Health complaints due to inhaling hazardous dust.
- B The driving knife can come loose.
- C Getting hit or caught by the saw or other moving parts.

11 Demolish activities

VCA covers risky operations in a hazardous environment. In this chapter demolition is treated. In performing this work you should pay extra attention to working safely.

Attention to risks is provided in the RI&E (risk assessment and evaluation), the TRA (Task Risk Analysis) and the LMRA (Last Minute Risk Analysis). The use of these instruments is a good and practical way to indicate the risks during the demolition, and how to reduce and make them manageable.



The demolition of an office building.

For the demolisher, the LMRA is very important. He assesses his own work with this to check if it is safe for demolition.

11.1 Demolition

The most common dangers and risks associated with demolition work are:

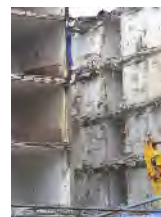
- * Tripping and stumbling.
- * Protruding parts of construction.
- * Working at height.
- * Instability in demolition front.
- * Falling material.
- * Collapse.
- * Release of hazardous substances and unpleasant dust.
- * Noise.

Standard safety measure by demolition work:

- * Fall arrest devices.
- * Take account of the load-bearing capacity of the remaining structure.
- * Do not work above or below each other without having a specific plan.
- * Use chutes to remove any material.
- * Personal protection equipment; helmet, safety shoes, overall, breathing apparatus, hearing protection.
- * Investigate the concentration of hazardous materials (asbestos inventory).

When removing ceramic fibres you should pay attention to the following:

- * Use a fixing agent.
- * Break as little as possible.
- * Use a P3 dust filter.
- * Wear a disposable overall.
- * Do not use a chute.



The demolition front can be unstable.



Who has to wear a helmet?!



The demolition site will be kept moist to prevent dust formation.

Examination questions: Demolish activities

The correct answers of these questions can be found in annex 1 of this book.

1 A hazard during demolition is?

- A Instability of demolition front.
- B Protruding parts of construction.
- C Both answers are correct.

2 During demolition work one must have:

- A A fall arrest device.
- B Sufficient break times.
- C Break as little as possible.

3 What must one do when removing ceramic fibres?

- A Break everything in little pieces for easy transportation.
- B Do not use a chute.
- C Wear your working overall.

12 Welding, cutting and burning

When working with metal one often has to weld and burn (electricity and radiation).

Welding is a process where one connects materials through pressure and/or heat. The material is turned into a liquid or pasty state at the connection spot. This allows the different parts to connect with each other.

The heat can be reached by using electric current or by burning gasses. Frequently used gasses are oxygen, acetylene and propane.

A gas cylinder is a metal cylinder in which a particular gas or gas mixture like oxygen, propane or butane is stored under high pressure.

When welding, harmful radiation may be released.



Performing welding activities.

12.1 Welding and cutting

Possible dangers/risks when electric welding are:

- * Electrocution.
- * Fire and explosion caused by heat formation and sparks.
- * Burnt skin caused by UV-radiation.
- * Burnt cornea caused by UV-radiation.
- * Blindness caused by infrared radiation.
- * Heat radiation.
- * Poisoning caused by welding smoke.
- * Lung complaints due to inhaling welding smoke.
- * Incorrect working posture.



Electric welding

During electric welding, the employee has to take the following safety measures:

- * There has to be a work permit.
- * Extraction of welding smoke.
- * Provide enough ventilation.
- * User personal protection: welding shield, welding clothing, safety shoes, air ventilated shield.
- * Use welding curtains to protect bystanders against UV- and infrared radiation.
- * Keep fire extinguishers within reach.



Welding goggles

Possible dangers/risks during gas welding, cutting and burning are:

- * Cylinder containing pressurised oxygen, promotes fire, oxidising.
- * Cylinder containing acetylene or propane, flammable and explosive.
- * Backdraft.
- * Fire or explosion.
- * Propane is heavier than air and remains hanging in wells, excavations and cellars.



Gas cylinders for autogenous cylinders

Safety measures for gas welding, cutting and burning with oxygen and acetylene are:

- * The acetylene cylinder can be used when standing straight or lying down in an angle of at least 30°.
- * When using acetylene take measures related to ventilation and if necessary gas measurements.
- * Flame extinguisher in hose between acetylene cylinder and torch.
- * Hose break protection.
- * Flow-back limiters on gas and oxygen hoses of the torch.



Hose break protection.

Examination questions: Welding, cutting and burning

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 What must one do when gas welding or using a cutting torch with oxygen and acetylene?
- ☐ A Acetylene cylinders must be stored upright or at an angle of at least 25°.
- ☐ B Acetylene cylinders must be stored upright or at an angle of at least 30°.
- ☐ C Acetylene cylinders must be stored upright or at an angle of at least 40°.
- 2 What is a risk of gas welding?
- A Air is heavier than propane.
- B Backdraft.
- C The cylinders are not pressurised.
- 3 What is a safety measure which has to be taken when electric welding is done?
- A Personal protection: welding mask, welding apron, welding clothing, safety shoes, air-ventilated welding mask.
- B Welding curtains to protect people in the vicinity from UV and infrared radiation.
- C Both answers are correct.
- 4 What is not a risk associated with electric welding?
- A Blinding due to infrared radiation.
- B Burnt cornea due to heat radiation.
- C Lung complaints due to inhaling welding smoke.

13 Digging and working at or in excavations

Making excavation pits like building wells for basements, underground car parks or even tunnels is becoming more common. Trenches are used for laying cables or sewers. In 20% of all these excavations, damages occur to cables and pipes. This causes many economical consequences and risks for safety.

Working in trenches and excavations brings many dangers. For example groundwater, collapsing walls, contaminated soil or bad ventilation.



Excavating near a leak in the water pipes.

Before starting an excavation, one should always check for cables and pipes with a Klic message. The detector will receive information about existing cables and pipes. If, during the dig, contaminated soil is found, experts need to be warned.

Damages on pipes or cables must be reported to the administrator of the network.

13.1 Digging and working at or in excavations

Careful digging and excavation is important for damage prevention because:

- * Safety risks.
- * Environmental risks.
- * Economic loss.
- * Security of delivery of utility companies

Possible hazards and risks when digging a trench:

- * Electrocution due to damaging live cables.
- * Fire and explosion due to damaging gas and oil pipes.
- * Being buried due to the trench caving in.
- * Flooding due to water flowing in.
- * Soil contamination due to damaging pipes carrying hazardous substances.
- * Suffocation due to damaging gas pipes.
- * Soil contamination due to hazardous products and/or biological substances.

When digging trenches or holes you should be careful to go. Note therefore always the following points:

- * Only excavate when the positional data are known.
- * Only excavate when told to do so by the person in charge or the contractor.
- * Excavate test trenches.
- * Follow up instructions of the grid operator.
- * Only excavate after the position of cables and pipes has been determined by test trenches within 1.5 metres of the reported position.
- * Report any different position or damage to the person in charge.
- * When using an excavator, use an excavator bucket with no teeth.



Deep holes for pouring foundations.



When trenching, you stand between the pipes and cables!

The greatest risk of excavations is caving in of the scope. There are various possibilities to reduce this risk:

- * Effective strutting and slope provisions.
- * Adjust the slope to the depth of the excavation and the nature of the subsoil.
- * Store excavated soil at a safe distance.
- * Ensure that the shoring is at the edge of the excavation.
- * Keep vehicles, materials and equipment at a safe distance.
- * Fencing around excavation.
- * No gas cylinders in the excavation.
- * Install two ladders.
- * Consider a deep excavation to be a confined space.

Examination questions: Digging and working at or in excavations

The correct answers of these questions can be found in annex 1 of this book.

1 Why is it important to excavate carefully?

- A Economic winnings.
- B Security of delivery of utility companies.
- C The digsite may collapse.

2 What is not a risk associated with excavating?

- A Suffocation due to damaging gas pipes.
- B Electrocution due to damaging live cables.
- C Explosion hazard.

3 What is a rule for carefully excavating?

- A Report any different position or damage to the person in charge.
- B Only excavate after the position of cables and pipes has been determined by test trenches within 1.5 metres of the reported position.
- C Both answers are correct.

4 When working in or near an excavation you must:

- A Install a ladder.
- B Use fencing at risky places.
- C Effective strutting and slope provisions.

14 Working at height

Most of the deadly victims in the Netherlands are in the construction sector. Many of those stayed dead after a fall from height or falling objects.

Working at height is very risky!

Working on ladders is being reduced because this is one of the biggest causes of falling.

Work as much as possible with fixed or mobile scaffolds. This makes work safer and more pleasant!



Prevent the risk of falling. Always mark unsafe workplaces.

According to the Arbowet, risk of falling takes place above 2.5 meters. This does not mean that a fall from a lower height cannot be deadly!

14.1 Working at height

According to Health and Safety legislation you are working at height when:

- * From a height of 2.5 metres above the floor.
- * When working where there is a risk of falling above a dangerous location such as moving parts or water.



Working at height, the unloading of concrete

Many accidents occur when working at height, some risks are:

- * Falling (for instance from the roof edge, an upper storey, etc.).
- * Falling through a floor opening.
- * Falling objects.

When working at a height of 2.5 meters or more, you should take the following measures:

- * Install safe scaffolding, a landing or a working platform.
- * Install effective fencing, rails or other similar provisions: minimum rail, interim rail and toe board.
- * Install safety nets.
- * Cover floor openings.
- * Use personal protection equipment.

When working on sloping roofs, the following security measures are required:

- * Use duckboards when roofing is not strong enough to allow movement over the sloping roof.
- * Install safety nets: when there are large openings in the roof and at the edges of the sloping roof.



Working on pitched roofs.

When working at less than four meters from the roof edge the following measures have to be taken:

- * Roof edge protection and safety nets have to be installed.
- * Use a safety harness if collective protection is impossible.



Roof edge protection

Ladders are used to bridge a certain height and to carry out light work. For heavier work or great heights it is better to use scaffolding or a hydraulic platform

Decide beforehand whether the work is indeed light. Take the following into consideration:

- * The ladder is approved and displays an inspection label.
- * When the use of scaffolding or a hydraulic platform is technically or economically infeasible.
- * Height less than 7.5 metres.
- * Standing time less than 4 hours.
- * Use of less force than 100N (10 kg).
- * Reach at the maximum one arm length.
- * Do not use ladders above wind force 6.



Take care of your ladders and stairs.

It is important to take sufficient safety measures, in order that you can safely work on the ladder. Observe the following guidelines when working on a ladder:

- * Position at an angle of approximately 75 degrees.
- * Ensure that the bottom of the ladder cannot sink into the surface or slide.
- * Ensure that the top of the ladder cannot slide sideways.
- * The ladder must extend at least 1 metre above the location to which the ladder provides access.
- * Put the ladder on a solid surface, not back to front or upside down.
- * Keep rungs and ladder safety shoes clean, do not climb the ladder with slippery or dirty soles.
- * Climb while facing the ladder.
- * Keep the access to the ladder free of obstacles.
- * Block any door or passage behind the ladder.
- * Place metal ladders at a distance of at least two metres from live parts.
- * Always have three points of contact with the ladder: two feet, one hand; two hands, one foot.



The ladder should be 1 meter above the roof.

Standard scaffolding is not intended to be moved. A scaffold can also be called a platform. For the erection of steel standard scaffolding the following requirements apply:

- * Authorised supervisor present during the erection activities.
- * Stability calculations for the scaffolding (platform) and instructions for erection or modification are available.

The following guidelines apply to the safe use of steel standard scaffolding:

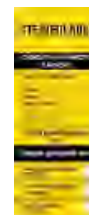
- * Do not change anything to the construction of the scaffold (platform).
- * Do not leave material or tools behind on the floor of the scaffold (platform).
- * Do not work on a stairs or ladder that is placed on the scaffold (platform).
- * Do not load the scaffolding more than the maximum load stated on the scaffold tag.

At all access points of the scaffolding a scaffold tag should be displayed. This tag displays the following:

- * The maximum allowable load.
- * Whether the scaffold (platform) can be accessed.
- * Whether the scaffold is inspected and until when this inspection is valid.



This is not allowed.



An example of a scaffold tag.

Suspended scaffolding is used, for instance, to clean the windows of a high building. To work in a safe manner with suspended scaffolding you should observe the following guidelines:

- * The operator must not leave the scene if there are people in the suspended scaffold and the operating device is not on the suspended scaffold itself.
- * In the event of fault leave the scaffold (platform).
- * Do not leave ready for use and unsupervised.
- * If there is no or limited eye contact between the operator and the people in the suspended scaffold use communication devices.
- * Do not exceed the maximum load.
- * Test before use.
- * Persons in the suspended scaffold must wear a safety harness that is attached to a special attachment point.
- * Cordon off the area under the scaffold using tape or fencing.

Rolling scaffolding has wheels in order to easily move the scaffolding. Rolling scaffolding is intended to be quickly erected and broken down. To work in a safe manner with rolling scaffolding the following guidelines apply:

- * Lock all wheels before climbing it.
- * Climb from the inside.
- * Hoist up materials and tools in a safe way.
- * Keep the working floor tidy.
- * Do not stand on the stabilizers.

To safely move rolling scaffolding the following guidelines apply:

- * Do not move it if there is someone or something on the rolling scaffold.
- * Stabilisers without wheels must remain as low as possible to the ground when moving the scaffold, stabilisers with wheels must remain on the ground.
- * Ensure that the stabilizers cannot shift when moving the scaffold.
- * The surface must be level and solid, otherwise install rails, etc.



Safe together (?) on a suspended scaffold in the port of Rotterdam.



Just so something real quick with a rolling scaffold...

Safety cages for people must meet the following:

- * They must be inspected and a written proof of the inspection must be present.
- * Suspended from a four-leg bridle.
- * Displays information on the outside specifying the: allowable load; dead weight; allowable number of persons.

The safety measures to take when using safety cages for people are:

- * The crane driver and the people in the safety cage must be able to see each other where possible.
- * The crane driver and the people must be able to communicate with each other.
- * One person in the safety cage gives instructions to the crane driver.
- * People in the safety cage must wear safety harnesses and be attached to a attachment point in the safety cage.

A hydraulic platform is a moveable piece of equipment to move people so they can work at height.

There are various types of hydraulic platforms, for example:

- * Self-propelled hydraulic platform.
- * Hydraulic platform on a trailer, truck or van.
- * Scissor lift.

Working with a hydraulic platform is not without its dangers. A number of dangers are:

- * Electrocution.
- * Collision.
- * Falling objects.
- * Getting trapped between an object and the platform.
- * Falling out of the safety cage.
- * Falling off the hydraulic platform.



The pruning of trees from an hydraulic platform.

Hydraulic platforms should meet the following requirements:

- * A valid inspection certificate.
- * A visible inspection sticker.
- * A plate showing the maximum load-bearing weight.
- * User instructions.
- * Clear signs on the operating panel.

The following rules apply to the safe use of a hydraulic platform:

- * Ensure that the platform stands horizontally on a flat surface.
- * Driving: only with retracted outriggers, the main arm down and the cage in the neutral position and according to the user instructions.
- * Driving over short distances: see user instructions.
- * Do not use for hoisting activities.
- * Do not use as passenger lift.
- * Persons in the cage must use a fall arrest system attached to the cage.
- * If the people in the cage are higher than 25 metres above the ground, they must use walkie-talkies.
- * When used in a high-risk environment, assistance at ground level must be present.



The bucket of a hydraulic platform.

Before you can work on a hydraulic platform you should have demonstrably tested expertise.

14.2 Working at wall and floor openings

Openings in the wall and floor can also cause accidents, for example:

- * Falling through a wall opening.
- * Falling through a floor opening.
- * Being struck by an object that falls through a wall or floor opening.

Safety regulations have been set out for working by openings in the wall and floor:

- * Cover with strong material fixed to the surrounding surface.
- * Install railing or fencing.
- * If protection is impossible, then mark the danger and make access difficult.

Examination questions: Working at height

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 From which height must the employer take legally required measures to prevent risks of falling?
 - ☐ C From 3.5 meters.
 - ☐ C From 1.5 meters.
 - ☐ C From 2.5 meters.
- ☐ 2 What safety measure can reduce the danger of falling for everyone working on a flat roof?
 - ☐ C A roof edge safety system.
 - ☐ C Walk boards.
 - ☐ C A safety harness.
- ☐ 3 What must be checked before a ladder may be used?
 - ☐ A Whether the ladder is no longer than 12 meters.
 - ☐ B If the ladder is set up at an angle of approximately 75 degrees.
 - ☐ C If the stiles of the ladder are provided with steel caps at the bottom.
- ☐ 4 What must persons in a suspended scaffold make use of?
 - ☐ A A retracting harness.
 - ☐ B A renal support belt.
 - ☐ C A safety harness.
- ☐ 5 On which side must a rolling scaffold be climbed?
 - ☐ A Front.
 - ☐ B Inside.
 - ☐ C Rear.
- ☐ 6 What should you preferably do with a floor opening?
 - ☐ A Keep your distance during work.
 - ☐ B Cover it with solid material.
 - ☐ C Wear a safety harness attached to a safety line.
- ☐ 7 What is not a hazard of working near wall and floor openings?
 - ☐ A Being struck by an object that falls through a wall or floor opening.
 - ☐ B Falling through a floor opening.
 - ☐ C Closing a large floor opening.
- ☐ 8 A ladder is used for:
 - ☐ A For carrying out minor activities.
 - ☐ B Bridging a certain height.
 - ☐ C Both answers are correct.
- ☐ 9 The reach at maximum one arm length is a rule for working with:
 - ☐ A Pillar drills.
 - ☐ B Ladders.
 - ☐ C Bench saws.

15 Electricity

Electricity can be lethal. Still, a lot of people have had an electric shock before. Lucky, but it could have gone wrong if the circumstances had been slightly less favourable.

One can survive even a direct hit from lightning. But this means you are very lucky...

When using electrical equipment, one needs to be well instructed or trained. Otherwise there is a high risk for accidents.

Electrical tools must be periodically checked/inspected according to NEN 3140.



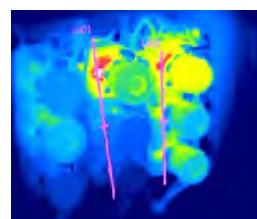
Maintenance work on electrical installations. Perform them safely!

When working with electrical equipment, always check the cables and the tools for possible damages. Deliver damaged tools to your supervisor.

15.1 The dangers of electricity

Although many people know electricity is dangerous, they often do not know why. The four most important dangers of electricity are:

- * Electrocution.
- * Injury caused by sparks and arcs.
- * Injury caused by electric shock and the reaction to an electric shock.
- * Risk of fire and explosion.



A thermographic image of a fuse box. The red parts are hot.

Factors on which an injury due to an electrical current passing through the human body depends are:

- * The route that the electrical current takes through the body.
- * Current intensity.
- * Duration of the current passing through the body.
- * The voltage and the current that are associated with it.
- * Physical condition.

When a current of more than 30 mA passes through the body, this results in an increased blood pressure and a difficult and irregular heartbeat. It will also cause convulsions. Alternating currents may cause ventricular fibrillation.

Differences between direct and alternating current:

- * Direct current is less dangerous than alternating current (in general).
- * 120 V direct current and 50 V alternating current are (under dry conditions) harmless (guideline).
- * In the event of short circuit, direct current causes larger arcs than alternating current.

The current intensity (the amount of current) depends on following factors:

- * Voltage.
- * Resistance of the surface including footwear (linoleum and a rubber mat have a high resistance, an earth floor, a paved or concrete floor is a good conductor and therefore has a low resistance).
- * Moisture level of the skin.
- * Thickness of the skin.
- * Contact surface (the larger the surface the higher the current).

The manners in which an employee can get a secondary injury when working with electricity are:

- * In the event of electrocution, the victim can be startled and as a result get injured.
- * In the event of a short circuit, one can get hit by flying material.
- * In the event of a short circuit one can be flung aside by the pressure wave.

A short-circuit occurs when two parts under differing voltages come into direct contact with each other. Earthing is direct contact between a live part and the earth.

The size of the spark depends on:

- * The potential difference (difference in voltage) between the parts.
- * Current intensity.

Fire and explosions can be caused by electrical current. Examples are:

- * Sparks or arcs.
- * Excessive temperature increase to parts of electrical installations, in spaces that pose an explosion risk or in spaces containing an explosive gas mixture.

Pay special attention to the following high-risk situations.

- * Defective and unsound machines, appliances and pipes/ cables.
- * Poor or missing earth connections.
- * Erroneous installation or assembly.
- * Direct or indirect contact with live parts.
- * Injudicious use of electrical installations and/ or materials.
- * Lack of attention, carelessness and ignorance.

15.2 Safety measures when working with electricity

Safety provisions when working with electricity are:

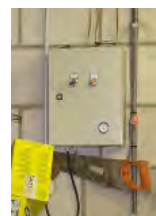
- * Physical protection.
- * Double Insulation.
- * Insulation.
- * Earth leakage protection.
- * Using very low voltages.
- * Earthing.

By way of physical protection you ensure that live parts cannot be touched. A physical protection is a cover forming a barrier between you and the danger.

Examples are a switch box and the housing of a washing machine.



In damp rooms, defective sockets are very dangerous.



The saw can damage the power cables in this cabinet.



Never leave electrical boxes open.

In the case of insulation live parts are made inaccessible by covering them with nonconductive material. You can only remove the material by breaking it.

Insulation is therefore a permanent covering of, for example:

- * Rubber.
- * Plastic.
- * Ceramics.

The function and operation of an earth leakage circuit breaker is to turn off the power as soon as a current above a certain value leaks to earth.

An earth leakage circuit breaker does not offer absolute safety. The limitations of a earth leakage circuit are:

- * The earth leakage circuit breaker should also be regularly tested.
- * It offers no protection against overcurrent, heating up and short circuits.

Safety earthing means the connection between the external metal parts of electrical appliances and earth.

When these parts do become live, the current immediately flows to the earth preventing external metal parts to become live. This reduces the dangers for the user.



A poorly connected socket.

When earthing, whenever possible the connection should be made to the existing earth network.

Scaffolding (platforms) must be earthed when there are electrical cables, wires and/or electrical equipment (including hand-held tools) on the scaffolding that carry an unsafe voltage.

When this is the case the steel scaffolding or platform should be earthed. If you do not do this the entire scaffolding could become live when the (exposed) lines or cables are damaged or broken.

15.3 Working on and in the vicinity of electrical installations

Laymen are people who have not followed any electro technical training and are also not aware of the dangers of electricity.

Laymen do not have any power of authority to work on live electrical installations.

Sufficiently trained persons are not electro technical experts, but they have been instructed by professional people in the dangers and know how they should work to minimise the risks. Under supervision they are allowed to carry out simple electro technical activities, for which they have received training.

The execution of carefully described activities for which demonstrably sufficient instruction has been given and under regular supervision: electrical activities requiring more in-depth knowledge require persons with far-reaching powers of authority, a professional person.



Work on electrical installations.

A professional person is a person who has been appointed and who has relevant (electrical) training and experience, allowing him/her to recognise and prevent hazards caused by electricity.

15.4 Temporary electrical equipment

Before you start working with electric equipment or tools you should always:

- * Visually check for damage and non-conformities
- * Report damage and do not use damaged equipment.

The electricity for a temporary installation is supplied by a construction site junction box. A construction site junction box should have:

- * 30 mA earth leakage circuit breaker (for each outgoing power supply line).
- * 300 mA earth leakage circuit breaker (general).



A leakage circuit breaker.

If a cable reel is not completely unwound when used, it can become very hot and cause a fire.

Safety measures to take when using cable reels are:

- * Always comply with the maximum allowable power of the reel in the wound up and unwound state.
- * Unwind the cable completely from the reel.



Limit the use of cable reels as much as possible.

The greatest danger when using extension cables is overload of the cable.

When using cable reels and extension cables one has to ensure that the cables are suitable for the power of the connected devices.



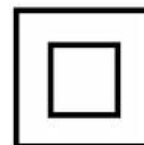
After use, clean-up extension cables.

Before you start working with electric equipment or tools you should always:

- * Carry out a visual check for any damage or any changes
- * Report any damage or changes
- * Not use the damaged equipment

Double insulation is no protection against dust or water. Therefore, an installation will be encased in a housing. Housings can be further protected against:

- * Double insulation (add sign) does not protect against the penetration of dust and moisture.
- * The penetration of moisture (e.g. rain).
- * The penetration of dust.



The symbol for double insulated tools.

15.5 Static electricity

Static electricity is the build-up of an electric charge outside an electrical installation. "Normal" electricity is always found inside a standard electrical system.

Static electricity occurs only on non-conductive surfaces, because the charge is unable to spread across such a surface (static means "without movement"). Static electricity can occur between:

- * Rising gas- or vapour bubbles that cause turbulence.
- * When spraying paint or similar activities.
- * Rubbing plastic.
- * During the pneumatic transport of powders and granulates in mixers, feeders of weighing bunkers, tanker trucks.
- * Friction of clothing against the skin.
- * Drive belts.
- * For some liquids, when the liquid flows through a plastic pipe or is stirred.
- * Walking over plastic floor covering.

Possible hazards of static electricity are:

- * Discharge of static electricity can result in a spark.
- * Sensitive electronic equipment can be damaged by static discharge.

Another danger is that sensitive electronic equipment can become defective as a result of static discharge. You can limit the build-up of static electricity within a process by:

- * Earthing pipelines, equipment and tanks.
- * Limiting the height that a product falls in a storage vessel or tank.
- * Limiting the speed of flow.
- * When earthing, wherever possible connect to the existing earth network.
- * Wearing antistatic shoes and clothing.



A managing decision to prevent static electricity is good earthing.



Jumping sparks from static electricity.

Examination questions: Electricity

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 Is a layman allowed, with the permission of an expert, to work on live electrical installations?
 - ☐ C Yes, provided the layman is over 21 years.
 - ☐ C Yes, but only if the expert is constantly available for consultation.
 - ☐ C No, never.
- ☐ 2 What does an earth leakage circuit breaker do?
 - ☐ C It diverts the high tension current to the earth wire.
 - ☐ C It disconnects the power at a certain leakage current.
 - ☐ C It works like an artificial earth, which is necessary in many installations.
- ☐ 3 Which safe voltage must you use in a confined space to work safely with electricity?
 - ☐ A Direct current (maximum 50 volts) or alternating current (maximum 130 volts).
 - ☐ B Direct current (maximum 12 volts) or alternating current (maximum 120 volts).
 - ☐ C Direct current (maximum 120 volts) or alternating current (maximum 50 volts).
- ☐ 4 What can cause accidents through electricity?
 - ☐ A The lack of earthing.
 - ☐ B Pressing the emergency stop.
 - ☐ C The blowing of a fuse.
- ☐ 5 What is the greatest danger from a flow of current through the body?
 - ☐ A The shock reaction.
 - ☐ B The occurrence of burns.
 - ☐ C Flow of current through the heart.
- ☐ 6 Electrocutation is a hazard related to:
 - ☐ A The use if hazardous substances.
 - ☐ B The use of electricity.
 - ☐ C The use of flammable substances.
- ☐ 7 Injury due to an electrical current passing through the human body does not depend on:
 - ☐ A The current intensity.
 - ☐ B The duration of the current passing through the body.
 - ☐ C The thickness of the skin.
- ☐ 8 The intensity of the current trough the body depends off:
 - ☐ A Resistance of the surface, excluding footwear.
 - ☐ B Voltage.
 - ☐ C Your lifestyle.

9 How does a short circuit occur?

- A When two parts at different voltages come in direct contact with each other.
- B When two parts at the same voltages come in direct contact with each other.
- C When a electric wire is not earthed.

10 The occurrence and size of an electric arc is determined by:

- A Current intensity.
- B The potential difference (difference in voltage) between the parts.
- C Both answers are correct.

11 Antistatic shoes are worn when:

- A Working in confined spaces.
- B Working with static electricity.
- C Working with ladders.

12 A hazard of static electricity is:

- A Discharge of static electricity can result in a spark.
- B Sensitive electronic equipment can be damaged by static discharge.
- C Both answers are correct.

13 In which of the situations stated will static electricity not occur?

- A In rooms with a low oxygen level.
- B For some liquids, when the liquid flows through a plastic pipe or is stirred.
- C Walking over plastic floor covering.

14 What is not a protection method for housings?

- A Double insulation does protect against the penetration of dust and moisture.
- B Protection against impact and falling objects.
- C Protection against the penetration of moisture (e.g. rain).

15 A hazard associated with using power cables and extension cables is:

- A The power cables are too short.
- B Tripping over cables.
- C Overloading of the cable.

16 When using a cable reel one must:

- A Unwind the cable completely from the reel.
- B Not use them in confined spaces.
- C Never completely unwind them from the reel.

17 When a cable reel is not completely unwind:

- A It can cause static electricity.
- B It can become very hot and cause a fire.
- C It can cause electric sparks.

18 The protection value of the earth leakage circuit breaker that must be added for extra safety in the temporary construction site junction box is:

- A 300 mA earth leakage circuit breaker (for each outgoing power supply line).
- B 25 mA earth leakage circuit breaker (for each outgoing power supply line).
- C 30 mA earth leakage circuit breaker (for each outgoing power supply line)

19 A safety measure to take before working with electrical equipment is:

- A Visually check for damage and non-conformities.
- B Report damage and do not use damaged equipment.
- C Both answers are correct.

20 When using electricity:

- A Scaffolding must be placed on rubber mats.
- B Scaffolding must be earthed.
- C Nothing special has to be done with scaffolding.

16 Confined spaces

Never seen or been to a confined place? This is almost impossible!

Tanks, crawling spaces, silos, manure cellars or sewers are all confined places. The dangers of confined places are often underestimated, resulting in many (fatal) accidents in confined places.

Work in confined spaces is therefore always risky and should never be performed individually.



Working in reservoirs is working in a confined spaces.

To perform these activities safely, you should use work permits. The work permit describes all required safety measures.

The security guard outside of the confined place should never enter this place.

16.1 Working in confined spaces

Hazards in a confined space are numerous and large. Therefore, when working in a confined space, there has to be a working permit. This identifies the risks and the required measures.

Think of how you can be saved when working in a confined space if things would go wrong.

Characteristics of a confined space are:

- * Narrow, small, wet, slippery and little room to manoeuvre.
- * Not intended as accommodation for people.
- * Hardly any natural ventilation.
- * Difficult to access.
- * Poor escape options.
- * Poorly lit.
- * Pipes and cables at unexpected locations.

Examples of confined spaces are:

- * Storage reservoirs or tanks.
- * Sewers and pipes.
- * Lift shafts, crawling spaces and cellars.
- * Tents in / near excavations, contaminated soil or pipelines.



These are also confined spaces.

There are many ways to reduce the risks before you start work. But also during the work, you can take measures to carry out the work as safely as possible:

- * The minimum oxygen percentage is 19%.
- * Any potentially explosive gas mixture present at less than 10% of the LEL.
- * Concentration of toxic substances or vapour must be lower than the threshold limit value.
- * Effective air ventilation.
- * Warning signs at the entrance to prevent unauthorised access.
- * Pipes connected to the space must be blanked off or disconnected.
- * A valid work permit or a written release.
- * People who enter the space are at least 18 years of age.
- * The room is clean and dry.
- * With a permanent attendant.

Sometimes it is not sufficient to only measure the concentrations beforehand. When there is a risk that the concentrations might change, it can be necessary to take measurements more often and in various positions in the confined space. Concentrations can change when hazardous substances evaporate, for example due to heat. But the work itself can also influence the concentrations in the space.



Never thrust your nose!

The supervisor ensures that all measures have been taken to be able to safely enter the confined space and to work there:

- * Carrying out supervision.
- * Maintain communication with the person in the confined space.
- * Warning the emergency services if the person in the confined space is in danger.
- * Checking the ventilation.

The person who is appointed to be the permanent attend should possess demonstrably tested expertise to be able to properly carry out this task.

Suffocation is the result of a lack of oxygen. The lack of oxygen in confined spaces can, for example, be the result of insufficient ventilation or be the result of an inert gas.

In many confined spaces there is an increased danger of fire and explosion. This involves several reasons:

- * Too little ventilation, as a result of which the lower explosion limit is reached rapidly.
- * Presence of flammable material in the space.
- * Presence of flammable substances in the material or when removing layers of corrosion.
- * Naked flame or sparks in the confined space.
- * Oxygen leakage from hoses or burners that have not been closed properly



A refinery burns down.

In confined spaces there is also an increased risk of poisoning. In a confined space the concentration of a hazardous gas can quickly increase because the space is so small and the ventilation often poor. This means that almost all substances, especially very volatile substances, can cause danger in a confined space.

When working in a narrow space with poor ventilation, you will be warm sooner and also start perspiring sooner. This increases the conduction through your body and therefore the danger of working with current. Always work with a safe voltage (lower than 50V for alternating current or lower than 120V for direct current).

For the person working in the confined space it is important that the time he remains in the confined space (the duration of stay) is as short as possible.

You should only start working in a confined space when the danger of poisoning, stupefaction, electrocution or injury has been removed. When, despite all measures, there is still danger present, you should use personal protective equipment. The standard equipment for working in confined spaces consists of:

- * Self-contained breathing apparatus (so: not filter masks) at concentrations of hazardous gases/ vapour that exceed the threshold limit value and when the oxygen percentage is too low.
- * Hearing protection (dependent on the type of activities and the nature of the confined space).
- * Life line, unless it poses an additional hazard.

A confined space sometimes has natural ventilation. However, in many cases this will not be sufficient. Ventilation of a confined space can be done by:

- * Natural ventilation.
- * Mechanical ventilation.
- * Local extraction when gases and vapours are released.

In narrow, damp spaces with walls made of a conductive material, there is an increased risk of electrocution (death as a result of an electric shock). Lowering the chance of electrocution can be done in the following ways:

- * Remove electricity from the electrical equipment in the correct manner.
- * Use safe voltages in confined spaces that have conducting walls.

Movable parts in confined spaces must be shut down or locked by an expert.



An agitator in a pit.

When working in a confined space where in gases are used, you should pay attention to the following:

- * Check the hoses for leaks in advance.
- * Keep gas cylinders outside of the space.
- * During breaks and at the end of the work, place the torches and hoses outside of the space.

The following also applies to welding and cutting:

- * Remove or cover all flammable materials.
- * Keep fire extinguishing equipment within reach.
- * Check for oxygen leaks.
- * Local extraction of welding fumes.
- * Use safe voltages when welding electrically.



Welding in a confined space.

Applying a coat of paint on the interior of a confined space poses specific risks. Especially when spraying these risks are greater. When applying paint you should pay attention to the following.

- * Use self-contained breathing apparatus.
- * Ventilation of the entire space.
- * All equipment must be earthed to prevent static electricity build up.
- * The space must be ventilated up to several days after the painting activities.
- * Do not shut off the oxygen supply required for curing the paint.

Examination questions: Confined spaces

The correct answers of these questions can be found in annex 1 of this book.

- 1 What is one feature of a confined space?
 - C Poor access.
 - C Good lighting.
 - C Enough work space.
- 2 Is a manhole guard compulsory while working in an enclosed space?
 - C Yes, but he may leave when the measurements have been completed.
 - C No, unless this is indicated in the contractor's license.
 - C Yes, for as long as there is someone in the enclosed space.
- 3 How can a suffocation hazard occur in confined space?
 - A Trough defects in the electrical installation.
 - B Trough high concentrations of oxygen.
 - C Through an insufficient concentration of oxygen.
- 4 Which of the types of voltage below is safe?
 - A 120 Volt alternating current.
 - B 240 Volt direct current.
 - C 120 volt direct current.
- 5 What is not an example of a confined space?
 - A Storage reservoirs or tanks.
 - B Welding or soldering tents in/ near deep excavations, contaminated soil or pipelines
 - C Narrow, small, wet, slippery and little room to manoeuvre.
- 6 A rule related to carrying out measurements in a confined space is:
 - A The measurements must be carried out by the supervisor.
 - B The measurements must be carried out by the employee who is working in the confined space.
 - C The measurements must be carried out by an expert person.
- 7 When supervising a confined space, this should be taken into account:
 - A Communication with the person in the confined space should not take place in danger of collapsing.
 - B Warning the emergency services if the person in the confined space is in danger.
 - C The supervisor should do other work while making sure nothing is wrong in the confined space.
- 8 The permanent attendant must:
 - A Be present when there are flammable materials in the confined space.
 - B Be on site all the time.
 - C Posses demonstrably tested expertise.
- 9 The duration of stay in a confined space:
 - A Is related to where the confined space is.
 - B Must be as short as possible.
 - C Does not matter.

10 A safety measure to reduce the risk of electrocution in a confined space is:

- A Both answers are correct.
- B Remove electricity from the electrical equipment in the correct manner.
- C Use safe voltages in confined spaces that have conducting walls.

17 Hoisting

In construction but also in big production halls, one can notice hoisting equipment is increasing. The use of a hoist is a form of source control to reduce the physical burdens of the employees.

When using hoisting equipment there are different risks like dropping the load or even failure or collapsing of the crane.



Hoisting activities at the Maasbrug in Gennep.

The crane operator is responsible for the safety while hoisting, but he can't do this without the people around him who load or give directions. Communication is a very important part of hoisting. All employees need to receive good instructions when working with hoisting equipment.

The use of good hoisting equipment is important. It is necessary that the equipment will be checked periodically and before every use.

17.1 Hoisting in general

Examples of hoisting equipment are cranes, tower cranes, rolling bridges or hand hoists.

Hoisting equipment is used for moving loads horizontally and vertically.

Hoisting accessories are accessories for hoisting loads (e.g. eye bolts, shackles, hoisting slings, chains, cables).

Hoisting equipment and accessories have to meet:

- * They must be inspected, proof of inspection must be present, the inspection details must be recognisably marked on the hoisting equipment.

Details regarding maximum load of the hoisting equipment can be found on a plate or ring, stamped into the hoist.



A picture with the data of the hoist.

General dangers while working with hoisting equipment are:

- * The hoisting equipment (e.g. crane) can topple over.
- * A load can fall.

General safety measures while working with hoisting equipment are:

- * Do not exceed the allowed working load.
- * Attach the load in the correct manner.
- * Regularly check the material visually.
- * Keep the angle between the legs of a two or three legged bridle as small as possible.

In the manual of the crane manufacturer's users instruction is stated when you can no longer hoist with cranes above a certain wind force.

The choice for hoisting accessories depends on:

- * Dimension and shape of the load.
- * Distance to be moved.

17.2 Hoisting equipment

Dangers when working with hoisting equipment are:

- * The hoisting equipment can fall over.
- * Being hit by equipment.
- * Getting hit within the turning cycle of the hoisting equipment.
- * Bad weather.



The crane is more stable when standing on blocks.

Safety measures while working with hoist equipment are:

- * Regularly check for damage and wear.
- * The operator of the hoisting equipment and assistant must be able to understand, see and hear each other well.
- * Use outriggers.
- * Good coordination when using multiple pieces of hoisting equipment.



The outriggers of a crane.

The following documents have to be present while working with hoisting equipment:

- * Crane logbook: This contains all signed inspections and maintenance activities.
- * Hoisting tables and graphs: This includes which weight can be moved over which distance.
- * Inspection certificates: Example: certificates for winches, hoists, gin pole derricks, additional chains, hoisting yoke and other hoisting equipment.

The professional knowledge of the hoisting equipment operator is shown by:

- * Proof of expertise (hoisting license) (obligatory for tower cranes, mobile cranes and piling rigs with a load moment equal to or greater than 10 metric tons).
- * Registration log (that includes how long the operator has worked and what types of hoisting equipment he has experience with).

Safety measures for working with chains are:

- * A link must only be loaded in the direction of the longitudinal axis.
- * Do not force chains.
- * Reject a chain that has stretched or that has elongated links.
- * Protect against sharp edges of the load.
- * Only use a special coupling link to extend the chain.
- * Completely screw in the bolts of shackles.
- * Never load the point of a hook.



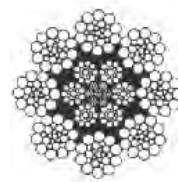
This is prohibited when using chains.

Safety measures while working with steel cables are:

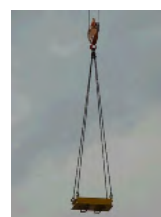
- * Store on a rack in a dry and well ventilated room.
- * Avoid contact with moisture and corrosive substances.
- * Regularly inspect for corrosion and wear.
- * Protect with square timber or corner protection if the load has sharp edges.
- * Do not knot.
- * Do not use if poor splices are present.
- * Do not use in the event of a break or damage at one place or a broken strand.
- * Do not use if severe rust is present.
- * Do not use if there is external damage.
- * Do not use if there are broken or cracked cable connections.
- * Do not use if the diameter has been reduced.

Some examples of hoisting equipment in assemblies are:

- * Hoisting Yoke.
- * Two legged bridle sling.
- * Three legged bridle sling.
- * Four legged bridle sling.



A section of a steel cable.



A load is lifted with a four leg bridle sling.

The maximum angle between slings in an assembly is 120 degrees.

Examples of situations in which a hoisting yoke is used:

- * To move a load with complicated dimensions exactly above the point of gravity.
- * If there is little room above the load.
- * When frequently moving similar loads.
- * Because the cables hang straight down, there is no chance that they will shift towards each other over the load.



A yoke is used to move a heavy load.

A manual hoist is a hoisting device which is operated by hand.

Examples of situations in which manual hoists are used:

- * To position loads in the vertical and horizontal direction during hoisting activities.
- * When using a crane is ineffective.



Manual hoists ready for use.

Possible dangers while working with manual hoists are:

- * A part of the hoist breaking.
- * The point to which the hoist is attached breaking.

Possible safety measures while working with manual hoists are:

- * Use sufficiently robust attachment points (never to landings, piping, etc.).
- * Do not load the point of the hook.
- * Do not overload.
- * Never extend the handle with a pipe.
- * Inspect before use, if defects are found return it for repair and report the defect to the supervisor.

Examination questions: Hoisting

The correct answers of these questions can be found in annex 1 of this book.

- 1 When should hoisting operations be suspended?
 - C Observe the crane manufacturer's user instructions for the value of the windforce.
 - C Above wind force 4.
 - C Above wind force 5.
- 2 Which is a hoisting tool?
 - C A sling.
 - C A retaining strap.
 - C A harness belt.
- 3 What are two slings on a steel ring called?
 - A A two legged bridle.
 - B A pair of slings.
 - C A duo.
- 4 What is recorded in a crane logbook
 - A Which inspection certificates are compulsory.
 - B Which weights may be moved over which distance.
 - C Which inspections a crane has had.
- 5 Which of the statements below about chains is CORRECT?
 - A With the use of chains along sharp edges you use edge protectors.
 - B Chains with stretched links can be used if the weight does not exceed the maximum chain limit.
 - C Chains may never be extended.
- 6 What is not an example of hoisting equipment?
 - A A tower crane.
 - B A rolling bridge.
 - C A pallet trolley.
- 7 What can one do with hoisting equipment?
 - A Move loads vertically.
 - B Move loads horizontally.
 - C Both answers are correct.
- 8 A general danger when working with hoisting material is:
 - A Vallen van de last
 - B Omvallen van het hijswerktuig (bijv. kraan)
 - C Both answers are correct.
- 9 Outriggers are used when working with:
 - A Pillar drills.
 - B Hazardous materials
 - C Hoisting equipment.
- 10 Steel cables:
 - A Must not be knot.
 - B Should be stored in a damp room.
 - C May still be used when they are very rusty.

18 Radiation

Radiation can be divided into ionising and non-ionising radiation.

We all know ionising radiation. Think of x-rays in hospitals or radioactive radiation in nuclear facilities. When living material is exposed to ionising radiation it may cause lasting effects (for example tumours) or even lead to death.

With non-ionising radiation you can think of the microwave at home, laser beams, regular UV light or radio waves. Each of these non-ionising radiation has its own risks.



Working with lasers is also risky.

Only work with (non-ionising) radiation sources if you have been instructed to.

18.1 Ionising radiation

Safety measures to prevent exposure to ionising radiation are:

- * Keep as far away as possible.
- * Avoid contamination with a radioactive substance through appropriate hygiene and by leaving packaging containing radioactive substances intact.
- * Cordon off the area around the source.
- * Wear the correct PPE.
- * Place warning signs.
- * Carry out (permanent) measurements.

Obligations of employee working with ionising radiation:

- * Measure and monitor the personal dose experienced by people.
- * To have a medical examination.

Tasks of a radiation expert within companies when working with ionising radiation:

- * To monitor radiation safety and hygiene.
- * To check whether employees, tools, equipment and/ or workplace are contaminated.
- * If required, to prescribe additional measures or halt the activities.



Radiation risks from radioactive substances.

18.2 Non-ionising radiation

Non-ionising radiation is weaker than ionising radiation. This radiation is not capable of creating ions and does not cause cell changes.

Examples of non-ionising radiation are:

- * Microwaves.
- * Ultraviolet radiation.
- * Sunlight.
- * Laser beams.
- * Infrared radiation.
- * Radio waves.



Protect your skin from the sun.

Safety measures to prevent risks by non ionising radiation are:

- * Pay attention to instructions and information displayed on appliances.
- * Respect the instructions related to the duration of use.
- * Respect safety distances.



Warning for laser beams.

Examination questions: Radiation

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 Infrared radiation is?
- ☐ A Non-ionizing radiation.
- ☐ B Ionizing radiation.
- ☐ C Natural background radiation.
- 2 A safety measure to take against ionising radiation is:
- A Keep as far away as possible.
- B Cordon off the area around the source.
- C Both answers are correct.
- 3 A medical examination is an obligation for employees who:
- A Work with hazardous substances over a short period of time.
- B Work with equipment or substances that emit ionising radiation.
- C Are ill once in a while.
- 4 Monitoring radiation safety and hygiene is a task of:
- A The supervisor.
- B The radiation expert.
- C The emergency response team.
- 5 What is the difference between ionising and non ionising radiation?
- A Non ionising radiation is the only radiation which creates ions.
- B Non ionising radiation is weaker than ionising radiation.
- C Non ionising radiation is stronger than ionising radiation.
- 6 A safety measure to take to prevent non-ionising radiation damaging health is:
- A Pay attention to instructions and information displayed on appliances.
- B Respect the instructions related to the duration of use.
- C Both answers are correct.

19 The ergonomic workplace

Ergonomics aims to design utilities, technical systems and tasks in a way which health, safety, comfort and effective functioning of people is promoted.

In the workplace, we usually see physical ergonomics. The physical ergonomics take the human anatomy into account. Relevant topics include working postures, manual handling, repetitive movements, work related complaints to muscles and tendons, workplace design and health and safety.



Lifting instructions are important for preventing physical complaints.

The European framework directive on health and safety at work creates a legal platform for ergonomics. In the first place, attention is going out to the prevention of overuse injuries caused by the working conditions.

19.1 Man and his working environment

The factors playing a role in ergonomics are:

- * Light and lighting. Depending on the work to be carried out by employees, light intensity must meet the current regulations.
- * Noise. Noise occurs at all workplaces. Machines operating, cars passing, cutting torches; it all creates noise.
- * Climate. Working comfort depends greatly on the climatic conditions. Extreme conditions result in an unpleasant working environment.
- * Vibrations. Vibrations can be harmful to your health, Exposure to vibrations with a certain frequency causes health damage.

You can never satisfy all employees with the internal climate. There will always be people with complaints.

Elements which determine the physical demands of a human are:

- * Effort: Should the employee make a big effort to perform the job?
- * Movement: Do you have to walk, bend or move a lot during work activities?
- * Posture: Do you have to stand or sit for long periods in wrong working positions?



At this stool you will quickly have a wrong working posture.

19.2 Vibrations and the workplace

We distinguish between two types of vibrations:

- * Vibrations of the hand/ arm: can arise when using mechanical (hand-held) tools.
- * Vibration of the body: can be transferred via a vehicle, large installation or moving floor.

Health problems that may result from vibrations of the hand/arm:

- * Pain in the hand and arms.
- * Damage the blood vessels and joints in the fingers and hand.
- * Numbness in the finger tips.
- * White fingers.



White fingers.

Health problems that may result from vibrations of the body:

- * Stomach ache.
- * Back ache.
- * Tiredness.
- * Headache.
- * Muscle pain.
- * Reduced concentration.
- * Disorders of the balance organ.

Exposure to vibrations with a certain frequency causes health damage. When wishing to reduce vibrations, it is best to start at the vibration source:

- * Pay attention to health hazards when purchasing machines.
- * Fit damping/ insulation, for instance fit hand-held tools with a damped handle or use vibration-damping gloves.
- * Use alternative techniques.
- * Limit the duration of exposure.



Break open a segment of road. A heavy load on the human body.

19.3 Climatological conditions and the workplace

Working comfort depends greatly on the climatic conditions. Extreme conditions result in an unpleasant working environment. The working environment depends of, for example:

- * Climate relates to the interior as well as the exterior climate. People can only work in an optimum manner when they feel comfortable. For example: rain, temperature, air flow (draught) of humidity.
- * Effort. When choosing work clothes you should also consider the physical effort that is required for the work. When carrying out heavy work your body will emit heat. When this heat cannot be disposed of sufficiently, the body will start to perspire. In order to let the perspiration evaporate, much energy is needed.
- * Clothing. When you are unable to control the climate, such as when working outside you can provide personal protective equipment, such as specific work clothing.

Factors that affect climatic conditions (indoor) in the workplace:

- * Ambient temperature.
- * Radiated heat.
- * Air humidity.
- * Air flows (draught).



Climate is important, specially in cleanrooms.

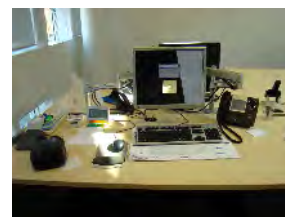
A good, comfortable work increased safety. Some measures for a good working climate are:

- * Keep the temperature as comfortable and even as reasonably possible and adjusted to the physical efforts.
- * No annoying air flows.
- * Limit exposure to an unfavorable indoor climate.
- * If necessary, provide specific PPE.
- * Maintain the moisture balance.

19.4 Lighting and the workplace

Light plays an important role in the working environment. With good lighting, the safety, quality but also the productivity will be improved. Points of attention for a correct lighting at the workplace are:

- * Adequate lighting to distinguish details.
- * Older employees need more light.
- * Contrasts in a working room must not be too great.
- * Light reflection must be prevented.
- * The light intensity of the lamps can reduce the longer they are used or due to age.
- * Dependent on the activities, the light intensity must meet the current regulations.



Even contrast differences in a computer monitor are annoying.

19.5 Physical and mental stresses during work

Elements that determine the physical impact on humans are:

- * Severity of the work.
- * The movements that must be made.
- * Working position/ posture.

The human body is vulnerable. Physical complaints can therefore rapid develop if the employee and the employer pays insufficient attention to the prevention of good working conditions. Working conditions posing a risk to the development of physical complaints are, for example:

- * Duration in the activity.
- * Prolonged sitting or standing.
- * Incorrect working positioning/posture.
- * Exposure to vibrations.
- * Carrying out many repetitive actions.
- * Excess effort such as heavy lifting.

Rest breaks at work are important. One of the functions of a break is to prevent overloading the employees.

Static stress means that you are persistently tensioning the muscles. Examples of static stress are sitting and standing.



Standing on sloped roofs is physically demanding.

Some health risk by static demands of the muscles for the employee are:

- * Reduced blood flow resulting in numbness.
- * Muscle ache and/or cramp.
- * Chronic tiredness.



Standing still over a long period of time is a static demand on the body.

Dynamic stress is caused by the alternate tensioning and relaxing of muscles. This makes you tired quickly and causes muscle ache.

Health complaints that can result from excess dynamic stresses:

- * Tiredness.
- * Muscle ache.
- * Injury.

Work also causes mental (psychological) stress. People can become mentally stressed through, for example:

- * A high workload.
- * Task versus capacity.
- * Relationship with colleagues.
- * Hierarchical relationship.
- * Physical working conditions.
- * The working hours scheme.
- * Job (in)security.
- * The infrastructure, equipment.

The individual stress during manual lifting is determined by the degree of physical lifting ability.

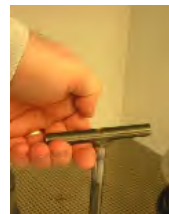
The factors that determine the risk level of manual lifting for the employee:

- * The horizontal distance between the body and the load.
- * The vertical distance between the load and the floor.
- * The vertical distance to move the load.
- * Frequency.
- * The degree to which the upper body is in a rotated position with respect to the lower body.
- * The contact area of the hands on the load.
- * The weight of the load.
- * The dimensions of the load in relation to its centre of gravity.

Organisational rules for safe manual lifting and moving:

- * Allow employees to determine the pace.
- * Plan short breaks.
- * Distribute lifting activities among several people.

In the last three months of their pregnancy, pregnant women must not lift any loads.



This handle is too small.

The safety measures that an employer must take to reduce the hazards of manual lifting: If the work can not be done safely in any conceivable manner, the employer must arrange that a different working method is used:

- * If the work can not be done safely in any conceivable manner, the employer must arrange that a different working method is used.
- * If the rules for safe lifting and moving do not sufficiently reduce the hazard, the employer must provide suitable aids and arrange for their use.
- * Providing training sessions and instructions for safe lifting and moving.
- * Monitoring health if there is a risk of back injury.

The correct working position/ posture during manual lifting and moving is:

- * Lift with a straight back and bend knees.
- * Keep the load as close as possible to the body.

When lifting, you put pressure on your spine. Possible hazards during manual lifting and moving are:

- * Back injuries due to incorrect lifting.
- * Load falling due to insufficient grip.
- * Getting your fingers trapped when putting the load down.

The safety measures to take by the employee during manual lifting and moving are:

- * The maximum weight to be lifted 25 Kg (advice).
- * Prevent lifting when seated.
- * Do not lift too high.
- * Do not move the load too far.
- * Take care on slippery, uneven floors, gaps and stairs.
- * Determine the pace yourself and take short breaks.
- * Lift calmly.

The safety measures that people who lift loads must take are:

- * Wear safety clothing and PPE.
- * Change position/ posture regularly.

Also seated work can be very stressfull to the body. Guidelines for a good sit during work are:

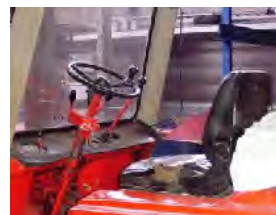
- * Regularly change position: walking, standing, sitting, various sitting positions.
- * Ensure an optimal sitting position.

Good posture is very important to work for a long time. In an optimal seating position during the following points are important:

- * Good support of the upper legs on the seat of the chair.
- * Good position of the back.
- * Relief of the shoulders, for instance arm supports.



Manual lifting of heavy loads.



Concrete slabs can cause a vibratory load on the forklift driver.

Seated work is mostly preferred but there are working conditions in which work when standing is preferred to seated work. Examples:

- * If there is insufficient leg room.
- * If forces exceeding 45 N (4.5 kg) must be used.
- * When frequently reaching down, up or far away from the body.
- * If one has to stand up frequently.
- * If downward force must be exerted.



There is not enough leg room underneath this bench.

Performing standing work is stressful because the legs and feet are heavily stressed. A tool to reduce this stress is a standing support.

- * Standing support: this relieves the legs and feet as the weight of the body is supported.



A standing support.

Examination questions: The ergonomic workplace

The correct answers of these questions can be found in annex 1 of this book.

1 What does not determine conditions of the working environment?

- A The supervisor.
- B The vibration.
- C The lighting

2 A type of physical vibrations is:

- A Vibrations of the hand/ arm: can arise when using mechanical (hand-held) tools.
- B Vibration of the body: can be transferred via a vehicle, large installation or moving floor.
- C Both answers are correct.

3 What is not a health complaint which can be a result of vibrations to the body?

- A Infections to the skin.
- B Muscle ache.
- C Tiredness.

4 Maintaining moisture balance is a control measure:

- A When working with hazardous substances.
- B To ensure there won't be fires.
- C To ensure working comfort.

5 A factor that influences physical stress is:

- A Working position/ posture.
- B Both answers are correct.
- C The movements that must be made.

6 A function of a break during the work is:

- A To have a nice informal talk with your co-workers.
- B To prevent overloading.
- C To have enough time for eating your lunch.

7 Chronic tiredness is a health hazard:

- A Working over time once a month.
- B Of static stress.
- C Of working with hazardous substances.

8 The relationship with your colleagues:

- A Influences the working hours scheme.
- B Should always be positive.
- C Influences mental stress.

9 The correct working position/ posture during manual lifting and moving is:

- A Keep the load as close as possible to the body
- B Lift with a straight back and bend knees
- C Both answers are correct.

10 What is not a working condition in which work when standing is preferred?

- A When frequently reaching down, up or far away from the body.
- B If forces exceeding 30 N (3.0 kg) must be used.
- C If there is insufficient leg room.

11 The optimum sitting position during work is:

- A Good support of the upper legs on the seat of the chair.
- B Relief of the shoulders, for instance arm supports.
- C Both answers are correct.

12 What is a guideline for correct sitting during work?

- A Ensure an optimal sitting position.
- B When you are in the optimal sitting position, keep working this way as long as possible.
- C Both answers are correct.

13 The advised maximum weight to be lifted of 25 Kg is a safety measure for:

- A Lifting when seated.
- B Manual lifting.
- C Lifting with a crane.

14 The correct posture during manual lifting is:

- A Lift with a straight back and bend knees.
- B Keep the load as close as possible to the body.
- C Both answers are correct.

20 Noise at the workplace

Noise induced deafness is a major occupational disease even though the risks from noise are well known. Too much noise does not hurt directly. People may notice the effects of noise exposure many years later.

Walkman's already cause hearing damage to young people. To limit the dangers of noise, you have to do the following according to the occupational hygiene strategy:

1. Source control;
2. General protective measures;
3. Individual protective measures;
4. Personal protective equipment.



Wearing hearing protection is recommended from 80 dB (A) and required from 85 dB (A).

Always use ear protection while working in noise. Even if you have to be exposed for a few minutes!

20.1 Noise and the workplace

The dangers of too much noise in the workplace include:

- * Loss of concentration.
- * Temporary and/or permanent hearing loss.
- * Reduced understanding of speech.

Noise has a great impact on your work. Too much noise can result in:

- * Disruption of the communication required during the activities.
- * Not hearing warnings or cries for help.
- * Permanent hearing damage.
- * Nervousness.
- * Reduced concentration.
- * Tiredness.
- * Headache.
- * Accelerated breathing.
- * Gastrointestinal complaints.
- * Increased blood pressure.

Hearing damage is final and cannot be rectified. You have permanent hearing loss when:

- * Having difficulty hearing high tones or soft sounds.
- * Having difficulty making phone calls.
- * Having difficulty following a conversation in a noisy environment.
- * Hearing whistles, peeps or buzzes not coming from the surroundings.

The total sound (pressure) level of two sound sources can be determined:

- * Two equal sound sources together increase the sound (pressure) level by 3 dB (logarithmic effect).
- * The result of two different sound (pressure) levels together can be determined by using a table.

The degree to which the sound (pressure) level reduces as the distance increases:

- * Doubling the distance gives a reduction of approximately 6 dB(A).



A sound meter.

The unit used to express the sound level is decibel (dB(A)).

Estimating if noise is harmful can be done by a general rule. Noise is harmful when you have to raise your voice when you are on a normal conversation distance.

Measures to take to control excess noise and the priorities placed on them are:

- * 1. Reduce the sound (pressure) level of the sound source.
- * 2. Screen off the sound source.
- * 3. Damp noise vibrations in the vicinity.
- * 4. Use personal hearing protection.



The walls of this studio are covered with insulating material.

There is legislation on noise in companies. On this basis, the employer has to take measures in his company to reduce noise.

Significant noise (pressure) levels to determine these measures are:

- * From 80 dB(A): providing the option to take a hearing test, making hearing protection available.
- * From 85 dB(A): a management plan including the measures to take, wearing hearing protection, employing warning signs.



Hearing protection is mandatory.

Examination questions: Noise at the workplace

The correct answers of these questions can be found in annex 1 of this book.

- 1 What is the effect of excessively loud noises in the workplace?
- C It increases risk in the workplace.
 - C If hearing protection is worn, loud noise has no effect.
 - C It's possible to work harder, since it's impossible to talk to each other.
- 2 Which factors determine, amongst other things, whether noise is harmful?
- A Exposure time and resistance of the ossicle.
 - B Pitch and strength of the noise.
 - C Distance and warmth development within the ear.
- 3 Two equal sound sources together increase the sound (pressure) level by:
- A 3 dB.
 - B 6 dB.
 - C 9 dB.
- 4 Exposure to excess noise causes:
- A White fingers
 - B Permanent brain damage.
 - C Loss of concentration.
- 5 Doubling the distance to noises gives a reduction of the sound (pressure) level of approximately
- A 3 dB(A).
 - B 6 dB(A).
 - C 9 dB(A).
- 6 Having difficulty making phone calls is a symptom of:
- A Temporary hearing loss.
 - B A noisy background.
 - C Permanent hearing loss.
- 7 The measurement unit used to express a sound (pressure) level is:
- A Lumen.
 - B Decibel.
 - C Amplitude.
- 8 The correct order to take control of excess noise is:
- A
 1. Reduce the sound (pressure) level of the sound source
 2. Use personal hearing protection
 3. Damp noise vibrations in the vicinity
 4. Screen off the sound source
 - B
 1. Use personal hearing protection
 2. Screen off the sound source
 3. Damp noise vibrations in the vicinity
 4. Reduce the sound (pressure) level of the sound source
 - C
 1. Reduce the sound (pressure) level of the sound source
 2. Screen off the sound source
 3. Damp noise vibrations in the vicinity
 4. Use personal hearing protection

9 Which of these measures must be taken by the employer with these sound levels?

- ☐ A From 85 dB(A), providing the option to have a hearing examination, making hearing protection available.
- ☐ B From 80 dB(A), providing the option to have a hearing examination, making hearing protection available.
- ☐ C From 75 dB(A), providing the option to have a hearing examination, making hearing protection available.

21 Personal Protective Equipment

Personal Protection Equipment (PPE) is a last resort. If an accident may happen, this will make sure the injury is as small as possible.

PPE must be matched to the risks at work and personal characteristics of the wearer. PPE with a low comfort is insufficiently worn by employees. All PPE must bear a CE-mark. Check before use if the PPE is still suitable for the job.

The employer must provide free PPE for every employee.



Mandatory sign: Wearing a face shield is required when using this sign.

The employer must also provide education and training on the use and maintenance of the PPE. Furthermore the employer must check that employees wear the PPE during work activities.

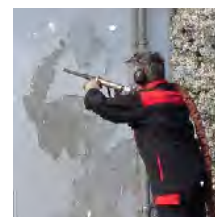
21.1 Application, responsibilities and obligations when using PPE

PPE (Personal Protection Equipment) must be used when:

- * If hazards cannot be removed at the source.
- * If hazards cannot be isolated or screened off.
- * If hazards cannot be removed by collective measures.

Personal Protective Equipment is:

- * Any piece of equipment that is intended to be worn or held by the employee to protect him from one or more hazards that can jeopardise his health or safety at work.
- * Supplementary equipment and accessories that must be used with the PPE.



PPE protects against noise, spatter, dust and water.

PPE must protect the wearer against dangers which can cause damages to one's health and safety.

The manufacturer of PPE must ensure that the PPE meets the following conditions:

- * Actually offer the promised protection.
- * Must be reliable.
- * Must be tested.
- * Must have a CE mark.

The person responsible for making PPE available is the employer or hiring company.

The employee has the following obligations when using PPE:

- * Check regularly.
- * Clean in accordance with the instructions.
- * Use correctly.
- * Store carefully.
- * Manage properly.



The employer determines which PPE is required or recommended. This sign says you need to wear a safety helmet.

21.2 Eye and face protection

Possible hazards for the eyes and face during work are:

- * Flying hard, sharp particles/ components.
- * Flying hot particles when grinding, welding, cutting.
- * Flying dust particles when hacking, drilling.
- * Flying splashes of liquid: corrosive, irritating, harmful.
- * Heat, light, radiation: ultraviolet (UV) and infrared (IR).

Several forms of eye and face protection are:

- * Safety glasses.
- * Safety goggles.
- * Face mask.
- * Welding goggles, welding masks, welding screens, welding helmets.



A face mask.

The most important properties of eye and face protection are:

- * Safety glasses: lenses of safety glass made of plastic, usually with side flaps.
- * Safety goggles: connects to the face, open or fully dust-tight ventilation openings in the upright edge.
- * Face mask: mask over the entire face, made of plastic or metal gauze (no protection against dust, gases, vapours, dust particles that come from below).
- * Welding goggles: clear window from safety glass or plastic and dark window (usually flip up).
- * Welding masks, welding screens: cover the entire face, with a dark window (sometimes in combination with a clear window).



A pair of safety glasses.

For the eye and face protections there are different fields of application:

- * Safety glasses: to protect against flying hard, sharp particles.
- * Safety goggles: in dusty surroundings, when hacking and drilling and to protect against splashes of hazardous liquids.
- * Face mask: protection against flying dust, hazardous liquids and flying hot parts in the event of short circuit, when working with high-pressure cleaners (no protection against substances, gases, vapour, dust particles that come from below).
- * Welding goggles: when welding using gas and against flying particles when chipping off slag and grinding.
- * Welding mask, welding screen or welding helmet: when welding using electricity and to protect against ultraviolet and infrared radiation, heat and flying metal particles and sparks.



These goggles protect against lasers.

21.3 Ear protection

Possible hazards to hearing during work are:

- * Noise-related deafness.
- * Other sounds are masked.
- * Difficulty to understand.

Different forms of ear protection are:

- * Plugs.
- * Foam ear plugs.
- * Plastic ear plugs.
- * Earplugs.
- * Otoplastics.
- * Ear muffs.

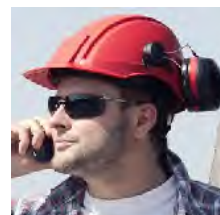
The most important properties of ear protection are:

- * Plugs or foam ear plugs: plasticised and give maximum protection of approximately 10 dB(A).
- * Plastic ear plugs: specially designed plastic rods or deformable foam rods that are worn in the ear, the maximum protection is approximately 10 to 15 dB(A).
- * Universal earplugs: specially formed, fixed to a band that is worn around the neck, dependent on the pitch they give protection between 10 and 15 dB(A).
- * Otoplastics: made using a cast of the ear, contains an exchangeable or adjustable filter that stops noise, maximum protection approximately 25 dB(A).
- * Earmuffs: look like large headphones, they shut off the ears from the surroundings, the protection factor depends on the type of muff and the pitch of the noise, maximum protection approximately 25 dB(A).

Hearing protection must be made available from a noise level of 80 dB(A). Hearing protection must be worn from a noise level of 85 dB(A).



Ear plugs.



A helmet with earmuffs.



This sign shows earprotection is mandatory.

21.4 Respiratory protection

Possible hazards to breathing during work are:

- * Too low oxygen concentration.
- * Too high concentration of vapours, gases or dust particles.
- * Annoyance due to vapour, gas, dust.

Several forms of breathing protection:

- * Filter masks with a particle filter.
- * Filter masks with a gas/vapour filter.
- * Self-contained breathing apparatus with air from cylinders or via an air line.
- * Self-contained breathing apparatus that draws in fresh air.

The most important properties of breathing protectors are:

- * Filter masks clean the air that is breathed in.
- * Self-contained breathing apparatus: the clean air coming from an external source is forced into the mask for instance via cylinders, an air line or an air unit.



The use of respiratory protection during welding.

Some important points of filter masks are:

- * Do not use when oxygen concentrations are too low.
- * Filters can only be used for a limited time dependent on the type of filter, the concentration of products in the air and the amount of air that is breathed in.
- * Particle filters are divided into the following classes: P1 irritating substance, P2 harmful substance, P3 toxic substance.
- * Gas vapour filters are classified according to absorption capacity and to the type of vapour/ gas that is trapped by the filter.
- * Particle filters do not trap gases or vapours.
- * Particle filters get clogged when used for a long time.
- * Gas vapour filters become saturated and allow gas and vapour through.

Self-contained breathing protection must be used when the concentration of the contamination is greater than the threshold limit value (when working or escaping in an emergency).



Always check the expiration date of filters.

Self-contained breathing protection must be used when:

- * If there is less than 19% oxygen in the air.
- * If there are high and unknown concentrations of hazardous substances in the air.

When one is in a confined space, self-contained breathing protection must be used when working with concentrations of substances above the threshold limit value.

The different respiratory protection filters have different fields of application:

- * Nose and mouth cover: only against particles, not for gases or vapours.
- * Semi and full face masks with insert: only against particles, not for gasses or vapours.
- * Semi and full face masks (usually made of rubber) with screw thread: against particles and gases or vapours.
- * Full face masks: cover the entire face.



Respiratory protection is mandatory.

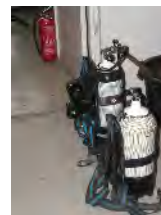
The different methods for application of self-contained breathing apparatus are:

- * Worn over the head with shoulder supports.
- * Air is extracted from the bottom of the cover at the front.
- * Air is supplied into the air hood without obstruction.
- * The air is supplied by a compressor via breathing air lines, via breathing air lines or via air lines connected to a fan.

When using full face masks with air cylinders as independant breathing protection, the air is drawn in via a lung demand valve from the cylinders into the breathing space.

The requirements a user of self-contained breathing protection devices must meet:

- * Instruction and drills for use.
- * Air for fresh air hoods, air lines and air cylinders filtered under pressure and of a good quality.
- * Correctly maintain and clean regularly.
- * Check whether the filter is suitable for the gas/ vapour or particles to be filtered.
- * Check whether the mask fits well (fit test).
- * No facial hair (can cause leaks and prevent correct operation).



Scuba tanks contain compressed air.

21.5 Head protection

Possible hazards for the head during work are:

- * Falling objects.
- * Impact.

A form of head protection is a safety helmet.

A helmet contains of 2 parts: interior and exterior.

- * Exterior: sufficiently robust to stop falling objects/ impacts to reach the head.
- * Interior: absorbs a shock and distributes it over the head.



A safety helmet is mandatory.

Safety helmets protect your head from falling or impact. The helmet must be used in the proper way and be maintained. Guidelines are:

- * Correctly adjust the interior of the helmet.
- * Replace after a certain period.
- * Replace after a fall, an impact or being hit by a falling object.
- * The use of metal helmets is prohibited in industrial situations: they conduct electricity.

21.6 Hand and arm protection

Possible hazards for hands and arms during work are:

- * Sharp objects.
- * Heat.
- * Cold.
- * Radiation.
- * Hazardous substances.

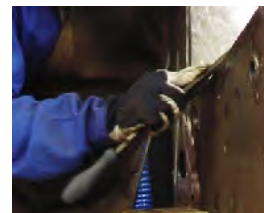


Safety gloves are mandatory.

Depending on the risks and the prescribed protection, hand and arm protection has different forms and is made from several different materials. Besides the normal gloves, there are also gloves with wrist and/or arm protection.

Hand and arm protectors have different fields of application:

- * Cut-proof gloves when cutting.
- * Insulating gloves when exposed to heat or cold.
- * Plastic or rubber gloves when working with hazardous substances.



Working gloves.

Leather or fabric gloves must not be worn when using hazardous substances. These gloves are not water proof.

No gloves may be worn near to moving parts in order to prevent being caught.

21.7 Feet and leg protection

Possible hazards for feet and legs during work are:

- * Falling objects.
- * Step onto sharp objects.
- * Hazardous substances.
- * Slipping.
- * Build-up of static electricity.



Safety boots are mandatory.

To protect feet and legs during riskfully activities safety boots of safety shoes must be worn.

Important parts of feet and leg protectors are:

- * Strengthened / steel nose to protect the toes from heavy falling objects.
- * Strengthened / steel mid sole to protect the feet against sharp objects that are stepped on.
- * Anti-slip sole.

Feet and leg protection is used on many risky workplaces. In the building sector, safety shoes with a strengthened /steel nose and steel mid sole are mandatory. Safety boots are obligatory where water or hazardous substances can occur. Antistatic shoes in areas with an explosion risk.



An employee wears feet protection.

Guidelines for the use of safety shoes are:

- * Regularly maintain and grease.
- * Do not dry wet shoes near a central heating radiator.
- * Replace when worn or after they have been in contact with hazardous substances.

21.8 Body protection

Possible hazards for the body during work are:

- * Contamination.
- * Hazardous substances.
- * Heat.
- * Cold.
- * Rain.
- * Poor visibility.

Several kinds of body protection are:

- * Overall.
- * Protective clothing.
- * Disposable clothing.
- * All-weather clothing.
- * Insulating underwear.
- * Rainproof clothing.
- * Warning clothing.

Body protection is used in many different fields of work. Some examples:

- * Overall: against contamination, when welding and grinding.
- * Protective clothing: when using hazardous substances, heat, cold, radiation.
- * Warning clothing: use to make the wearer visible.
- * All-weather clothing and insulating underwear: at low temperatures.
- * Anti-static clothing: in environments with explosion risk.

Rules of use for body protection are:

- * Do not blow clean using compressed air.
- * Immediately clean or exchange dirty clothing.
- * Wear done up near moving and rotating parts.
- * Have damaged clothing repaired or replaced immediately.
- * Throw away disposable clothing after wearing it once.

21.9 Fall protection

Different forms of fall protection are:

- * Systems that prevent falling: safety harness, a safety line and an anchor point.
- * Systems that arrest the fall: safety harness, fall damper, life line and anchor point.

Fall protection is used for different purposes:

- * Use the positioning system in such a way that falling is impossible as there is always sufficient distance to the edge of the working area (drop).
- * Use the fall arrest system in such a way that the fall height is limited as far as possible and that the shock when falling is damped as much as possible.
- * Personal fall protection devices: use when the other protection devices such as railings, edge protection or safety net cannot be used.



Body protection is mandatory.



A work overall in combination with breathing protection.



A system to arrest the fall.

The rules of use for various forms of fall protection are:

- * Have fall protection devices that have arrested a fall inspected.
- * Fall protection systems and safety harnesses must be inspected at least once a year.
- * Life lines must not be dirty.
- * Shock absorbers and safety harnesses must be stored in a dry and clean location.
- * Safety harnesses must fit well and be adjusted to the wearer.
- * The user must check the harness for among other things wear and frayed ends.

Working with a safety harness is not without risk. A fall can cause unconsciousness because the bloodflow can be restricted. There is a possibility of death within 10-20 minutes. Possible safety measures after a fall are:

- * Always try to keep moving the legs and try to pull yourself up on the life line to reduce the pressure on the legs.
- * Work with at least 2 people to be able to call for help.



A safety harness.

Examination questions: Personal Protective Equipment

The correct answers of these questions can be found in annex 1 of this book.

- 1 Why should people in industry not wear metal safety helmets?
 - A A metal safety helmet can injure the head.
 - B A metal safety helmet cannot sufficiently cushion the impact of a falling object.
 - C A metal safety helmet can conduct electricity.
- 2 For which of the activities below is a dust filter mask a form of breathing protection?
 - A When spraying paint.
 - B When welding.
 - C When breaking and hacking.
- 3 In which of the situations below do safety goggles offer a good protection?
 - A For activities by which acid splashes occur.
 - B With work above the head.
 - C For work activities in confined spaces.
- 4 Which of the statements below is CORRECT? When working with a noise level higher than 85 DB (a)
 - A Hearing protection is recommended.
 - B The employee may choose to use hearing protection.
 - C Hearing protection is compulsory.
- 5 PPE must be:
 - A Available in one size.
 - B On site at every workplace.
 - C Tested.
- 6 The difference between safety glasses and a welding mask are:
 - A Welding masks are used to protect eyes from infrared radiation and safety goggles are used to protect eyes from flying particles.
 - B Safety glasses are used to protect eyes from UV radiation. Welding masks are used to protect the face against flying particles.
 - C Safety glasses cover the entire face and welding masks do not.
- 7 Too low oxygen concentration is a hazard for:
 - A The environment.
 - B Fires.
 - C Breathing.
- 8 Self-contained breathing protection must be used:
 - A If there are high and unknown concentrations of hazardous substances in the air.
 - B If there is less than 19% oxygen in the air.
 - C Both answers are correct.
- 9 What is not a rule for using various forms of fall protection?
 - A Safety harnesses must fit well and be adjusted to the wearer.
 - B Fall protection devices that have arrested a fall can be used again.
 - C Life lines must not be dirty.

10 Warning clothing is used:

- A At low temperatures.
- B To make the wearer visible.
- C Against contamination, when welding and grinding.

11 Slipping is a hazard to:

- A Hands and arms.
- B Feet and legs.
- C Head and face.

12 Various forms of hand and arm protection are:

- A Gloves with anti-slip sole.
- B Gloves with extended wrist and/or arm protection in various materials.
- C Steel nose to protect the fingers from heavy falling objects

13 A main property of a safety helmet is:

- A Interior: absorbs a shock and distributes it over the head.
- B Exterior: absorbs a shock and distributes it over the head.
- C Interior: sufficiently robust to stop falling objects/ impacts to reach the head.

14 Hazards for the head during work are:

- A Impact
- B Falling objects
- C Both answers are correct.

15 A nose and mouth cover is used for:

- A Against particles and gases or vapours.
- B Covering the entire face.
- C Only against particles, not for gases or vapours.

22 Signs and markings

Sometimes, measures can't prevent or limit the risks for health and safety of the employees. A sign which points attention to an object, activity or situation can be necessary.

Dangerous situation can always be present. It is not always possible to reduce the risks, there is always a risk. To get people's attention, there are warning signs.

If the use of Personal Protective Equipment is required, this will be stated on mandatory signs.



A warning sign for lasers.

Hazardous places can (temporarily) be marked with a red and white ribbon.

Signs are designed to show hazards as good as possible. They are also intended to show escape routes or emergency response resources during emergencies. Signs and markings are an important tool for working safely!

In the appendix there are examples of different signs.

22.1 Signs and markings

The use of signs and markings is a part of the H&S legislations. The functions of signs and markings are to indicate for:

- * Possible hazards.
- * Imposed measures.
- * The use of certain mandatory personal protective equipment.
- * The point of first aid and the fire-fighting equipment.
- * Evacuation routes.

The signs are divided in five groups:

- * Prohibitive signs.
- * Mandatory signs.
- * Warning signs.
- * Signs for safety provisions.
- * Signs for fire-fighting equipment.

Requirements for signs and markings:

- * They must be displayed in such a way that they are noticed by the people for whom they are intended,
- * The design must be such that their meaning is clear to everyone involved.
- * They must be unambiguous.
- * Preferably pictograms with as little explanatory text as possible.
- * The size depends on the distance at which they must be seen.

A prohibitive sign is designed in the following way:

- * Circular white sign with a red edge and a red diagonal.
- * The white field contains a black symbol.



Make sure signs are unambiguous.



No smoking.

Examples of prohibitive signs:

- * No smoking.
- * No transport vehicles.
- * No unauthorised access.
- * Do not drink the water.
- * Do not stop the fire with water.
- * No fire, smoking or open fire.
- * No pedestrians.

A Mandatory sign is a round blue sign with a white symbol.

The meaning of mandatory signs are: obligatory use of: eye protection, safety helmet, hearing protection, respiratory protection/breathing apparatus, safety shoes, safety gloves, body protection, face protection, individual safety harness.

- * The meaning of mandatory signs are: obligatory use of: eye protection, safety helmet, hearing protection, respiratory protection/breathing apparatus, safety shoes, safety gloves, body protection, face protection, individual safety harness

Facial protection is mandatory

A safety suit is mandatory

Ear protection is mandatory

Safety helmet is mandatory

Respiratory protection is mandatory

Fall protection is mandatory

Some important signs are: Eyeprotection is mandatory



Oogbescherming verplicht.

Warning signs are triangular shaped and yellow with a black border. In the yellow field there is a black symbol.



Beware! Corrosive substances.

The general meaning of the warning signs are:

- * Danger.
- * Warning.
- * Caution.

Examples of warning signs:

- * Toxic hazard.
- * Fire hazard.
- * General warning.
- * Radioactive material.
- * High temperatures.
- * Strong magnetic field.
- * Irritating hazard.
- * Low temperatures.
- * Corrosive hazard.
- * Biological hazard.
- * Transport vehicle hazard.
- * Hazard of falling goods.
- * Explosion hazard.
- * Laserbeams.
- * Falling hazard.
- * Electricity.
- * Danger for the environment.
- * Stumble hazard.
- * Ionising hazard.
- * Oxidizing hazard.

The meaning of the following warning signs are: toxic substances, suspended loads, low temperatures, laser beam, non-ionising radiation.

The meaning of the following warning signs are: flammable materials or high temperature, oxidising substances, radioactive substances, harmful or irritating substances, stumbling, transport vehicles, falling due to a difference in height.

The signs for safety provisions make clear where the safety provision can be found in an emergency situation. These signs (for example a sign for a First Aid station or an evacuation route) are square or rectangular and green. The symbol is white.

Safety provision signs show:

- * Emergency exits or escape routes.
- * The route to first aid and evacuation means.

Examples of signs for safety provision:

- * First aid telephone.
- * Eye flush.
- * Emergency shower.
- * Emergency exit.
- * First aid.
- * Escape route.
- * Emergency exit or escape route.

Signs for fire-fighting equipment are square or rectangular and red with a white symbol.



A warning sign for combustibles.



A sign for emergency exits or escape routes.



First aid telephone.

They indicate:

- * The way to fire-fighting equipment.
- * The location of the fire-fighting equipment.



A sign for small fire extinguishers.

Some examples of signs for fire-fighting equipment:

- * Location of the fire ladder.
- * Route to fire-fighting equipment.
- * The location of the fire extinguisher.
- * The location of the fire hose.
- * Location of the telephone for fire alarm.

Besides signs a workplace can also have markings which indicate a possible hazard. These could be for example tape or markings on the floor.



A red and white marking.

Tapes are mostly red and white or black and yellow. Tape does not form a barrier. Tape warns you for a hazard or risk.

Yellow-and-black markings are used for example to:

- * Point out narrow or low passages
- * To point out objects that can cause an impact or to which the person can bump into, for instance a crane block
- * Signs indicating areas containing hazardous substances



Bump hazard against the crane blocks.

The objective of marking (striped) of a different colour (usually yellow or white) near stairs are:

- * Marking (stripe) of a different colour (usually yellow or white) to indicate the top step of the stairs.
- * Marking (stripe) of a different colour (usually yellow or white) to indicate the bottom step of the stairs.
- * Mark locations where there is a danger of tripping.
- * Indicate if there is little distinction between the floor and the steps.

The objectives and format of markings of passages and stacking areas are:

- * White or yellow stripes.
- * Keep passages free; do not store or stack.
- * Store or stack within the marked area.

Examination questions: Signs and markings

The correct answers of these questions can be found in annex 1 of this book.

- ☐ 1 Which characteristics apply to a mandatory sign?

☐ A It is round, blue, and it has a white symbol.
☐ B It is triangular and yellow with a black symbol.
☐ C It is rectangular and green with a white symbol.
- ☐ 2 Which requirements apply to warning signs at the workplace?

☐ A The design must be such that the meaning is clear to all.
☐ B They must be attached at a safe height.
☐ C They preferably wear a text in the Dutch language.
- ☐ 3 A group of signs is:

☐ A Mandatory signs.
☐ B First Aid signs
☐ C Exit signs.
- ☐ 4 A sign that prohibits something is:

☐ A Circular red sign with a white edge and a red diagonal.
☐ B Triangular yellow sign with a black edge.
☐ C Circular white sign with a red edge and a red diagonal.
- ☐ 5 A triangular yellow sign with a black edge is the design of:

☐ A A provision sign.
☐ B A warning sign.
☐ C A mandatory sign.
- ☐ 6 The general meaning of warning signs is:

☐ A Prohibiting something.
☐ B The route to provisions.
☐ C Caution, danger.
- ☐ 7 A square or rectangular green sign with a white symbol is the design for:

☐ A Mandatory signs.
☐ B Warning signs.
☐ C Safety provision signs.
- ☐ 8 Fire-fighting equipment is located at signs which are:

☐ A Square or rectangular and green with a white sign.
☐ B Square or rectangular and red with a white sign.
☐ C Square or rectangular and black with a white sign.
- ☐ 9 The general meaning for fire-fighting equipment:

☐ A Indicates where the fire-fighting equipment can be found: the route to it.
☐ B Indicates where the fire-fighting equipment can be found: the place itself.
☐ C Both answers are correct.
- ☐ 10 Yellow or white stripes are used for:

☐ A Marking the top and bottom step of the stairs.
☐ B Markings of passages and stacking areas.
☐ C Mandatory signs.

Annex 1 Answers

This explanation of the practice questions VILVCU is an attachment to the textbook VILVCU. This book contains the correct answers to the test questions (they are checked) together with an explanation why the other answers are wrong.

- ☐ This explanation has the same chapter structure as the textbook.
- ☐ This explanation of the practice questions is intended as an annex to the textbook for classroom training and is somposed with the utmost care.

We hope you enjoy learning and have a succesfull exam.

Tips for the exam and practice questions:

- Read each question properly and calmly. Do you not know the answer immediately, skip the question and finish it at the end.
- You will have a pencil with eraser at the exam. You can correct yourself but do not rush this. Usualy the first answer is correct.
- Just make 1 box with each question black with your pencil. If you tick none or multiple boxes, the answer is always wrong.
- Are you really uncertain about an answer, check which answers are clearly wrong and then make a selection of the remaining answers. Always fill in an answer!
- Make sure you are on time for the exam. If you are late, you will not be atmitted to the exam.
- In cause of fraude you will be removed from the exam. Follow the examination regulation exactly!

The exam consists of 70 multiple choice questions. The student must choose from 3 possible answers with each question. The student must choose the best answer. At least 49 questions have to be answered correctly to pass the exam.

The exam lasts 105 minutes. Practice shows this is more then enough time.

1 Legislation

1 What measure can the Inspection Service take?

- ☐ C ☐ The Labour Inspectorate can do research at companies about Health and Safety.
- ☐ C ☒ This is not the job of the Labour Inspectorate.

2 What is the objective of the Health and Safety legislation?

- ☒ A ☐ This is what will be achieved by providing the best possible working conditions.
- ☐ C ☐ The atmosphere is not an objective from the Health and Safety legislation.

3 Which of the measures below can the Inspection Service take?

- ☐ A ☐ This is not the task of the Inspection Service. Only employers can fire employees.
- ☒ B ☐ This is not the task of the Inspection Service. Only employers can fire employees.

4 What does the CE mark on an article indicate?

- ☒ A ☐ It meets the minimum European requirements.
- ☐ C ☐ It meets the minimum European requirements.

5 What is one of the objectives of the environmental management act?

- ☐ A ☐ This is only one way of protecting the environment.
- ☒ B ☐ The environmental management act has measures towards the environment and humans.

6 What does 'ARBO' describe?

- ☒ A ☐ ARBO describes the working conditions (occupational health and safety).
- ☐ C ☐ ARBO describes the working conditions (occupational health and safety).

7 What is the applicability of the Health and Safety legislation?

- ☐ A ☐ The H&S legislation is restricted to the workplace.
- ☒ B ☐ It's also about the employer, not only about the employee.

8 At which locations does the H&S legislation apply?

- ☒ A ☐ All locations where work is carried out.
- ☐ C ☐ All locations where work is carried out.

9 The H&S legislation applies to:

- ☐ A ☐ Both answers are correct.
- ☐ B ☐ Both answers are correct.
- ☒ C ☐

2 Legislation

2

10 The H&S legislation does not relate to:

- A ☐ Health is a part of the H&S legislation
- B ☐ Welfare is a part of the H&S legislation.
- C ☒

11 The H&S legislation applies to:

- A ☐ The legislation applies to both.
- B ☐ The legislation applies to both.
- C ☒

12 What is not an assumption behind the H&S legislation?

- A ☐ This is an assumption behind the H&S legislation.
- B ☒
- C ☐ This is an assumption behind the H&S legislation.

13 What is an important task of the experts or services in the area of prevention?

- A ☒
- B ☐ They supervise employees who are ill.
- C ☐ They perform occupational health examinations,

14 Important objectives of the working hours legislation are:

- A ☐ Both answers are right.
- B ☐ Both answers are right.
- C ☒

15 Principles of the CE identification symbols are:

- A ☐ Only one can do this.
- B ☐ Only CE marked equipment can be marketed and used.
- C ☒

2 Dangers, risks and prevention

2

1 How is the concept of risk described?

- A ☒ Risk = probability X effect.
- B ☐ Risk = probability X effect.
- C ☐ Risk = probability X effect.

2 Hazards at the workplace are:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

3 What is not a source of dangers and risks at the workplace?

- A ☐ This is a source
- B ☐ This is a source
- C ☒ This is a source

4 In which way can prevention be tackled practically?

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

5 What is an unsafe situation?

- A ☐ An unsafe situation is a situation in which work is carried out without complying with the conditions for safe working and that can lead to an accident.
- B ☐ This is an unsafe situation, but not a the general meaning of 'unsafe situation'.
- C ☒ This is an unsafe situation, but not a the general meaning of 'unsafe situation'.

6 Which of the actions stated below is an unsafe action?

- A ☐ This considered a safe action.
- B ☒ This considered a safe action.
- C ☐ This considered a safe action.

7 What is the best way you can deal with unsafe situations?

- A ☒ This is not the best way to do.
- B ☐ This is not the best way to do.
- C ☐ This is not the best way to do.

8 The various steps in an RI&E are:

- A ☐ This is the wrong order.
- B ☒ This is the wrong order.
- C ☐ This is the wrong order.

9 What does LMRA stand for?

- A ☐ Last Minute Risk Analysis.
- B ☒ Last Minute Risk Analysis.
- C ☐ Last Minute Risk Analysis.

10 What is the objective of the work planning?

- A ☐ To determine a working method that is safe and does not cause any health damage.
- B ☒ To determine a working method that is safe and does not cause any health damage.
- C ☐ To determine a working method that is safe and does not cause any health damage.

3 Dangers, risks and prevention

3

11 What is an approach of the intermediary in the event of posting employees for high-risk tasks?

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

3 Accidents: causes and prevention

1 What is the cause of most accidents?

- A ☒ This is mostly an effect of human error.
- B ☐ Tool maintenance should be done by humans. Poor maintenance is an effect of human error.
- C ☐

2 According to Lateiner, which factors play a role in the occurrence of an accident?

- A ☒
- B ☐ Background instead of surroundings
- C ☐ Background instead of environment

3 Drawing up procedures for taking action when accidents occur, is a measure towards:

- A ☐ This is a measure towards organization
- B ☒
- C ☐ This is a measure towards organization

4 Which parts, at least, have to be included in the final report of an accident investigation

- A ☐ These things are handled in the investigation itself. The final report describes the facts and details.
- B ☒
- C ☐ The accident investigation itself contains these things.

5 The two direct causes of an accident or incident are:

- A ☐ Unsafe actions or unsafe situations.
- B ☒
- C ☐ Unsafe actions or unsafe situations.

6 What is not an examples of person-related factors that can lead to an incident?

- A ☒
- B ☐ This is an example of person-related factors.
- C ☐ This is an example of person-related factors.

7 To which person must an accident be reported to ?

- A ☐ The immediate supervisor.
- B ☐ The immediate supervisor.
- C ☒

8 What must one do with all accidents or near accidents?

- A ☐ All accidents and near accidents must be recorded
- B ☒
- C ☐ All accidents and near accidents must be recorded

9 Who is responsible for the accident investigations?

- A ☒
- B ☐ They are not responsible for the investigation.
- C ☐ They should do the investigation if it is an severe incident.

10 Who is responsible for the absence due to illness?

- A ☐ This is incorrect.
- B ☒
- C ☐ This is incorrect.

4 Accidents: causes and prevention

4

11 What is the objective of the Wet Verbetering Poortwachter (Eligibility for Permanent Invalidity Benefit (Restrictions) Act)?

- A ☐ This act is for sick employees.
- B ☒
- C ☐ This answer is incorrect.

12 What is an obligations that the temporary employment agency has within the framework of the Eligibility for Permanent Invalidity Benefit (Restrictions) Act?

- A ☒
- B ☐ Both answers are correct.
- C ☐ Both answers are correct.

4 Safety conduct

1 Which of the examples below is a cause of an accident by falling and stumbling?

- A ☐ An anti-slip floor must reduce accidents.
- B ☒
- C ☐ This will reduce accidents.

2 What is not a cause of unsafe behaviour?

- A ☐ This is a cause of unsafe behaviour.
- B ☐ This is a cause of unsafe behaviour.
- C ☒

3 Which policy element influences human behaviour so that work can be done safely?

- A ☐ There also has to be a reward side. A sanction and reward policy.
- B ☒
- C ☐ The policy has to be focused on better safety performance.

4 Why is the role of the supervisor important for human behaviour?

- A ☐ This is not the correct answer.
- B ☐ The supervisor determines the working method. Not the employees.
- C ☒

5 A general rule of behaviour for safe working is?

- A ☒
- B ☐ Employees should intervene immediately .
- C ☐ One has to take these opinions into account. Storing them does not help with safe working.

6 Which of the statements below is not behaviour that promotes safe working?

- A ☐ This is behaviour that promotes safe working.
- B ☒
- C ☐ This is behaviour that promotes safe working.

7 The influence of alcohol and drugs on safe working causes:

- A ☒
- B ☐ There will be behaviour changes when using alcohol or drugs.
- C ☐ Drugs or alcohol will lead to less effective assessments of situations.

8 What is expected with respect to alcohol and drug use in regards to working safely?

- A ☐ Even if the supervisor would know, the use of alcohol or drugs is not allowed.
- B ☒
- C ☐ The supervisor must be informed about drug or alcohol use.

9 A task of the intermediary related to the promotion of safe working by the temporary agency workers is?

- A ☒
- B ☐ Both answers are correct.
- C ☐ Both answers are correct.

5 Tasks, rights, duties and discussions**5**

1 What are the obligations of employers according to the Health and Safety legislation?

- A ☒
 B ☐ This is a task of the employees.
 C ☐ The Labour Inspectorate has to be informed about serious work accidents.

2 What are the obligations of employees according to the working conditions act? They are obliged to

- A ☐ The Labour Inspectorate has to be informed about serious work accidents.
 B ☒
 C ☐ This is a task of the employer.

3 When do employees have the right to interrupt their work?

- A ☐ If PPE is not available, employees should not start their work. This has to be reported to their direct supervisor immediately.
 B ☐ The Labour Inspectorate wants to see how people work. Work activities should not be interrupted unless the Inspectorate says differently.
 C ☒

4 Who participates in the works council consultation?

- A ☐ Only the employer and representatives of the employees.
 B ☐ Only the employer and representatives of the employees.
 C ☒

5 What is the difference between SCC* and SCC** certification?

- A ☐ SCC* focuses on the direct safety control, SCC** also includes the safety structures.
 B ☒
 C ☐ SCC* focuses on the direct safety control, SCC** also includes the safety structures.

6 What must one do in the event of immediate danger?

- A ☐ The dangers have to be reported to the supervisor. He will take measures.
 B ☐ One has to stop working immediately to ensure his own safety.
 C ☒

7 What is a benefit of work place inspections and observation rounds?

- A ☐ Both answers are correct.
 B ☐ Both answers are correct.
 C ☒

8 A statutory liability with respect to health and safety is:

- A ☐
 B ☐ This are tasks of the temporary employment agency.
 C ☐ This is not the right anser.

9 The difference between the formal and the actual employer is:

- A ☐ Both answers are correct.
 B ☒
 B ☐ Both answers are correct.

5 Tasks, rights, duties and discussions

5

10 Who is responsible for the working conditions in the event of temporary work?

- A ☐ This answer is incorrect.
- B ☐ This answer is incorrect.
- C ☒

11 A statutory liability with respect to health and safety is arranged is:

- A ☐ The Labour Inspectorate is not a part of this relationship.
- B ☒
- C ☐ The temporary employment agency is liable with respect to the duty to pass on information.

12 A Health and Safety aspect that is important when assigning a temporary worker is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

13 What must be discussed with the user enterprise regarding medical examination?

- A ☒
- B ☐ Medical examinations can only be done by authorised people.
- C ☐ This information has to be given to the temporary employee. Not to the agencies.

14 Elements that the temporary worker's dossier must contain are:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

15 What is not a necessary point of focus to be recorded when the user enterprise submits an application?

- A ☐ This is a necessary point of focus.
- B ☐ This is a necessary point of focus.
- C ☒ It is not about VOL VCA but about Basic Safety.

16 What must be done in relation to verifying the agreements made with the user enterprise?

- A ☐ This is not the correct answer.
- B ☒
- C ☐ This is not the correct answer.

17 An aspect that must be focused upon when evaluating the temporary work with the temporary worker is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☐

18 What is important in the meeting with respect to health and safety within the temporary employment agency?

- A ☐ This is not the correct answer.
- B ☐ This is not the correct answer.
- C ☒

6 Tasks, rights, duties and discussions

6

19 What is the involvement of the temporary agency worker in the HSE meetings (toolbox meetings)?

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

6 Procedures, instructions and signs

6

1 The procedure for reporting a fire falls under the.

- A ☐ This is stated in the General Safety Rules
- B ☒ This is stated in the General Safety Rules
- C ☐ This is stated in the General Safety Rules

2 What must be described AMONGST OTHER THINGS in a work permit?

- A ☐ This is not required for a work permit
- B ☐ This is not required for a work permit
- C ☒ This is not required for a work permit

3 General safety rules are intended for:

- A ☐ General safety rules are for the entire company.
- B ☒ General safety rules are for the entire company.
- C ☐ In particular, the personnel of these two.

4 General safety rules:

- A ☒ General safety rules are for the entire company.
- B ☐ General rules are not specific rules.
- C ☐ Foreign people must also understand the general safety rules.

5 Specific safety rules do not relate to:

- A ☒ Specific safety rules do relate to this.
- B ☐ Specific safety rules do relate to this.
- C ☐ Specific safety rules do relate to this.

6 A guideline concerning the use of specific safety rules is:

- A ☒ They should be available for the employees involved in the matter.
- B ☐ They should be available for the employees involved in the matter.
- C ☐ A check must establish whether this is true.

7 A function a connection flange is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

8 What is not a section of the work permit concerning the requested activities?

- A ☐ This is a section of the work permit concerning the requested activities.
- B ☐ This is a section of the work permit concerning the requested activities.
- C ☒ This is a section of the work permit concerning the requested activities.

9 What is a section in the work permit concerning the measures to be taken by the operational employee.

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

10 What is the duty to pass on information entails?

- A ☒ Both answer are correct.
- B ☐ Both answer are correct.
- C ☐ Both answer are correct.

7 Preparations for emergencies

7

1 Which of the tasks below belongs to the role of the emergency response officers?

- A ☒
- B ☐ The emergency response team only takes control of starting accidents. The investigation is done by a different team.
- C ☐ This is not a task of the emergency response team.

2 What is a concept of an emergency situation?

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

3 A company emergency plan is:

- A ☒
- B ☐ This is not what an emergency plan is for.
- C ☐ A description of the measures and provisions that a company has prepared in case emergency situations arise.

4 Limit effects of possible emergency situations is the purpose of:

- A ☐ An (internal) company emergency plan.
- B ☒
- C ☐ The emergency response team .

5 What is not a phase in controlling emergency situations?

- A ☐ This is a phase in controlling emergency situations.
- B ☒
- C ☐ This is a phase in controlling emergency situations.

6 The first alarm in emergency situations is:

- A ☐ This is not the correct answer.
- B ☒
- C ☐ This is not the correct answer.

7 A component of the termination of the emergency situation is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

8 One of the minimum requirements with respect to first aid is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

9 Which sequence is correct in the event of an evacuation signal?

- A ☐ This is not the right sequence.
- B ☒
- C ☐ This is not the right sequence.

8 Hazardous substances

1 You are reading the label of a hazardous product. How can you tell which risks the product presents?

- ☐ C ☐ These are Safety phrases.
- ☐ C ☐ R phrases describe the risks of a substance.
- ☐ C ☒

2 What does the yellow surface of the diamond hazard warning sign indicate?

- ☒ C ☐
- ☐ C ☐ Blue section (left): Health
- ☐ C ☐ Red section (top): Flammability

3 Which poisonous effect does carbon monoxide have?

- ☐ A ☐ Carbon monoxide prevents blood to absorb oxygen.
- ☒ B ☐
- ☐ C ☐ Carbon monoxide prevents blood to absorb oxygen.

4 What information is contained in a material safety data sheet?

- ☐ A ☐ The purpose of the material safety data sheet is to provide information about the hazards of the substances and information regarding the safety measures to take.
- ☐ B ☐ The purpose of the material safety data sheet is to provide information about the hazards of the substances and information regarding the safety measures to take.
- ☒ C ☐

5 Under normal conditions, the oxygen in the air is:

- ☒ A ☐
- ☐ B ☐ 21%
- ☐ C ☐ 21%

6 Lack of ventilation:

- ☐ A ☐ You have to ventilate to achieve this.
- ☒ B ☐
- ☐ C ☐ This is not always true.

7 A white shoulder on a cylinder means:

- ☐ A ☐ Grey shoulder.
- ☐ B ☐ Light green shoulder.
- ☒ C ☐

8 When storing gas cylinders:

- ☒ A ☐
- ☐ B ☐ Gas cylinders should not be stored in cellars.
- ☐ C ☐ Cylinders should not be stored at the workplace.

9 What is the meaning of 'hazardous substances'?

- ☐ A ☐ Materials that pose a danger to health and/or to the environment.
- ☒ B ☐
- ☐ C ☐ Materials that pose a danger to health and/or to the environment.

10 Hazardous substances at the workplace are indicated with:

- ☐ A ☐ The background is orange.
- ☒ B ☐
- ☐ C ☐ The background is orange.

8 Hazardous substances

8

11 Vaccinations is a precautionary measure for working with:

- A ☒
 B ☐ Biological substances.
 C ☐ Biological substances.

12 Biological substances can be present in:

- A ☐ Both answers are correct.
 B ☐ Both answers are correct.
 C ☒

13 Professional removal of spilt product is a preventive measure to take against:

- A ☐ Leaks.
 B ☐ Leaks.
 C ☒

14 What is not a cause of leaks?

- A ☐ This is a cause of leaks.
 B ☐ This is a cause of leaks.
 C ☒

15 Slipping is a consequence of:

- A ☐ Leaks.
 B ☒
 C ☐ Leaks.

16 The hazard of organic solvents is:

- A ☒
 B ☐ Flammable or highly flammable.
 C ☐ Flammable or highly flammable.

17 An example of cyclic compounds is:

- A ☐ This is a heavy metal.
 B ☒
 C ☐ This is an organic solvent.

18 The chance of mesothelioma is a health effect related to:

- A ☐ Asbestos.
 B ☐ Asbestos.
 C ☒

19 What is not a possible location of asbestos?

- A ☒
 B ☐ This can be a location of asbestos.
 C ☐ This can be a location of asbestos.

20 What is not a legal stipulation with respect to asbestos?

- A ☒
 B ☐ This is a legal stipulation with respect to asbestos.
 C ☐ This is a legal stipulation with respect to asbestos.

21 A concept and objective of the periodic medical examination is:

- A ☒ it should be hazardous substances.
 B ☒
 C ☐ Periodicity depends on the product and the exposure, for most materials it is once a year.

9 Hazardous substances

9

22 Safety recommendations on the new labels describe:

- A ☐ This is not the correct answer.
- B ☒
- C ☐ This is not the correct answer.

23 The numbers on a hazard diamond go from:

- A ☐ 0 to 4.
- B ☒
- C ☐ 0 to 4.

24 The measures to take to avoid risks when working with the product are described in:

- A ☐ The S phrases.
- B ☐ The S phrases.
- C ☒

9 Fire and explosions

9

1 What is a disadvantage of extinguishing a fire with water?

☒ C

☐ B The water may react fiercely with different substances.

☐ C The water may react fiercely with different substances.

2 What must you know before you start extinguishing a fire?

☐ A Depending on what substance is burning, you know what kind of extinguishing agent you can use.

☐ B It is more important to know what kind of substance is burning.

☒ C

3 What must be done first when a fire is detected?

☐ A This is the last thing you have to do.

☐ B This action has to be taken after you have reported the fire.

☒ C

4 What does an explosion limit signify?

☐ A Explosion limits are about a concentration of gas.

☒ B

☐ C Explosion limits are about a concentration of gas.

5 What is meant by the flash point?

☐ A It is about the lowest temperature of a liquid

☐ B It is about the lowest temperature of a liquid.

☒ C

6 What result can an excess of oxygen have during a fire?

☒ A

☐ B A high dose of oxygen is a positive catalyst. This will cause flammable substances to promote the reaction speed of a fire.

☐ C Fire needs oxygen to burn.

7 A flammable substance:

☒ A

☐ B This is not true.

☐ C This is not true.

8 What is an explosion?

☐ A This is not an explosion.

☒ B

☐ C A fire is not an explosion.

9 What does a class A fire involve?

☐ A Solid metals.

☐ B Solid metals.

☒ C

10 The sign that indicates an explosion risk zone:

☒ A

☐ B This is not correct.

☐ C This is not correct.

10 Fire and explosions

10

11 An explosion risk environment is:

- A ☒
- B ☐ An environment where special measures must be taken to prevent the ignition of explosive mixtures that may be present.
- C ☐ An environment where special measures must be taken to prevent the ignition of explosive mixtures that may be present.

12 A requirement for the hot work watch is:

- A ☐ A specific training.
- B ☒
- C ☐ A specific training.

13 When treating burns:

- A ☐ Flush with water for at least 15 minutes.
- B ☒
- C ☐ Flush with water for at least 15 minutes.

14 When extinguishing a fire one must not:

- A ☐ This is what one must do.
- B ☐ This is what one must do.
- C ☒

15 Fire blankets:

- A ☐ Exclude oxygen.
- B ☐ Exclude oxygen.
- C ☒

16 A disadvantage of extinguishing with a fire blanket is:

- A ☐ This is a disadvantage of carbon dioxide.
- B ☒
- C ☐ This is a disadvantage of extinguishing powder.

17 A disadvantage of extinguishing with carbon dioxide is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

18 An extinguishing characteristic of sand is:

- B ☐ This is a characteristic of carbon dioxide.
- B ☒
- C ☐ This is a characteristic of water.

10 Equipment

1 What must you look out for when using chisels?

- A ☒
- B ☐ Only use chisels that have no burrs on the cutting edge.
- C ☐ Only use chisels that have no burrs on the cutting edge.

2 Why may you never wear gloves when working with machines that have rotating parts?

- A ☐ The danger is getting caught in the machine.
- B ☐ A damaged machine is less of a problem then when a hand is dragged into the machine.
- C ☒

3 What does working safely with pillar drills mean?

- A ☐ There is no riving knife on a pillar drill.
- B ☒
- C ☐ A pilar drill does not work with counter pressure.

4 What must hammers be provided with to prevent accidents? They must be provided with

- A ☐ The head must be firmly fixed on the handle.
- B ☐ This will cause too much vibrations when using the hammer.
- C ☒

5 What is the purpose of a dead man's switch on a pneumatic tool?

- A ☐ A dead man's switch will stop the tool immediately on release.
- B ☐ A dead man's switch will stop the tool immediately on release.
- C ☒

6 What must be present on a table circular saw?

- A ☐ A glass shield is not required on a table circular saw.
- B ☒
- C ☐ A blade guard must be present.

7 Electrical hand tools for a voltage of 230 volts

- A ☒
- B ☐ Earthing is not required.
- C ☐ There is no special training required before working with powered manual tools.

8 A periodic inspection has to be done:

- A ☒
- B ☐ The periodic inspection will be set by the company. It reaches over a longer period of time then one week.
- C ☐ This is not an inspection.

9 A machine will stop as quickly as possible when:

- A ☒
- B ☐ The machine will stop, but not immediately.
- C ☐ This will block the machine from running. It does not stop the machine.

10 What is a pallet trolley?

- A ☐ A pallet trolley transports pallets.
- B ☐ This is a forklift truck
- C ☒

10 Equipment

11 A requirement that is placed on a forklift truck driver is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

12 What is not a safety measure to take when working with forklift trucks?

- A ☐ This is a safety measure.
- B ☒
- C ☐ One is a safety measure.

13 When working with saws:

- A ☒
- B ☐ The blade must be sharp otherwise one cannot use the saw properly.
- C ☐ One does not need a special training for using saws.

14 When working with knives:

- A ☐ Saws have teeth.
- B ☒
- C ☐ This is a requirement of a saw.

15 When working with small work pieces:

- A ☐ One must clamp the workpiece.
- B ☒
- C ☐ This is not correct.

16 Trousers and gloves made of special fibre are used when:

- A ☒
- B ☐ Working with a chain saw.
- C ☐ Working with a chain saw.

17 A nailer/stapler must have:

- A ☐ This is not a requirement of a nailer.
- B ☒
- C ☐ This is not a requirement of a nailer.

18 A riving knife is used with:

- C ☐
- A ☒
- B ☐ A hand-held circular saw.
- C ☐ A hand-held circular saw.

19 Which information has to be present on a grinding disc?

- A ☐ The maximum rotational speed must be present.
- B ☒
- C ☐ This is not information which needs to be present.

20 When working with pneumatically powered hand tools one must:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

11 Equipment

11

21 When working with bench grinders, what is not a possible hazard?

- A ☐ This is a possible hazard.
- B ☒ This is not a possible hazard.
- C ☐ This is a possible hazard.

11 Demolish activities

11

1 A hazard during demolition is?

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

2 During demolition work one must have:

- A ☒
- B ☐ This is not a safety measure.
- C ☐ Demolition work is based on destruction. One has to break enough to keep the remaining structure strong.

3 What must one do when removing ceramic fibres?

- A ☐ Asbestos should not be broken.
- B ☒
- C ☐ A disposable overall has to be worn.

12 Welding, cutting and burning

12

1 What must one do when gas welding or using a cutting torch with oxygen and acetylene?

- A ☐ The angle must at least be 30°.
B ☒
C ☐ The angle must at least be 30°.

2 What is a risk of gas welding?

- A ☐ Propane is heavier than air.
B ☒
C ☐ The cylinders are pressurised.

3 What is a safety measure which has to be taken when electric welding is done?

- A ☐ Both answers are correct.
B ☐ Both answers are correct.
C ☒

4 What is not a risk associated with electric welding?

- A ☐ This is a risk. A burnt cornea is caused by UV radiation.
B ☒
C ☐ This is a risk. A burnt cornea is caused by UV radiation.

13 Digging and working at or in excavations

13

1 Why is it important to excavate carefully?

- A ☐ It is important to excavate carefully to not have any economic loss.
- B ☒
- C ☐ This is a subject within the safety risks.

2 What is not a risk associated with excavating?

- A ☐ This is a risk associated with excavating.
- B ☐ This is a risk associated with excavating.
- C ☒

3 What is a rule for carefully excavating?

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

4 When working in or near an excavation you must:

- A ☐ You must install two ladders.
- B ☐ The fences must be around the excavation.
- C ☒

14 Working at height

1 From which height must the employer take legally required measures to prevent risks of falling?

- ☐ C ☐ From 2.5 meters
☐ C ☐ From 2.5 meters
☐ C ☒

2 What safety measure can reduce the danger of falling for everyone working on a flat roof?

- ☐ C ☒
☐ C ☐ A roof edge safety system is the best solution.
☐ C ☐ A roof edge safety system is the best solution.

3 What must be checked before a ladder may be used?

- ☐ A ☐ The length of the ladder does not matter as long as it's stable and meets all other requirements.
☐ B ☒
☐ C ☐ This is not a requirement on ladders.

4 What must persons in a suspended scaffold make use of?

- ☐ A ☐ A safety harness is mandatory when working on a suspended scaffolding.
☐ B ☐ A safety harness is mandatory when working on a suspended scaffolding.
☐ C ☒

5 On which side must a rolling scaffold be climbed?

- ☐ A ☐ The weight distribution is better when climbing from the inside. Also you will not bump your head on environmental items outside of the scaffold.
☐ B ☒
☐ C ☐ The weight distribution is better when climbing from the inside. Also you will not bump your head on environmental items outside of the scaffold.

6 What should you preferably do with a floor opening?

- ☐ A ☐ This is not the best thing to do when there are openings in floors.
☐ B ☒
☐ C ☐ It is best to cover the openings with a solid material.

7 What is not a hazard of working near wall and floor openings?

- ☐ A ☐ This is a hazard when working near wall and floor openings.
☐ B ☐ This is a hazard when working near wall and floor openings.
☐ C ☒

8 A ladder is used for:

- ☐ A ☐ Both answers are correct.
☐ B ☐ Both answers are correct.
☐ C ☒

9 The reach at maximum one arm length is a rule for working with:

- ☐ A ☐ Ladders.
☐ B ☒
☐ C ☐ Ladders.

15 Electricity

1 Is a layman allowed, with the permission of an expert, to work on live electrical installations?

- ☐ C ☐ No, laymen are never allowed to work with electrical installations.
☐ C ☐ No, laymen are never allowed to work with electrical installations.
☐ C ☒

2 What does an earth leakage circuit breaker do?

- ☐ C ☐ It makes sure power is turned off when a certain leakage value is reached.
☐ C ☒
☐ C ☐ It does not work as an artificial earthing method.

3 Which safe voltage must you use in a confined space to work safely with electricity?

- ☐ A ☐ Direct current (maximum 120 volts or alternating current (maximum 50 volts).
☐ B ☐ Direct current (maximum 120 volts or alternating current (maximum 50 volts).
☐ C ☒

4 What can cause accidents through electricity?

- ☐ A ☒
☐ B ☐ Pressing the emergency stop must prevent accidents from happening.
☐ C ☐ A blown fuse is a prevention of electrical accidents.

5 What is the greatest danger from a flow of current through the body?

- ☐ A ☐ The current through the heart can cause an irregular heartbeat.
☐ B ☐ Current through the heart can cause an irregular heartbeat.
☐ C ☒

6 Electrocutation is a hazard related to:

- ☐ A ☐ This is not the correct answer.
☐ B ☒
☐ C ☐ This is not the correct answer.

7 Injury due to an electrical current passing through the human body does not depend on:

- ☐ A ☐ Injury due to an electrical current does depend on this.
☐ B ☐ Injury due to an electrical current does depend on this.
☐ C ☒

8 The intensity of the current through the body depends on:

- ☐ A ☐ Resistance of the surface, including footwear.
☐ B ☒
☐ C ☐ Voltage.

9 How does a short circuit occur?

- ☐ A ☒
☐ B ☐ The voltages have to be different for this to happen.
☐ C ☐ This is not how a short circuit occurs.

10 The occurrence and size of an electric arc is determined by:

- ☐ A ☐ Both answers are correct.
☐ B ☐ Both answers are correct.
☐ C ☒

16 Electricity

11 Antistatic shoes are worn when:

- A ☐ This is not the correct answer.
- B ☒
- C ☐ This is not the correct answer.

12 A hazard of static electricity is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

13 In which of the situations stated will static electricity not occur?

- A ☒
- B ☐ In this situation static electricity will occur.
- C ☐ In this situation static electricity will occur.

14 What is not a protection method for housings?

- A ☒
- B ☐ This is a protection method.
- C ☐ This is a protection method.

15 A hazard associated with using power cables and extension cables is:

- A ☐ This is not a hazard.
- B ☐ This is a hazard, but not the most important one.
- C ☒

16 When using a cable reel one must:

- A ☒
- B ☐ This is not the correct answer.
- C ☐ You should always completely unwind the cable from the reel.

17 When a cable reel is not completely unwind:

- A ☐ This is not the correct answer.
- B ☒
- C ☐ This is not the correct answer.

18 The protection value of the earth leakage circuit breaker that must be added for extra safety in the temporary construction site junction box is:

- A ☐ For each outgoing power supply line it has to be 30 mA.
- B ☐ This has to be 30 mA.
- C ☒

19 A safety measure to take before working with electrical equipment is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

20 When using electricity:

- A ☐ The scaffolding must be earthed.
- B ☒
- C ☐ The scaffolding must be earthed.

16 Confined spaces

1 What is one feature of a confined space?

- ☒ C
- ☐ C Mostly, confined spaces have very bad lighting.
- ☐ C There is rarely enough work space in confined spaces.

2 Is a manhole guard compulsory while working in an enclosed space?

- ☐ C This person must not leave as long as there is someone in the confined space.
- ☐ C Yes. There must be a guard when someone is working in a confined space.
- ☒ C

3 How can a suffocation hazard occur in confined space?

- ☐ A This can not cause suffocation.
- ☐ B Suffocation is caused by insufficient oxygen.
- ☒ C

4 Which of the types of voltage below is safe?

- ☐ A Alternating current is more dangerous than direct current.
- ☐ B This is only true under dry circumstances.
- ☒ C

5 What is not an example of a confined space?

- ☐ A This is an example of a confined space.
- ☐ B This is an example of a confined space.
- ☒ C This is a characteristic, not an example of a confined space.

6 A rule related to carrying out measurements in a confined space is:

- ☐ A The measurements have to be carried out by an expert.
- ☐ B The measurements have to be carried out by an expert.
- ☒ C

7 When supervising a confined space, this should be taken into account:

- ☐ A Communication with the person in the confined place must take place to make sure everything is ok.
- ☒ B
- ☐ C The supervisor has to supervise the work activities within the confined space.

8 The permanent attendant must:

- ☐ A Posses demonstrably tested expertise.
- ☐ B Posses demonstrably tested expertise.
- ☒ C

9 The duration of stay in a confined space:

- ☐ A As short as possible
- ☒ B
- ☐ C As short as possible

10 A safety measure to reduce the risk of electrocution in a confined space is:

- ☒ A
- ☐ B Both answers are correct.
- ☐ C Both answers are correct.

17 Hoisting

17

1 When should hoisting operations be suspended?

- ☒ A ☐ Observe the crane manufacturer's user instructions for the value of the windforce.
- ☐ B ☐ Observe the crane manufacturer's user instructions for the value of the windforce.

2 Which is a hoisting tool?

- ☒ A ☐ This is not a lifting tool. A retaining strap is meant to keep a load on it's place.
- ☐ B ☐ This is not a hoisting tool but Personal Protection

3 What are two slings on a steel ring called?

- ☒ A ☐ This is called a bridle. Depending on the amount of slings, this one is called a two legged bridle.
- ☐ B ☐ This is called a bridle. Depending on the amount of slings, this one is called a two legged bridle.

4 What is recorded in a crane logbook

- ☐ A ☐ The crane logbook contains the inspections and maintenance activities the crane has had.
- ☐ B ☐ This is stated in the tables and graphs.
- ☒ C

5 Which of the statements below about chains is CORRECT?

- ☒ A ☐ Chains with stretched links should be disgarded from any use.
- ☐ B ☐ They may only be extended with a special coupling chain.

6 What is not an example of hoisting equipment?

- ☐ A ☐ This is an example of hoisting equipment.
- ☐ B ☐ This is an example of hoisting equipment.
- ☒ C

7 What can one do with hoisting equipment?

- ☐ A ☐ Both answers are correct.
- ☐ B ☐ Both answers are correct.
- ☐ C

8 A general danger when working with hoisting material is:

- ☐ A ☐ Both answers are correct.
- ☐ B ☐ Both answers are correct.
- ☒ C

9 Outriggers are used when working with:

- ☐ A ☐ Outriggers are not used when working with pillar drills.
- ☐ B ☐ Outriggers are not used when working with hazardous materials.
- ☒ C

10 Steel cables:

- ☒ A ☐ This will cause corrosion.
- ☐ B ☐ This can cause them to break sooner.

18 Radiation

18

1 Infrared radiation is?

- A ☒
- B ☐ Infrared radiation is not capable of creating ions. For this reason it is non-ionizing.
- C ☐ This only gets caused by naturally occurring substances.

2 A safety measure to take against ionising radiation is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

3 A medical examination is an obligation for employees who:

- A ☐ People who work with hazardous substances over a longer period have to get a medical examination.
- B ☒
- C ☐ This is not the correct answer.

4 Monitoring radiation safety and hygiene is a task of:

- A ☐ This is not a task of the supervisor.
- B ☒
- C ☐ This is not a task of the emergency response team.

5 What is the difference between ionising and non ionising radiation?

- A ☐ Only ionising radiation creates ions.
- B ☒
- C ☐ This is the other way around.

6 A safety measure to take to prevent non-ionising radiation damaging health is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

19 The ergonomic workplace

1 What does not determine conditions of the working environment?

- A ☒ This does not determine the conditions of the working environment.
- B ☐ This does determine the conditions of the working environment.
- C ☐ This does determine the conditions of the working environment.

2 A type of physical vibrations is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

3 What is not a health complaint which can be a result of vibrations to the body?

- A ☒ This is a health complaint with vibrations.
- B ☐ This is a health complaint with vibrations.
- C ☐ This is a health complaint with vibrations.

4 Maintaining moisture balance is a control measure:

- A ☐ To ensure working comfort.
- B ☐ To ensure working comfort.
- C ☒ To ensure working comfort.

5 A factor that influences physical stress is:

- A ☐ Both answers are correct.
- B ☒ Both answers are correct.
- C ☐ Both answers are correct.

6 A function of a break during the work is:

- A ☐ To prevent overloading.
- B ☒ To prevent overloading.
- C ☐ To prevent overloading.

7 Chronic tiredness is a health hazard:

- A ☐ Of static stress.
- B ☒ Of static stress.
- C ☐ Of static stress.

8 The relationship with your colleagues:

- A ☐ Influences mental stress.
- B ☐ Influences mental stress.
- C ☒ Influences mental stress.

9 The correct working position/ posture during manual lifting and moving is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

10 What is not a working condition in which work when standing is preferred?

- A ☐ This is a working condition.
- B ☒ This is a working condition.
- C ☐ This is a working condition.

11 The optimum sitting position during work is:

- A ☐ Boths answers are correct.
- B ☐ Boths answers are correct.
- C ☒ Boths answers are correct.

20 The ergonomic workplace

20

12 What is a guideline for correct sitting during work?

- A ☒
- B ☐ One must change position regularly.
- C ☐ This is not correct.

13 The advised maximum weight to be lifted of 25 Kg is a safety measure for:

- A ☐ One should never lift when seated.
- B ☒
- C ☐ When lifting with a crane, one can lift more then 25 Kg.

14 The correct posture during manual lifting is:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

20 Noise at the workplace

1 What is the effect of excessively loud noises in the workplace?

- A ☒ This depends on which kind of hearing protection is used and how loud the noise really is.
- B ☐ When you can't hear others, risks will only increase.
- C ☐ When you can't hear others, risks will only increase.

2 Which factors determine, amongst other things, whether noise is harmful?

- A ☐ Whenever you have to raise your voice, noise is harmful. This is caused by the strength of the noise.
- B ☒ Whenever you have to raise your voice, noise is harmful. This is caused by the strength of the noise.
- C ☐ Whenever you have to raise your voice, noise is harmful. This is caused by the strength of the noise.

3 Two equal sound sources together increase the sound (pressure) level by:

- A ☒ 3 dB.
- B ☐ 3 dB.
- C ☐ 3 dB.

4 Exposure to excess noise causes:

- A ☐ This is a cause of vibrations.
- B ☐ Excess noise can cause permanent hearing loss.
- C ☒ Excess noise can cause permanent hearing loss.

5 Doubling the distance to noises gives a reduction of the sound (pressure) level of approximately

- A ☐ 6 dB(A).
- B ☒ 3 dB(A).
- C ☐ 6 dB(A).

6 Having difficulty making phone calls is a symptom of:

- A ☐ The correct answer is permanent hearing loss.
- B ☐ This is not the correct answer.
- C ☒ This is not the correct answer.

7 The measurement unit used to express a sound (pressure) level is:

- A ☐ Decibel.
- B ☒ Decibel.
- C ☐ Decibel.

8 The correct order to take control of excess noise is:

- A ☐ 1. Reduce the sound (pressure) level of the sound source
2. Screen off the sound source
3. Damp noise vibrations in the vicinity
4. Use personal hearing protection
- B ☐ 1. Reduce the sound (pressure) level of the sound source
2. Screen off the sound source
3. Damp noise vibrations in the vicinity
4. Use personal hearing protection
- C ☒ 1. Reduce the sound (pressure) level of the sound source
2. Screen off the sound source
3. Damp noise vibrations in the vicinity
4. Use personal hearing protection

9 Which of these measures must be taken by the employer with these sound levels?

- A ☐ From 80 dB(A)
- B ☒ From 80 dB(A)
- C ☐ From 80 dB(A)

21 Personal Protective Equipment

1 Why should people in industry not wear metal safety helmets?

- A ☐ It will protect against falling objects, but it can also conduct electricity.
- B ☐ This is not correct.
- C ☒

2 For which of the activities below is a dust filter mask a form of breathing protection?

- A ☐ Dust filter masks do not protect against gases or fumes.
- B ☐ Dust filters masks do not protect against gases or fumes.
- C ☒

3 In which of the situations below do safety goggles offer a good protection?

- A ☒
- B ☐ When working above the head, it is better to wear a safety helmet for falling parts.
- C ☐ It is depending on which activities have to be doen within these confined spaces.

4 Which of the statements below is CORRECT? When working with a noise level higher than 85 DB (a)

- A ☐ Hearing protection is mandatory from 85 dB.
- B ☐ Hearing protection is mandatory from 85 dB.
- C ☒

5 PPE must be:

- A ☐ PPE must be tailored to the specific employee.
- B ☐ This depends on which work activities are carried out.
- C ☒

6 The difference between safety glasses and a welding mask are:

- A ☒
- B ☐ This is the other way around.
- C ☐ This is the other way around. Welding masks cover the entire face.

7 Too low oxygen concentration is a hazard for:

- A ☐ Breathing.
- B ☐ Breathing.
- C ☒

8 Self-contained breathing protection must be used:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

9 What is not a rule for using various forms of fall protection?

- A ☐ This is a rule for using fall protection.
- B ☒
- C ☐ This is a rule for using fall protection.

10 Warning clothing is used:

- A ☐ This is why insulating underwear is used.
- B ☒
- C ☐ This is why overalls are used.

22 Personal Protective Equipment

22

11 Slipping is a hazard to:

- A ☐ Feet and legs.
- B ☒ Feet and legs.
- C ☐ Feet and legs.

12 Various forms of hand and arm protection are:

- A ☐ These gloves do not exist.
- B ☒ These gloves do not exist.
- C ☐ These gloves do not exist.

13 A main property of a safety helmet is:

- A ☒ The interior absorbs a shock.
- B ☐ The interior absorbs a shock.
- C ☐ The exterior is sufficiently robust.

14 Hazards for the head during work are:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒ Both answers are correct.

15 A nose and mouth cover is used for:

- A ☐ Only against particles, not for gases or vapours.
- B ☐ Only against particles, not for gases or vapours.
- C ☒ Only against particles, not for gases or vapours.

22 Signs and markings

1 Which characteristics apply to a mandatory sign?

- A ☒
- B ☐ Warning signs are designed in this way.
- C ☐ Safety signs are designed in this way.

2 Which requirements apply to warning signs at the workplace?

- A ☒
- B ☐ They should be positioned in such a way that they will be noticed by those for whom they are intended.
- C ☐ Warning signs should be understood by people from different countries. This means the text cannot only be in Dutch.

3 A group of signs is:

- A ☒
- B ☐ These are signs categorized under safety provision signs.
- C ☐ These signs are categorized under information signs.

4 A sign that prohibits something is:

- A ☐ The colours are organized the other way around.
- B ☐ This is a warning sign.
- C ☒

5 A triangular yellow sign with a black edge is the design of:

- A ☐ A warning sign.
- B ☒
- C ☐ A warning sign.

6 The general meaning of warning signs is:

- A ☐ Caution, danger.
- B ☐ Caution, danger.
- C ☒

7 A square or rectangular green sign with a white symbol is the design for:

- A ☐ Safety provision signs.
- B ☐ Safety provision signs.
- C ☒

8 Fire-fighting equipment is located at signs which are:

- A ☐ Square or rectangular and red with a white sign.
- B ☒
- C ☐ Square or rectangular and red with a white sign.

9 The general meaning for fire-fighting equipment:

- A ☐ Both answers are correct.
- B ☐ Both answers are correct.
- C ☒

10 Yellow or white stripes are used for:

- A ☐ Markings of passages and stacking areas.
- B ☒
- C ☐ Markings of passages and stacking areas.

